



D2.2

Social Acceptability Workshops

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Abstract

Deliverable D2.2 presents the outcomes of Task 2.2 within WP2 of the AntifragiCity project, reporting on a series of Social Acceptability Workshops conducted across three demonstration sites: Bratislava, Larissa, and Thessaloniki. These workshops applied a citizen assembly-inspired framework to explore public perspectives on urban mobility interventions under both everyday and crisis conditions. The analysis draws on complementary quantitative and qualitative data: pre- and post-workshop surveys captured participants' baseline perceptions, shifts in priorities, and social acceptability judgments, while structured facilitation outputs and thematic synthesis of open-ended responses provided in-depth insight into reasoning and deliberative dynamics. This dual approach enabled assessment of both process-related dimensions and substantive outcomes.

Findings indicate consistently high support for interventions that strengthen system-level capacity, accessibility, safety, and resilience. Measures imposing strict behavioral, spatial, or economic constraints were less accepted, highlighting the importance of visible, inclusive, and coordinated approaches. Cross-site comparisons reveal context-specific engagement patterns but also convergent principles underpinning socially acceptable crisis mobility planning, including fairness, transparency, and protection of vulnerable users. Overall, the deliverable provides empirically grounded insights to guide the design of antifragile urban mobility strategies. By combining deliberative processes with rigorous data collection, D2.2 strengthens the societal and governance basis for policy development, modelling, and decision-making in both routine and crisis-oriented mobility planning.

Keywords

Antifragility, Mobility Resilience, Urban Disruption, Citizens' Assemblies, Social Acceptability, Urban Mobility, Deliberative Workshops, Mobility Prioritization, Behavioral Responses

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CO Confidential to ANTIFRAGICITY project and Commission Services

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*** Deliverable types:**

R: document, report (excluding periodic and final reports).

DEM: demonstrator, pilot, prototype, plan designs.

DEC: websites, patent filings, press and media actions, videos, etc.

OTHER: software, technical diagrams, etc.

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Abbreviations

ACF	AntifragiCity Citizens' Forum
AUTH	Aristotle University of Thessaloniki
BAF	Bratislava AntifragiCity Forum
CA	Citizens' Assembly
DEPP-NE	Department of European Programs & Promotion of New Entrepreneurship
DMP	Deliberative Mini-Public
IED	Institute for Entrepreneurship Development
LAF	Larissa AntifragiCity Forum
LISER	Luxembourg Institute of Socio-Economic Research
MIB	Metropolitan Institute of Bratislava
OECD	Organization for Economic Cooperation and Development
T	Task (e.g., T2.2 refers to Task 2.2)
TAF	Thessaloniki AntifragiCity Forum
WP	Work Package (e.g., WP2 refers to Work Package 2)

1 Executive summary

This deliverable 2.2 (D2.2) presents the outcomes of Task 2.2 (T2.2) within Work Package 2 (WP2) of the AntifragiCity project, focusing on the social acceptability of urban mobility interventions under conditions of everyday use and crisis or disruption. It reports on a series of Social Acceptability Workshops conducted across three demonstration sites, Bratislava, Larissa, and Thessaloniki, implemented through a citizens' forum approach that enabled structured, informed, and facilitated deliberation among diverse groups of residents.

The methodology combined preparatory expert discussions to define key thematic areas with deliberative workshops integrating expert presentations and facilitated participatory discussions. Participants were recruited through an open communication strategy, aiming for diversity in socio-demographic characteristics, mobility practices, and lived experiences. Pre- and post-workshop surveys and qualitative outputs provided complementary insights, capturing both process-related dynamics (e.g., inclusiveness, legitimacy, and quality of deliberation) and substantive outcomes (e.g., preferred mobility regimes, conditions for social acceptability, and acceptable trade-offs).

Key findings indicate strong public support for interventions that enhance system-level capacity, safety, accessibility, and resilience. Measures imposing strict behavioral, spatial, or financial constraints were less accepted, emphasizing the importance of visible, inclusive, and coherent systemic solutions. Across all sites, citizens evaluated interventions holistically, prioritizing fairness, transparency, institutional responsibility, and the protection of vulnerable users.

Methodologically, the deliverable demonstrates the value of deliberative processes as both participatory and analytical instruments. By enabling participants to engage with expert knowledge, exchange perspectives, and reflect on trade-offs, the forums provide insight into how preferences are formed and potentially evolve through deliberation. This goes beyond conventional approaches based on static preference measurement by capturing the reasoning processes underlying citizens' judgments.

Overall, this deliverable provides empirically grounded insights to inform the design of antifragile urban mobility strategies. It strengthens the societal and governance foundations for subsequent modelling, scenario development, and policy design activities within the AntifragiCity project. By embedding citizens' perspectives into the analysis of urban mobility systems, D2.2 contributes to ensuring that future interventions are not only technically robust, but also socially acceptable, inclusive, and aligned with the lived realities of urban populations across diverse European contexts.

2 Introduction

Deliverable 2.2 (D2.2), “Social Acceptability Workshops,” constitutes the primary output of Task 2.2 (T2.2) within Work Package 2 (WP2) of the AntifragiCity project. T2.2 investigates the social acceptability of urban mobility interventions by directly engaging citizens through structured, deliberative processes, the AntifragiCity Citizens’ Forums (ACFs), implemented across three demonstration sites: Bratislava, Larissa, and Thessaloniki. The deliverable aims to generate empirically grounded insights into public perceptions, behavioral responses, and normative expectations regarding urban mobility systems under both routine conditions and disruptive or crisis scenarios.

The AntifragiCity project addresses the growing need to understand urban mobility as a socio-technical and experiential system shaped by interactions between infrastructure, technologies, institutions, and everyday practices. In this context, the concept of antifragility provides a guiding framework, emphasizing not only the capacity of systems to withstand shocks, but also their ability to adapt and improve under conditions of stress and uncertainty. Social acceptability is therefore not treated as a static constraint on policy design, but as a dynamic and evolving process that plays a critical role in shaping the feasibility and success of mobility interventions.

This deliverable contributes to WP2 by complementing event-based analyses of urban mobility disruptions with a societal and governance-oriented perspective. While other components of the project focus on mapping and modelling urban events and system responses, T2.2 focuses on how citizens perceive, evaluate, and reason about mobility interventions in these contexts. Hence, rather than focusing on technical optimization or predictive modelling, D2.2 documents how citizens reason about urban mobility under both everyday and disruptive conditions. By doing so, it provides an essential layer of understanding regarding the human and behavioral dimensions that underpin mobility system performance and transformation.

The deliverable addresses four core objectives aligned with T2.2 and the workshop design:

Objective 1: Design, implement, and evaluate a deliberative, citizens’ assembly-based methodology to elicit informed public perspectives on urban mobility, encompassing both routine, day-to-day mobility experiences and adaptive measures required during disruptive or crisis events.

Objective 2: Examine process-related dimensions of deliberative engagement – such as perceived legitimacy, inclusiveness, trust, and quality of deliberation – alongside substantive outcomes, including preferred mobility regimes, conditions for social acceptability, and governance expectations.

Objective 3: Evaluate urban mobility interventions and policy alternatives, focusing on prioritization, triage, and trade-offs affecting diverse user groups in normal and crisis conditions.

Objective 4: Distill shared principles and context-specific concerns to inform socially robust models, scenarios, and policy recommendations for antifragile urban mobility planning in subsequent work packages.

The methodology adopted in this deliverable combines preparatory expert consultations with deliberative workshop sessions. Expert inputs were used to define thematic areas and ensure that participants were exposed to balanced and relevant information. The workshops themselves provided structured environments for facilitated group discussions, enabling participants to reflect collectively on mobility challenges, exchange perspectives, and articulate reasoned judgments. Importantly, the workshops were conceived as deliberative citizen forums rather than conventional survey instruments or consultation exercises. Participants were not only asked to express opinions, but also to engage with information, interact with other participants, and refine their views through discussion. This emphasis on deliberation enables the capture of how citizens reason about complex and contested policy issues, including how they interpret trade-offs and evaluate competing priorities.

Participants were recruited through open and locally adapted communication strategies designed to ensure diversity in socio-demographic characteristics, mobility behaviors, and lived experiences. This approach reflects the objective of capturing a wide range of perspectives rather than statistical representativeness. A mixed-methods design was applied, integrating pre- and post-workshop surveys with facilitation outputs. This combination allows for a comprehensive assessment of both deliberative processes and their outcomes.

The primary contribution of this deliverable lies in its application of deliberative methods to the study of urban mobility across multiple European contexts within an antifragility framework. While participatory governance approaches and social acceptability research are well established individually, the structured use of deliberation to explore mobility disruption, prioritization, and trade-offs across multiple European contexts remains a novel contribution. This deliverable establishes a critical societal baseline that complements the urban event mapping outputs of WP2 and supports the co-development of publicly legitimate, antifragile urban mobility strategies in later stages of the AntifragiCity project. The insights generated form the foundation for subsequent work packages, where social acceptability findings will be translated into quantitative models, decision-support tools, and policy frameworks to guide the development of antifragile, inclusive, and socially robust urban mobility strategies.

3 AntifragiCity Citizens' Forums

The ACFs were implemented at three demonstration sites – Bratislava, Larissa, and Thessaloniki – and constitute the primary empirical component underpinning D2.2. Their primary objective was to elicit informed and deliberative insights into citizens' perspectives on urban mobility, both under routine conditions and in the context of disruptions or crisis situations. In this sense, the ACFs functioned as structured spaces in which socially grounded understandings of urban mobility could emerge through dialogue, reflection, and interaction.

Beyond gathering citizens' views, the forums were explicitly designed as an applied implementation of deliberative democratic principles. They therefore also served as an experimental setting to assess the quality, feasibility, and relevance of deliberative processes for urban mobility governance, particularly within research-oriented and multi-site contexts. In doing so, they contribute to ongoing discussions on how deliberative approaches can support complex socio-technical transitions, such as those required in the field of urban mobility.

3.1 Forum design and implementation

3.1.1 A note on citizens' assemblies

In recent years, there has been a notable rise in new forms of citizen participation around the globe (Smith, 2009; Scarrow, 2001). According to the Organization for Economic Cooperation and Development (OECD), more than 700 deliberative processes have been implemented worldwide, with numbers continuing to grow (OECD, 2023). This expansion has led scholars to describe a broader 'deliberative wave', reflecting the increasing use and visibility of deliberative processes (Paulis et al., 2021; OECD, 2020; Escobar & Elstub, 2019; Courant & Sintomer, 2019). These developments are grounded in the concept of deliberative democracy, which marks a shift away from purely vote-centric models of democratic decision-making toward approaches emphasizing dialogue, mutual respect, and reasoned argumentation (Chambers, 2003). Rather than simply aggregating pre-existing preferences, deliberative processes aim to create conditions in which preferences are formed, examined, and revised through interaction. In this sense, deliberation is not only a decision-making procedure but also a transformative process of collective reasoning.

Deliberative democracy should be understood as an umbrella concept encompassing a wide range of institutional designs. These include deliberative mini-publics (DMPs), citizens' juries, deliberative polls, and participatory budgeting (Dryzek, 2010; Fung, 2003). Among them, deliberative mini-publics have emerged as particularly prominent instruments, as they combine representational diversity, typically achieved through random or stratified selection, with intensive, facilitated

deliberation (Dryzek, 2009; Chwalisz, 2015). This combination enables a depth of reflection that is rarely attainable in large-scale participation formats.

Within this category, citizens' assemblies (CAs) are widely regarded as one of the most comprehensive and influential formats (Elstub & McLaverty, 2014). Per Fournier et al. (2011, p. 10), CAs represent "the most extensive modern form of collective decision-making by common folk." Since their first appearance, they have been implemented across multiple governance levels – from local to transnational contexts – to address complex policy challenges. Prominent examples include the Belgian G1000, the French *Convention Citoyenne pour le Climat*, the Irish Constitutional Convention, the UK Climate Assembly, the Conference on the Future of Europe, and the Global Assembly. These cases demonstrate the versatility of CAs in terms of scale, design, and institutional role. While many involve a small number of participants (Grönlund et al., 2014), some designs can accommodate significantly larger groups (Caluwaerts & Reuchamps, 2015). Their roles in the broader policy process also vary widely, functioning anywhere from consultative bodies to decision-making entities with substantial authority (Smith, 2009).

Despite their diversity, CAs typically share several core defining features. Participants are usually selected through random or stratified sampling to reflect the socio-demographic composition of the broader population. They are provided with balanced and accessible expert input and then engage in equal and respectful discussions within facilitated working groups (Bächtiger & Parkinson, 2019; Owen & Smith, 2015; Goold et al., 2012; Bächtiger et al., 2010; Mansbridge et al., 2010; Dryzek, 2002). Facilitation plays a central role in ensuring that deliberation remains inclusive, structured, and oriented toward mutual understanding rather than adversarial debate. Ultimately, participants collectively develop judgments or recommendations on societal issues to inform decision-making processes and ideally play a critical role in shaping policy (Curato et al., 2021; Elstub, 2014; Grönlund et al., 2014; Ryan & Smith, 2014; Smith, 2012; Steiner, 2012; Dryzek, 2009; Fishkin, 2009).

CAs typically address complex, cross-cutting societal challenges such as climate change, energy transition, and urban development. These issues are characterized by uncertainty, long-term implications, and competing values, requiring the integration of technical expertise with societal perspectives. In such contexts, CAs provide a structured environment for navigating complexity and articulating trade-offs, where traditional expert-driven or purely representative mechanisms may be insufficient on their own.

Advocates of deliberative processes argue that they can enhance both the quality and legitimacy of decision-making. By fostering inclusive participation and informed discussion, they contribute to more representative, accountable, and well-reasoned outcomes (Lacelle-Webster & Warren, 2021; Elstub & Escobar, 2019; Curato & Böker, 2016; Bächtiger et al., 2014; Niemeyer, 2014; MacKenzie & Warren, 2012; Dryzek, 2010; Smith, 2009; Fung, 2007, 2006; Goodin & Dryzek, 2006; Saward, 2000). Empirical research further suggests that deliberation improves the epistemic quality of decisions by enabling participants to engage with evidence, confront diverse perspectives, and revise their views.

Importantly, deliberative settings do not merely aggregate preferences but actively shape how they are formed and transformed through processes of reflection and argumentation (Bächtiger & Parkinson, 2019; Bächtiger et al., 2018; Mansbridge et al., 2010; Fishkin, 2009; Cohen, 2005, 1997; Gutmann & Thompson, 2004; Chambers, 2003; Dryzek, 2000; Elster, 1998). In other words, through deliberation, participants not only deepen their knowledge but may also reconsider and transform their preferences, ultimately shaping strategies and policies in a more informed and collaborative manner.

This dynamic is particularly relevant in the context of urban mobility. Mobility systems are increasingly understood as socio-technical and experiential configurations shaped by interactions between infrastructure, technologies, institutions, and everyday practices (Graham & Healey, 2007; Vigar, 2017). As a result, purely technical planning approaches are insufficient to capture how mobility systems are experienced and used in practice. Deliberative processes can complement these approaches by incorporating experiential knowledge and revealing how policies are perceived and evaluated by citizens. Empirical evidence indicates that preferences related to mobility policies are not fixed but evolve through deliberation, as individuals reflect on information, engage with alternative viewpoints, and reconsider initial positions (Allert, 2025).

Closely related to this is the concept of social acceptability, defined as individuals' evaluative orientation toward policy measures based on perceived effectiveness, fairness, and personal implications (Schade & Schlag, 2003; Schuitema & Steg, 2008). These perceptions are dynamic and shaped through information, lived experience, and social interaction. Deliberative settings therefore provide a valuable context for examining how acceptability is formed and negotiated. More broadly, such processes contribute to collective learning, the articulation of trade-offs, and the negotiation of competing values, thereby enhancing the adaptive capacity of socio-technical systems under conditions of uncertainty (Ensor et al., 2018; Wardekker et al., 2020). In this regard, deliberation can be understood as supporting the development of antifragile systems – systems that improve through stress and adaptation rather than merely resisting disruption (Taleb, 2012).

Building on Scharpf (1999), participation processes can be understood as serving two complementary purposes: improving the quality of policy inputs and enhancing the legitimacy and acceptance of policy outputs. In deliberative contexts, these dimensions are closely intertwined, as inclusive and reasoned discussion can simultaneously generate more robust inputs while fostering public understanding and acceptance of outcomes. This dual function is particularly important in mobility transitions, where public support is a critical precondition for implementing policy measures (Banister, 2008). Engaging citizens can improve policy quality, legitimacy, and implementability, helping to align solutions with societal values, while also addressing issues of social inclusion and diversity (Sagaris, 2014; Lindenau & Böhler-Baedeker, 2014; Giering, 2011; Grant, 2005).

These dynamics are especially salient given that mobility practices are deeply embedded in social norms and everyday routines, making behavioral change both necessary and contested. Deliberative formats that facilitate active engagement with information and diverse perspectives are therefore considered promising tools for supporting socially just transition processes. They enable participants to reflect on their own positions, develop arguments, and reconsider viewpoints, while fostering a sense of collective agency in addressing complex challenges. In this regard, deliberative approaches, and particularly CAs, have been proposed as a way to strengthen the role of citizens as active contributors with rights, voices, and responsibilities in shaping mobility futures (Sørensen & Paulsson, 2020).

Recent applications of CAs in the field of urban mobility and urban development illustrate both their growing relevance and methodological diversity. Examples include assemblies in Kraków (2023) and Oxfordshire (2025) on transportation, a smaller-scale process in Burgwedel (Hanover, 2024) on sustainable mobility, the Zhytomyr assembly (2025) focusing on safety and accessibility, and the Érd Citizens' Assembly (2022) addressing the transformation of a commuter town. Additional initiatives in Vilnius (2025) and Esch (2026) further highlight the use of deliberative processes to explore broader questions of urban livability and sustainability. Across these cases, mobility is treated not as an isolated sector but as intrinsically linked to wider urban systems.

In this broader context, the ACFs build on the methodological foundations of CAs to support the development of antifragile urban mobility systems. Their design reflects an effort to operationalize deliberative principles in a research-oriented, multi-site setting, positioning deliberation not only as a consultative tool but as a mechanism for collective learning, preference formation, and the co-production of knowledge.

3.1.2 Adaptations in the AntifragiCity context

The adaptations introduced in the AntifragiCity context can be understood as pragmatic trade-offs between the normative ideals of deliberative mini-publics and the practical constraints of multi-site, research-oriented implementation. This is particularly relevant given the complexity of urban mobility systems and the importance of capturing evolving citizen perspectives, as discussed in Section 3.1.1.

Such trade-offs are widely acknowledged in the literature, which emphasizes that deliberative designs must remain context-sensitive while preserving core principles such as diversity, informed discussion, and facilitated reasoning (Verhasselt, 2025). Rather than representing a limitation, these adaptations can be understood as an inherent feature of applying deliberative methodologies in real-world settings, where institutional, logistical, and socio-political conditions shape implementation.

In this sense, the ACFs reflect an approach consistent with recent deliberative practice, where methodological robustness is balanced with feasibility, and where the objective is not to replicate

ideal-type CAs, but to meaningfully operationalize their underlying principles in context-specific environments.

3.1.2.1 Terminology

The AntifragiCity Forums were conceptually inspired by the CA model, particularly its emphasis on diversity, informed deliberation, and collective reasoning. In line with this approach, the forums convened diverse groups of residents (aged 18 and above) to deliberate on urban mobility challenges and co-develop context-sensitive recommendations.

However, the term “citizens’ assembly” carries relatively rigid methodological expectations, including assumptions regarding scale, representativity, duration, and formal policy linkage. These expectations are often associated with processes that have a clear institutional mandate and a direct connection to decision-making.

Given the practical constraints of the AntifragiCity project – such as limited timeframes, resource considerations, and the need for adaptability across multiple demonstration sites – a more flexible approach was required. This is consistent with findings from previous deliberative processes, which demonstrate considerable variation in scale (e.g., Burgwedel, 14 participants; Vilnius, 43; Kraków, 80) and design, depending on context and objectives.

For this reason, the term “citizens’ forum” was adopted to describe the participatory workshops initially referred to as “citizens’ assembly workshops” in T2.2. This revised terminology reflects a pragmatic adaptation of deliberative principles to local contexts, while avoiding the prescriptive expectations typically associated with full-scale CAs.

Importantly, this adaptation concerns format rather than methodological intent. The AntifragiCity Forums retained the defining features of deliberative processes, including:

- Diversity of perspectives
- Provision of balanced and accessible information
- Structured and facilitated deliberation
- The co-creation of collective outputs

By adopting the “citizens’ forum” format, the project was able to leverage existing local expertise and organizational capacities, ensuring feasibility without compromising methodological integrity. This approach also facilitated smoother integration with established workshop practices at the demonstration sites – particularly crucial as they were not experienced with the CA methodology.

Crucially, this flexibility aligns with the broader understanding that deliberative processes should be adapted to the socio-technical and institutional contexts in which they are applied. In the domain of urban mobility – where issues are embedded in everyday practices and local conditions – such adaptability is essential for capturing meaningful, context-sensitive insights (Graham & Healey, 2007; Vigar, 2017).

In essence, while the format was adapted to suit project-specific constraints, the overarching objective remained consistent: to capture citizens' informed preferences, priorities, and reasoning regarding urban mobility, particularly in the face of uncertainty and disruption, through structured and reflective deliberation. In this sense, the ACFs can be understood as context-sensitive DMPs, designed to balance methodological robustness with feasibility and local adaptability, while remaining aligned with the principles outlined in Section 3.1.1.

3.1.2.2 Delay

The delay in D2.2 reflects the methodological demands associated with implementing deliberative processes across multiple sites.

CAs require careful design to ensure meaningful participation, balanced information provision, and high-quality facilitation. For all project partners, CAs represented a novel approach, requiring additional time for preparation, alignment, capacity building – particularly in establishing a shared understanding of deliberative principles and practices.

Key factors contributing to the delay included:

- Methodological alignment: Establishing a shared understanding of deliberative principles (e.g., facilitation, inclusiveness, neutrality).
- Scheduling and coordination: Aligning availability of local partners, facilitators, and venues, while avoiding conflicts with local events, elections, and public holidays.
- Participant recruitment: Ensuring diversity, consistent with deliberative ideals.
- Administrative and ethical approvals: Certain sites required additional time for internal or ethical clearances, particularly regarding data protection and consent.
- Training needs: High-quality deliberation depends on skilled moderation, necessitating dedicated training.
- Cross-site harmonization: Aligning materials and processes to ensure comparability across demonstration sites.

Mitigation measures included:

- Structured facilitator training to build shared methodological capacity
- Provision of standardized materials and templates (guidelines, milestones, facilitation tools, consent forms, participant documentation)
- Additional coordination calls and tailored support sessions to address site-specific challenges
- Proactive adjustment of local schedules to maintain inclusiveness and feasibility

From a deliberative perspective, these preparatory steps are not ancillary but central to ensuring process quality. Effective deliberation depends on participants' ability to engage with information, reflect on differing perspectives, and articulate reasoned judgments (Bächtiger & Parkinson, 2019; Fishkin, 2009). As such, investment in preparation and facilitation can be understood as a prerequisite for achieving meaningful outcomes.

Essentially, a quality-first approach was adopted, prioritizing the integrity of the deliberative process over strict adherence to the initial timeline.

The forums were successfully conducted in Months 9–10 (January–February 2026). The revised schedule ensured methodologically consistent deliberative processes, in line with the principles outlined in Section 3.1.1., and well-documented implementation across all demonstration sites.

3.1.2.3 Odesa

T2.2 initially envisaged a Social Acceptability Workshop at a fourth demonstration site, Odesa. Preparatory work included meetings and exchanges with local individuals and organizations experienced in participatory and deliberative processes. However, given the ongoing situation in Ukraine and associated risks for participants, it was deemed unsafe to convene an in-person workshop in Odesa within the timeframe of this deliverable. The workshop at this site was therefore suspended to prioritize participant safety.

This decision reflects a fundamental requirement of deliberative processes: they depend on conditions that enable safe, inclusive, and meaningful participation. In contexts affected by conflict or instability, such conditions cannot be guaranteed, and proceeding with deliberation may risk excluding participants, distorting participation, or undermining the integrity of the process. From a broader perspective, this highlights the importance of context sensitivity and ethical responsibility in the implementation of deliberative methodologies. It also underscores that deliberative processes are inherently embedded in socio-political environments and must remain adaptable to changing conditions.

In line with the project's focus on antifragility, this adaptation can be understood not merely as a disruption but as a necessary adjustment to external stressors, reinforcing the need for flexibility in the design and implementation of participatory processes. The suspension of the Odesa forum therefore illustrates both the limitations and the resilience of deliberative approaches in real-world settings.

Although the in-person workshop in Odesa could not be conducted, alternative steps will be undertaken within the project to capture relevant insights into social acceptability and establish a contextual social baseline. These include activities under Task T2.4, as well as in-depth sentiment analysis. Together, these complementary approaches aim to ensure that the perspectives, attitudes, and contextual dynamics relevant to urban mobility in Odesa are still reflected in the broader analytical framework of the project, despite the absence of a physical deliberative forum at this site.

3.1.3 Participant recruitment and selection

The recruitment strategy was designed in alignment with both the objectives of T2.2 and established principles of DMPs (Section 3.1.1). It also took into account the research-oriented EU context, local governance structures, and operational constraints. The aim was not statistical representativity per se, but the creation of diverse groups capable of informed and high-quality deliberation.

Importantly, this approach reflects a broader understanding within deliberative research that the value of participation lies not only in mirroring population distributions, but in enabling the interaction of diverse perspectives through structured dialogue. In this sense, diversity was conceived not as a purely statistical objective, but as a condition for generating meaningful deliberation, collective learning, and the articulation of trade-offs (Ensor et al., 2018; Wardekker et al., 2020).

3.1.3.1 Governance context and implications for participation

Unlike many CAs, which are directly linked to policymaking processes, the ACFs were primarily research-oriented. Their purpose was to generate insights at the EU level rather than to inform binding decisions at the local level.

This distinction is important from a deliberative perspective. Participation in mini-publics is often higher when there is a clear link to decision-making and perceived impact. In other words, deliberative processes generally achieve higher participation rates when there is a clear line of accountability and a tangible expectation that outcomes will feed directly into decisions made at a governance level close to participants' everyday lives.

At the same time, participation processes serve dual functions: they can both improve the quality of policy inputs and enhance the legitimacy and acceptance of outputs (Scharpf, 1999). In the absence

of a direct decision-making mandate, the ACFs primarily emphasized the former, focusing on the generation of informed, reflective, and context-sensitive insights.

This is particularly relevant in the domain of urban mobility, where policy measures often depend on public acceptance and alignment with everyday practices (Banister, 2008; Schade & Schlag, 2003). Even in a research-oriented setting, deliberative engagement provides valuable insights into how such acceptability is formed, contested, and potentially transformed through interaction and reflection. In this case, success was therefore assessed in terms of deliberative quality and diversity of perspectives, rather than scale or formal representativity.

3.1.3.2 Recruitment approach

Recruitment for the AntifragiCity Forums relied on an open communication approach, inviting interested individuals to register voluntarily. This open, voluntary registration model was designed to maximize reach and feasibility across all pilot cities. Calls for participation were disseminated through multiple channels, including project and partner platforms, local partner networks, and where available, official communications such as city newsletters.

Direct invitations, such as formal household invitations, were not used. Such invitations are typically associated with deliberative processes mandated by public authorities with a clearly defined policy remit and explicit commitment to consider participant recommendations. They also require access to population-level datasets, which were not uniformly available to local partners and could introduce implementation asymmetries across sites. The chosen approach therefore represents a pragmatic adaptation, consistent with the flexible application of deliberative principles discussed in Section 3.1.2.

Insights from prior deliberative processes indicate that response rates to direct invitations are generally low, often in the range of 3–6%, even when formal household letters and financial incentives are provided. Low participation reflects structural barriers such as time constraints, competing responsibilities, perceived relevance of the topic, and uncertainty regarding the tangible impact of participation.

The AntifragiCity recruitment approach hence represents a deliberate trade-off. While voluntary registration can introduce self-selection bias – potentially favoring the so-called ‘usual suspect’ – this was consciously accepted to ensure feasibility and methodological consistency across sites with differing institutional capacities.

Importantly, this approach also aligns with the objective of fostering high-quality deliberation. Participants who self-select are more likely to engage actively, reflect on complex issues, and sustain participation over the duration of the process. Given that deliberation relies on interaction, argumentation, and the reconsideration of viewpoints (Allert, 2025; Bächtiger & Parkinson, 2019),

such engagement is a critical condition for achieving meaningful outcomes. Furthermore, in the context of mobility transitions, where practices are deeply embedded in everyday life, recruiting participants with lived experience and interest in the topic enhances the ability to capture experiential knowledge and context-sensitive perspectives (Graham & Healey, 2007; Vigar, 2017). This is particularly relevant for examining social acceptability, which emerges through reflection, interaction, and the negotiation of perceived trade-offs (Schuitema & Steg, 2008).

The success of recruitment was therefore evaluated not by absolute participant numbers or statistical representativity, but by the ability to convene diverse groups capable of sustained, high-quality deliberation. This approach also ensured that participants had a genuine interest in urban mobility, enhancing the depth and relevance of discussions, an important consideration given the two-day duration of the forums.

While the adopted recruitment approach is less likely to produce statistically representative samples – a limitation that applies, to varying degrees, to all deliberative processes regardless of outreach strategy – efforts were made to ensure variation in socio-demographic characteristics, mobility practices, and lived experiences. These included differences in age, gender, employment status, mobility needs, and everyday travel patterns. In line with findings from previous deliberative processes, such diversity is essential not only for representational reasons but also for enabling the confrontation of perspectives, the articulation of competing values, and the co-production of more robust and context-sensitive insights. However, as with all deliberative processes involving a limited number of participants, it is not possible for the ACFs to produce statistically representative samples of the broader urban population.

3.1.3.3 Summary

Overall, the recruitment and selection strategy reflects a deliberate balance between normative ideals of deliberative democracy and practical feasibility constraints. The resulting forums should therefore be understood as deliberative research instruments, designed not only to capture citizens' preferences but also to examine the processes through which these preferences are formed, negotiated, and transformed through structured interaction.

In this sense, they remain consistent with the core logic of CAs while being adapted to a research-oriented context. At the same time, they contribute to a growing body of work exploring how deliberative methodologies can be applied in real-world settings characterized by institutional constraints, limited resources, and varying governance conditions. By prioritizing deliberative quality, experiential knowledge, and context sensitivity, the ACFs provide insights into how citizen participation can support more adaptive, inclusive, and socially grounded approaches to urban mobility transitions.

3.2 Thematic areas

The AntifragiCity Forums were designed to enable participants to provide informed responses and actionable recommendations to the central question:

“What urban mobility will make your city more liveable today and prepared to respond to future crises?”

This guiding question reflects the growing use of deliberative processes to address complex urban mobility challenges through citizen engagement and builds directly on established practices in CAs on urban mobility, which typically frame transport not as an isolated technical issue but as a means to achieve broader societal goals. For instance, the Kraków Transport Assembly (2023) asked how both the city and its residents could shape the development of sustainable transport measures, while the Oxfordshire Citizens’ Assembly focused on the long-term question of how transport systems can support health, economic development, and environmental sustainability by 2050. Processes such as the Burgwedel (Hanover) assembly (2024) explored what sustainable mobility should look like in everyday contexts, and the Cologne Citizens’ Assembly (2025) explicitly linked mobility to well-being and neighborhood-level quality of life.

Other assemblies further demonstrate how mobility-related questions are often embedded within broader urban development and social considerations. In Vilnius (2025), the focus was on encouraging shifts toward public transport, walking, and cycling across different parts of the city, while in Zhytomyr (2025), the emphasis was placed on accessibility, safety, and inclusiveness for all population groups. Similarly, the assembly in Érd (2022) addressed the transformation of a commuter town into a more livable and vibrant urban environment, highlighting the close relationship between mobility, urban form, and community life. The Citizens’ Assembly in Esch (2026), although framed around public space and social cohesion, further illustrates how mobility is intrinsically linked to broader questions of urban livability and social interaction.

Taken together, these examples demonstrate that deliberative processes on urban mobility rarely focus on transport in isolation. Instead, they typically situate mobility within wider discussions on livability, sustainability, accessibility, and social cohesion. The central question of the ACFs builds on this established approach while extending it by explicitly incorporating uncertainty, disruption, and crisis preparedness. In doing so, it reflects the project’s focus on antifragility, emphasizing not only how mobility systems function under normal conditions but also how they adapt under stress. It then also extends existing deliberative practice by incorporating uncertainty and disruption, key dimensions that are often implicit but less frequently addressed explicitly in comparable processes.

The question was co-developed with project partners and local stakeholders to ensure alignment with overarching project objectives while remaining sensitive to the specific social, political, and

urban contexts of each demonstration site. The formulation also reflects insights from deliberative research indicating that citizens' preferences evolve through engagement with complex issues, interaction with diverse perspectives, and collective reflection (Allert, 2025). Furthermore, it acknowledges that urban mobility systems are embedded in broader socio-technical and experiential contexts, requiring approaches that capture not only technical performance but also lived experiences and societal expectations (Graham & Healey, 2007; Vigar, 2017).

Given the complexity of the question and the limited duration of the two-day forums, the design followed established best practices in deliberative processes by structuring discussions around clearly defined thematic areas (see Figure 1). This approach is consistent with the design of many CAs, where complex policy questions are broken down into manageable and complementary dimensions to enable focused and in-depth deliberation. Each topic represented a distinct perspective on urban mobility, providing participants with a clear framework to guide their deliberations while allowing for in-depth, context-sensitive discussion. The thematic areas were defined as follows:

Liveable cities – people, space, and quality of life: Designing urban spaces and transport systems that ensure a city for everyone. How urban mobility shapes everyday life, public spaces, and citizens' well-being. Includes air quality, noise, pedestrian/cycling infrastructure, and social inclusivity.

Crisis-responsive mobility – priorities during emergencies : Ensuring mobility systems can adapt and reprioritize during emergencies (e.g., floods, heatwaves, pandemics, energy/fuel shortages). The emphasis is on resilience, preparedness, and rapid reconfiguration of transport systems.

Resilient and diverse mobility – multiple options for a trip: Building redundancy, flexibility, and choice into mobility networks. Ensuring that when one mode fails, others remain viable. Includes walking, cycling, public transport, shared mobility, micro-mobility, and car-sharing.

Sustainable mobility transition – Low-carbon and green: Transforming mobility to promote long-term sustainability. Shaping long-term transitions to low-carbon mobility systems. Exploring how cities can shift to sustainable transport (walking, cycling, public transport, EVs, shared systems) while ensuring fairness, affordability, and adaptability to ecological and social challenges.

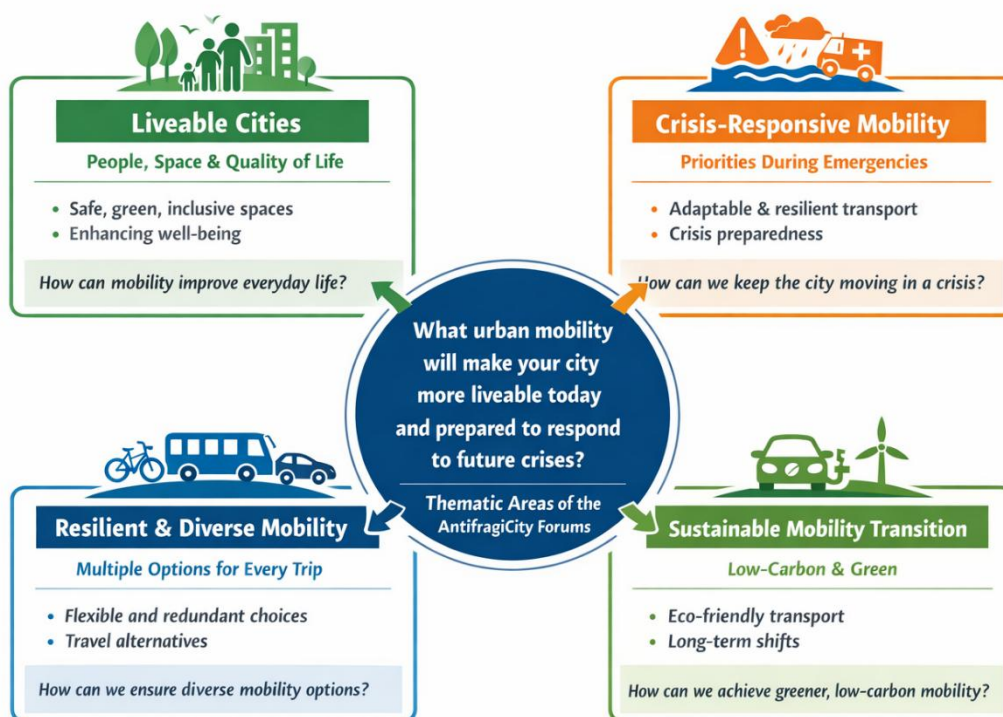


Figure 1 Thematic areas of the AntifragiCity Forums

This thematic structure ensured that mobility was approached as a means to achieve broader urban objectives rather than an end in itself. Deliberation began with participants' visions of desirable urban futures and then explored how mobility systems could support these goals. This approach aligns with deliberative practices that emphasize collective envisioning and the articulation of shared values.

At the same time, the framework remained flexible. Facilitators and experts adapted discussions to local contexts, ensuring relevance while maintaining cross-site comparability. This balance between structure and adaptability is consistent with best practices in multi-site deliberative research.

Overall, the thematic design enabled the systematic exploration of public preferences, priorities, and values across sites, while generating actionable insights to support more livable, resilient, equitable, and sustainable urban mobility systems.

3.3 Methodological approach

Beyond adopting a deliberative approach through citizens' forums, the ACFs were designed to actively engage participants in structured collaboration, enabling them not only to express individual

perspectives but also to collectively work toward shared conclusions and actionable recommendations. Accordingly, participants were guided through a process that encouraged both individual reflection and group synthesis. Through their engagement, participants were expected to:

- Reflect on current mobility challenges and articulate personal values and priorities
- Explore potential emergency scenarios and consider adaptive responses
- Integrate expert insights with peer perspectives to broaden understanding
- Surface insights that might otherwise remain unheard, especially from marginalized or underrepresented groups
- Anticipate community reactions to proposed mobility changes
- Build trust in public decision-making by engaging in meaningful, outcome-oriented dialogue

By balancing individual reflection with structured group activities, participants moved from personal opinion toward collective understanding, ensuring that recommendations reflected a shared perspective rather than isolated viewpoints.

Accordingly, the combined two complementary modes of thinking to support the collaborative process:

- **Divergent thinking:** Encouraging participants to explore multiple perspectives, identify challenges, and generate a wide range of potential solutions
- **Convergent thinking:** Supporting the refinement of ideas into actionable, collectively agreed-upon proposals

This process was facilitated through a combination of plenary sessions, small-group discussions, and expert contributions. Plenary sessions allowed participants to share ideas broadly, while smaller working groups enabled deeper exploration and negotiation, fostering consensus and synthesis

3.3.1 The Three Horizons Framework for collective sense-making

To structure discussions and support shared conclusions, the forums adopted the **Three Horizons: Patterns of Change framework**. This framework (see Figure 2) helped participants situate present realities, emerging innovations, and long-term aspirations within a coherent temporal perspective, allowing collective sense-making across multiple timeframes:

- **Horizon 1 (H1) – Current Reality:** Groups identified dominant mobility systems, assessing strengths, weaknesses, and limitations. This step created a shared understanding of existing challenges.
- **Horizon 2 (H2) – Innovation & Transition:** Participants explored experimental approaches and adaptive strategies that could bridge current systems and future visions. Through dialogue, the group negotiated which ideas had potential and how they could be applied.

- **Horizon 3 (H3) – Transformative Future:** The forums encouraged participants to envision long-term systemic change, identifying innovative mobility solutions aligned with equity, sustainability, and resilience goals.

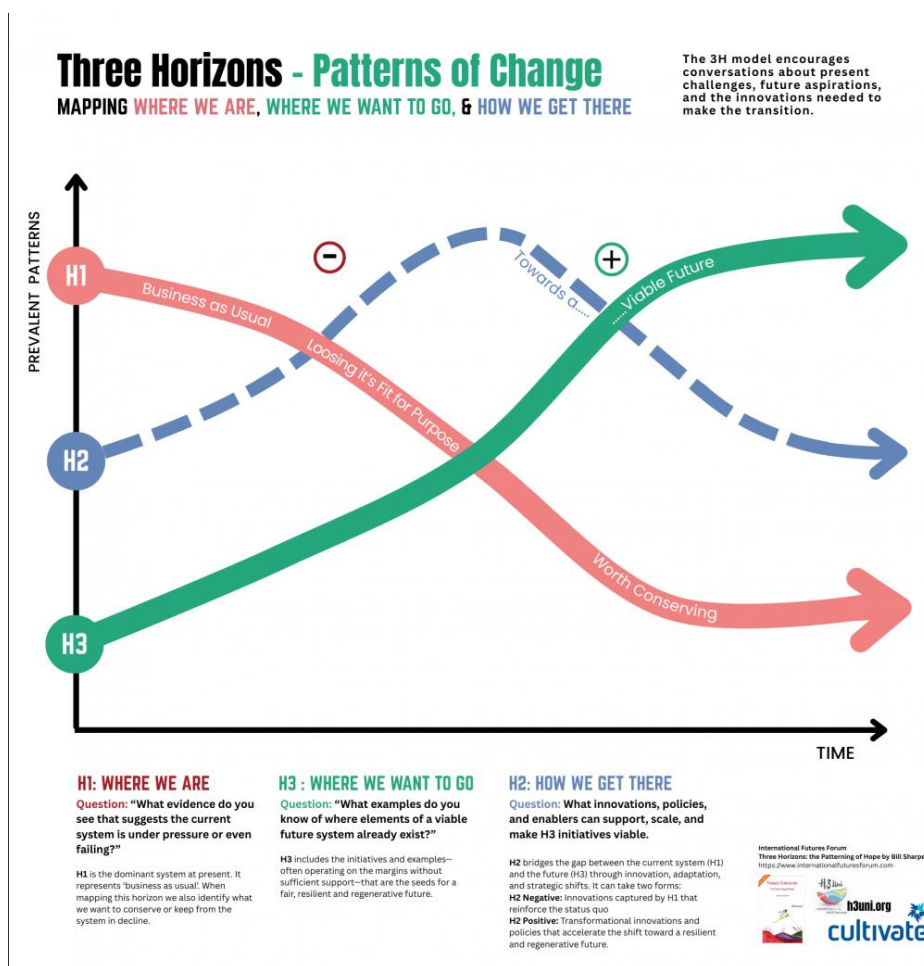


Figure 2 Three Horizons Framework (a)

By moving sequentially through these horizons, participants collectively analyzed the present (H1), envisioned possible futures (H3), and identified pathways (H2) to reconcile the two (see Figure 3). This iterative process of discussion, reflection, and negotiation enabled the group to converge on conclusions that balanced feasibility, ambition, and inclusivity.

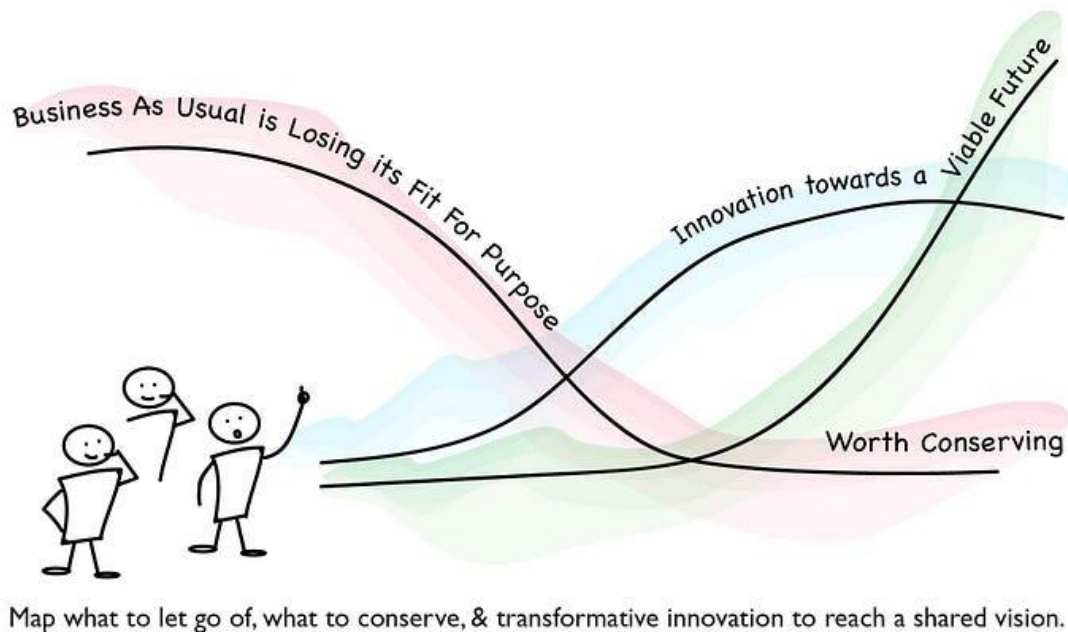


Figure 3 Three Horizons Framework (b)

3.3.2 Achieving shared understanding and recommendations

The combined use of thematic working groups and the Three Horizons framework ensured that participants did not simply share opinions, but actively constructed shared knowledge and recommendations:

- Insights from experts were integrated to ground discussions in evidence and broaden possibilities
- Participants first articulated individual perspectives and concerns
- Through structured dialogue, they debated priorities, negotiated trade-offs, and considered diverse viewpoints
- Collective reflection and synthesis led to actionable proposals that represented consensus or broadly supported ideas

In this way, the Forums facilitated a process in which participants co-created outcomes, moving from individual experiences to collective judgments, and ultimately producing recommendations that are both grounded in local realities and oriented toward sustainable, resilient, and equitable urban mobility futures.

4 Data Analysis Methodology

The data analysis methodology for D2.2 adopts a mixed-methods approach, integrating quantitative and qualitative analytical strategies to generate a comprehensive and nuanced understanding of urban mobility perceptions, deliberative processes, and conditions of social acceptability. This approach reflects the conceptual framing of urban mobility as a socio-technical and experiential system, and aligns with the project's objective of capturing both measurable patterns and the reasoning processes underlying citizens' perspectives.

By combining structured survey data with qualitative insights derived from deliberative forums, the methodology enables the systematic examination of both outcomes (e.g., preferences, priorities, acceptability conditions) and processes (e.g., deliberative quality, learning, and perspective change). In doing so, it supports the analysis of social acceptability not as a static variable, but as a dynamic construct shaped through interaction, information exchange, and collective reflection.

4.1 Research design

The research design for D2.2 adopts a mixed-methods approach, combining qualitative and quantitative analyses to capture both the breadth and depth of participant experiences and perceptions. Central to this design is a pre-post evaluation structure, which enables the assessment of how engagement in the ACFs influences participants' understanding, attitudes, and priorities.

This design reflects the deliberative nature of the forums, where participants are exposed to expert knowledge, engage in facilitated discussions, and collectively reflect on complex policy issues. As such, the research design is not limited to capturing pre-existing preferences, but also seeks to analyze how these preferences are shaped, refined, or reconsidered through the deliberative process.

The key objectives of the research design are:

- Understanding perceptions and experiences: To capture participant insights, lived experiences, and perceptions regarding urban mobility challenges, opportunities, and system performance under both routine and crisis conditions.
- Assessing deliberative impact: To evaluate changes in knowledge, attitudes, priorities, and evaluative criteria as a result of participation in the forums, thereby examining the role of deliberation in shaping social acceptability.
- Enabling comparative analysis: To support systematic comparison across the three demonstration sites – Bratislava, Larissa, and Thessaloniki – in order to identify both context-specific dynamics and shared patterns.

The mixed-methods design ensures that both measurable patterns (through surveys) and nuanced perspectives (through forum discussions), as well as cross-site synthesis inform the development of event mapping and antifragile urban mobility strategies. This integration is particularly important in the context of antifragile urban mobility, where system performance depends not only on technical optimization but also on socially embedded behaviors, perceptions, and adaptive capacities.

4.2 Data sources

The data collected for this deliverable comprise both qualitative and quantitative sources, summarized as follows:

- Qualitative Insights:
 - Observations, facilitated discussions, and deliberative exchanges from the three demonstration sites (Bratislava, Larissa, Thessaloniki) were systematically documented
 - These data capture how participants interpret mobility challenges, negotiate trade-offs, and articulate preferences within a structured deliberative setting
 - The forum outputs provide rich, context-specific insights into citizens' perspectives, including experiential knowledge, normative considerations, and collective reasoning processes
- Pre- and post-forum surveys:
 - Surveys were administered to participants before and after each forum to capture baseline conditions and post-deliberation outcomes
 - The surveys included:
 - Quantitative items (e.g., single- and multiple-choice questions, Likert scales) addressing mobility behaviors, perceptions, and evaluations
 - Qualitative open-ended questions allowing participants to elaborate on their views and experiences
 - This design enables:
 - Assessment of changes in perceptions, priorities, and knowledge
 - Analysis of deliberative effects on social acceptability
 - Comparison across sites and participant groups

These combined data sources ensure a comprehensive understanding of participant experiences while enabling cross-city comparison and triangulation.

4.2.1 Technical approach

The questionnaire items were developed in alignment with the OECD (2021) Evaluation Guidelines for Representative Deliberative Processes, as well as established practices in urban mobility

research. Questionnaire development was conducted in consultation with project partners to ensure methodological robustness and contextual relevance.

For each demonstration site, two questionnaires were prepared: a pre-survey and a post-survey. Both were translated into the respective local language – by the demonstration sites – and underwent iterative validation with partners to ensure clarity, cultural appropriateness, and contextual relevance. Following this validation process, the questionnaires were implemented using the EU Survey platform, including a GDPR-compliant data protection notice and consent form, again in the relevant local language.¹ This version was sent back to the demonstration sites for final review, both in terms of content and translation. After implementing the last round of adjustments, the questionnaires were officially published on the EU Survey portal.

Dissemination of the surveys was coordinated through the demonstration sites. For the pre-survey:

- In Bratislava, pre-selected participants received the survey in advance via email and were asked to complete it prior to the forum; additional on-site completion was facilitated where necessary.
- In Larissa and Thessaloniki, where recruitment followed an open format, participants completed the pre-survey on-site upon arrival.
- Paper-based versions were made available in all locations to ensure accessibility.

For the post-survey, participants in all three sites were asked to complete it immediately after the conclusion of the Forum. Paper-based alternatives were again provided to accommodate different preferences.

The pre-survey consisted of four sections, comprising a total of 16 questions:

- Demographics
- Mobility habits and behavior
- Perceptions of urban mobility
- Initial expectations and process-related perceptions

The post-survey included seven sections, – with a total of 31 questions, or 60 items when counting individual components of multi-part Likert scales:

- Demographics
- Mobility habits and behavior
- Perceptions of urban mobility

¹ Ethical considerations concerning participation in the AntifragiCity Forums were managed collaboratively by LISER and the respective local partner. Participants provided informed consent prior to participation in the forums, and were informed about the voluntary nature of participation and their right to withdraw at any time. All involved ensured that these procedures were followed in accordance with local regulations and project guidelines.

- Perceived knowledge gains
- Process evaluation (e.g., inclusiveness, deliberative quality, trust)
- Representation
- Mobility application and digital engagement

4.2.2 Analytical methods

The analytical approach integrates quantitative and qualitative methods to generate both descriptive patterns and interpretive insights:

- Quantitative analysis
 - Descriptive statistics: Frequency distributions, percentages, and mean values were calculated to identify patterns in participant characteristics, mobility behaviors, perceptions, and evaluations
 - Comparative analysis: Survey responses collected before and after the forums were compared to assess changes in:
 - Knowledge and awareness
 - Attitudes toward mobility interventions
 - Perceived priorities and trade-offs
 - Cross-City comparison: Quantitative results were analyzed across Bratislava, Larissa, and Thessaloniki to identify:
 - Context-specific differences
 - Common trends and shared patterns of acceptability
- Qualitative analysis
 - Thematic coding: Open-ended survey responses and forum discussions were coded to identify recurring themes, insights, and challenges related to urban mobility and participatory processes
 - Pattern identification: Analysis focused on identifying differences and similarities in stakeholder priorities, perceived challenges, and adaptive strategies across the three cities
- Integration Strategy
 - Synthesis of comparative insights: Pre-post comparisons and cross-city analyses were integrated to identify robust patterns and derive transferable insights relevant to antifragile urban mobility planning
 - Triangulation: Quantitative and qualitative findings were systematically combined to validate and enrich interpretations

Overall, this methodological approach ensures that both measurable trends and in-depth citizen perspectives are captured and analyzed in a coherent framework. Hence, the methodology supports

a comprehensive understanding of how social acceptability is formed and how it can inform the design of resilient, adaptive, and socially robust urban mobility strategies.

4.1 Presentation of findings

It is important to note that this deliverable does not integrate academic or policy references when presenting the findings from the ACFs. This constitutes a deliberate and explicit methodological choice. The primary objective of the report is to foreground citizens' perspectives, the deliberative process, and the outcomes as they emerge from participants themselves, rather than interpreting or framing them through existing academic or policy literature.

In conventional research practice, empirical findings are typically situated within established theoretical frameworks. While this approach facilitates comparison and generalization, it can also shift analytical emphasis away from the primary data by filtering participants' contributions through pre-existing concepts, debates, and normative expectations. In the context of deliberative processes, such framing risks diluting the distinct epistemic value of citizens' reasoning by translating it into categories that are external to the deliberative setting itself.

Moreover, integrating findings into existing bodies of literature may introduce implicit hierarchies of knowledge, whereby citizens' inputs are evaluated against academic standards or policy paradigms rather than being treated as valid and self-standing forms of knowledge. This is particularly relevant in deliberative contexts, where the aim is not only to collect opinions, but to document processes of reasoning, negotiation, and collective meaning-making.

For this reason, the present deliverable adopted a descriptive and process-oriented approach to reporting findings. It sought to preserve the integrity, originality, and contextual richness of participants' contributions by presenting them as they were expressed within the forums. The emphasis was thus placed on capturing how citizens articulate priorities, evaluate trade-offs, and construct judgments under conditions of structured deliberation.

This approach does not preclude theoretical interpretation; rather, it deliberately separates the documentation of deliberative outcomes from their subsequent conceptual or analytical positioning. In doing so, the deliverable ensures that the empirical insights generated through the ACFs remain transparent, traceable, and grounded in participants' own perspectives.

5 Bratislava

This section of the report presents the findings from the Bratislava AntifragiCity Forum. The objectives of this evaluation encompasses two main aspects:

- Learning about the Bratislava Forum proceedings: The evaluation assesses its success as a deliberative process, gaining valuable insights into its strengths and areas for improvement
- Understanding the Bratislava Forums' impact: The evaluation seeks to delve into its impact on its members, as well as delve into the recommendations

5.1 Introduction

The Bratislava AntifragiCity Forum was held over two Saturdays: 24 January 2026 and 31 January 2026. On both days, participants met from 9:00 to 15:00.

The recruitment of participants was designed to maximize diversity and ensure that the BAF membership reflected, as closely as possible, the demographic and attitudinal composition of Bratislava's population. The recruitment and selection process was conducted by LISER in close collaboration with MIB.

A public call for participation was launched in early October 2025 via MIB and the AntifragiCity project channels, including both social media and mainstream media outlets. This approach generated substantial interest, with 229 individuals registering to take part in the BAF. The large response allowed LISER to implement a random selection process (sortition) to form a representative panel.

To conduct the selection, the online sortition tool Panelot was used. Panelot requires as input the pool of volunteers, the desired panel size, and quotas to ensure representativity. It outputs quota-compliant panels with assigned selection probabilities. The algorithm ensures a leximin-optimal distribution of selection probabilities, meaning it first maximizes the probability of selection for the volunteer with the lowest chance, then the second lowest, and so on. This approach ensures fairness in selection across all volunteers. The optimization problem underlying Panelot is highly complex due to the large number of possible panels and is solved using state-of-the-art techniques (Flanigan et al., 2021).

To ensure a balanced and diverse panel, the following quotas were applied:

- Gender
- Age
- Nationality
- Educational attainment
- Employment status
- Work sector
- Geographical location
- Primary mode of transportation

- Satisfaction with transportation in Bratislava
- Overall life satisfaction
- Physical or health conditions affecting mobility
- Childcare or caretaking responsibilities
- Exclusion criteria: active membership in political parties, lobby groups, advocacy organizations, or professional groups with a direct stake in the topics under discussion

As the extensive list of quotas demonstrates, the BAF participant selection process was designed not only to ensure diversity across traditional socio-demographic characteristics (e.g., age, gender, education, employment status, and geographic location) but also to capture diversity in attitudes, experiences, and practical ‘possibilities’ affecting mobility in Bratislava. To achieve this, additional questions were included in the registration survey that went beyond standard demographic information:

- Primary mode of transportation: Participants were asked which modes of transport they primarily use (e.g., walking, cycling, public transport, private car). This ensured that the panel included individuals with different mobility behaviors, which could influence perspectives on urban transport planning.
- Satisfaction with transportation in Bratislava: Participants rated their level of satisfaction with the city’s transport system. Including this variable helped ensure that the panel reflected a spectrum of experiences, from satisfied to dissatisfied users, capturing diverse viewpoints on transport challenges and priorities.
- Overall life satisfaction: This measure served as a proxy for broader subjective well-being, allowing for further attitudinal diversity within the panel. Individuals with different levels of life satisfaction may perceive mobility and urban issues differently, and including this dimension helped avoid skewing the deliberations toward a homogenous perspective.
- Physical or health conditions affecting mobility: Participants were asked whether they have conditions that may influence their ability to use certain types of transport. This ensured representation of individuals with different levels of physical ability, highlighting accessibility challenges that are critical to inclusive urban planning.
- Childcare or caregiving responsibilities: Recognizing that caregiving responsibilities impact mobility needs and constraints, participants were asked to indicate whether they have dependents requiring care. Including this factor ensured that the panel represented the practical realities of participants who may rely on specific modes of transport for family-related tasks.

Furthermore, to ensure full accessibility and inclusivity, all respondents were asked to indicate any specific requirements needed to facilitate their participation, such as mobility aids, dietary requirements, or assistance for hearing or visual impairments. This proactive approach ensured that the BAF process was open to all interested individuals, regardless of physical ability or other constraints, and reinforced the integrity of the deliberative process.

Using Panelot, 40 participants and 10 substitutes were initially selected to form a representative and quota-compliant panel. Prior to the first day of the BAF, 7 of the initially selected participants were unable to attend and were replaced by substitutes. One substitute did not respond, resulting in 39 confirmed participants.

On the first day, 36 participants attended, with the remaining three unable to join due to illness. The second day was attended by 34 participants, reflecting normal attrition in deliberative processes. As the deliberations on the second day directly built on discussions and recommendations from day one, it was not possible to bring in additional substitutes at that stage.

5.2 Evaluation

5.2.1 Representativity

This section summarizes the demographic, socio-economic, geographic, and experiential characteristics of the 40 participants selected for the AntifragiCity Citizens' Forum in Bratislava. Comparisons are made with both the broader city population (demographics insights into Bratislava residents aged 18+, $n = 391,784$) to understand representativeness and how selection aligns with existing findings on democratic innovations. Additionally, given the large number ($n = 229$) of respondents to the registration survey, this section also compares the selected panel of participants to the survey respondent pool.

Relative to the city population, men are slightly overrepresented and women slightly underrepresented. However, the panel brings gender distribution closer to the population compared to the survey respondents, which skewed more male. While the panel does not perfectly mirror the city, the selection strategically corrected survey bias by increasing female participation, improving representativity along the gender dimension.

	Selected panel (%)	Survey respondents (%)	General population (%)
Female	47.5%	42.36%	53%
Male	52.5%	56.77%	47%

Table 1 Gender

The panel covers all adult age groups and aligns more closely with city demographics than the survey respondents, which overrepresented younger adults. Younger adults (18–34) are now closer to the actual population share, reducing the survey bias toward this group. Furthermore, middle-aged and

older adults (45–74) are proportionally higher in the panel than in the survey, better reflecting city demographics. While the 35–44 group remains slightly underrepresented, the panel achieves a more balanced age distribution, improving population representativity. Overall, the selection process effectively mitigated age-related survey bias.

	Selected panel (%)	Survey respondents (%)	General population (%)
18-24	5%	8.3%	5.5%
25-34	20%	32.3%	16.8%
35-44	15%	31.0%	23.5%
45-54	22.5%	12.2%	16.1%
55-64	15%	4.8%	15%
65-74	15%	6.55%	14.4%
75+	7.5%	4.8%	8.6%

Table 2 Age

The panel captures the major nationalities of Bratislava, with Czech and Hungarian participants slightly above city proportions. This is, however, primarily a consequence of the small sample size: in a panel of 40 participants, a single person accounts for 2.5%. Small absolute differences therefore translate into noticeable percentage shifts. Overall, the panel is reasonably aligned with city demographics for nationality, though very small minority groups remain unrepresented.

	Selected panel (%)	Survey respondents (%)	General population* (%)
Slovak	95%	95.63%	92.8%
Hungarian	2.5%	1.31%	2.9%
Czech	2.5%	0.44%	1.3%

Other	0%	2.62%	3%
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Table 3 Nationality

**Note: City percentages shown for known nationalities*

With regard to individuals' economic activity, the panel better reflects city demographics than the survey respondents:

- Retired adults are proportionally aligned with the city population
- Middle-aged and older adults (45+) are more prominent in the panel, improving representativity relative to survey bias toward younger employed participants
- Students and homemakers remain slightly underrepresented, reflecting a trade-off in the small panel

Overall, the selection has moved the panel closer to the overall population structure, improving representativity across economic activity.

	Selected panel (%)	Survey respondents (%)	General population* (%)
Employed	65%	69.87%	64.1%
Unemployed	2.5%	5.24%	3.1%
Homemaker	2.5%	0.44%	5.2%
Student / pupil	2.5%	8.73%	3.2%
Retired	25%	12.23%	22%
Maternity leave	2.5%	3.49%	2.5%

Table 4 Economic activity / status

**Note: City percentages shown for known nationalities*

Furthermore, the panel successfully preserves and even enhances sectoral diversity compared to survey respondents. Large sectors such as state/municipal administration, construction, education, health, and transport remain well represented. Smaller sectors such as craftsmanship, associations/NGOs, and consulting/legal are included in the panel, even if these sectors represented

a small fraction of survey respondents (0.63–3.77%). The selected panel avoids overrepresentation by any single sector (max 25%), while smaller sectors are still present, ensuring a breadth of professional perspectives.

	Selected panel (%)	Survey respondents (%)
State or municipal administration	25%	21.38%
Construction / civil engineering	12.5%	8.81%
Other business services	8.33%	5.66%
Retail trade	8.33%	4.40%
Transport	8.33%	6.29%
Education	8.33%	6.92%
Health / social care / care homes	8.33%	3.77%
Associations / NGOs	4.17%	3.77%
IT / ICT	4.17%	6.92%
Industry	4.17%	3.14%
Craftsmanship	4.17%	0.63%
Consulting / Auditing / Legal	4.17%	3.77%

Table 5 Sectoral diversity

The panel is highly educated, exceeding city averages. Compared to survey respondents, extreme overrepresentation of Master's degree holders are slightly reduced, partially moderating survey bias.

While highly educated participants dominate, the panel is closer to city educational structure than the original survey respondents.

	Selected panel (%)	Survey respondents (%)	General population* (%)
Incomplete primary	0%	0.44%	0.13%
Primary school	0%	2.18%	5.39%
Lower secondary	7.5%	2.18%	12.45%
Upper secondary	35%	17.03%	28.14%
Post-secondary	0%	0.87%	6.28%
Bachelor's	7.5%	10.48%	4.11%
Master's	45%	61.14%	39.04%
Doctorate	5%	5.68%	4.45%

Table 6 Education

**Note: City percentages shown for known nationalities*

The panel spans both central and peripheral districts, improving geographic representativity relative to the survey respondents, which concentrated heavily in central areas. Some outer districts remain unrepresented due to the small sample size, but overall, the panel captures a broader geographic distribution, aligning more closely with city demographics.

	Selected panel (%)	Survey respondents (%)	General population* (%)
Petržalka	22.5%	14.85%	24.4%
Ružinov	20%	22.27%	17%
Staré Mesto	10%	16.16%	9.7%

Nové Mesto	10%	10.92%	9.2%
Dúbravka	2.5%	5.24%	7.6%
Karlova Ves	7.5%	8.30%	7.6%
Rača	5%	4.37%	5.4%
Podunajské Biskupice	7.5%	4.37%	4.9%
Vrakuňa	5%	5.68%	4.4%
Devínska Nová Ves	2.5%	3.06%	3.6%
Devín	2.5%	0.44%	0.4%
Lamač	2.5%	0.87%	1.6%
Záhorská Bystrica	0%	2.18%	1.2%
Čunovo	0%	0.44%	0.3%
Jarovce	0%	0.87%	0.5%
Rusovce	2.5%	0.87%	0.9%
Vajnory	0%	0.44%	1.2%

Table 7 Geographic distribution

The selected panel maintains and even enhances diversity in mobility compared to the survey respondents. While survey respondents were heavily concentrated in walking + public transport (21%) and public transport only (10%), the panel distributes participants more evenly across all modes, including mixed modes and less common combinations (car + motorcycle, bicycle + public transport). This ensures deliberations are informed by a broad spectrum of mobility experiences, rather than dominated by a few prevalent modes.

	Selected panel (%)	Survey respondents (%)
Public transport only	20%	10.48%
Foot + Public transport	15%	20.96%
Car only	10%	6.55%
Foot + Car	10%	6.11%
Public transport + Foot	7.5%	13.54%
Car + Public transport	7.5%	10.48%
Public transport + Bicycle	5%	2.62%
Car + Motorcycle	5%	1.31%
Car + Foot	2.5%	2.62%
Bicycle only	2.5%	1.31%
Bicycle + Public transport	2.5%	3.49%
Train + Public transport	2.5%	2.62%
Car + Bicycle	2.5%	1.31%
Foot + Bicycle	2.5%	2.62%

Table 8 Mobility and transport

Notably, the panel broadens the range of transport satisfaction, deliberately including participants who are somewhat or very dissatisfied. While survey respondents were mostly satisfied (66%), the panel ensures critical perspectives are represented, enhancing the diversity of opinions and experiences.

	Selected panel (%)	Survey respondents (%)
Very satisfied	12.5%	12.66%
Satisfied	20%	20.95%
Somewhat satisfied	35%	25.33%
Somewhat dissatisfied	17.5%	3.06%
Dissatisfied	12.5%	2.18%
Very dissatisfied	2.5%	1.31%

Table 9 Transportation satisfaction

Similarly, the selected panel enhances diversity in life satisfaction: survey respondents were highly skewed toward satisfied participants (93%), whereas the panel includes more participants who are somewhat or very dissatisfied (32.5%).

	Selected panel (%)	Survey respondents (%)
Very satisfied	12.5%	12.66%
Satisfied	37.5%	55.02%
Somewhat satisfied	20%	25.33%
Somewhat dissatisfied	15%	3.06%
Dissatisfied	10%	2.18%
Very dissatisfied	5%	1.31%

Table 10 Living conditions satisfaction

Health diversity is also increased, with more participants reporting health needs, and caretaking responsibilities are represented as well.

	Selected panel (%)	Survey respondents (%)
No health issues	87.5%	93.45%
Health needs	12.5%	6.55%
No caretaking responsibilities	80%	72.93%
Caretaking responsibilities	20%	27.07%

Table 11 Health and caretaking considerations

Overall, the panel ensures a broader range of personal circumstances is included, capturing experiences that might otherwise be underrepresented in deliberations.

Summary

The selected panel for the AntifragiCity Citizens' Forum in Bratislava preserves and enhances diversity compared to the survey respondents. Relative to the survey pool, participants are:

- More balanced in gender, with increased female representation
- More representative across age groups, including middle-aged and older adults, reducing the survey bias toward younger adults
- Major nationalities (Slovak, Hungarian, Czech) are represented. Small absolute differences reflect the small panel size rather than systematic bias
- Professionally and socio-economically diverse, with representation across large and smaller employment sectors
- Highly educated participants dominate, but the panel slightly moderates overrepresentation of Master's degree holders compared to survey respondents, maintaining diversity across educational levels
- Geographically broader, including participants from both central and peripheral districts
- Diverse in mobility patterns, life satisfaction, health status, and childcare responsibilities, capturing perspectives underrepresented in the survey

The selected panel achieves a more balanced and inclusive mix of perspectives than the broader registration survey, effectively addressing the biases identified among survey respondents. In other words, the panel selection strategically corrected for over- or underrepresentation in the survey, creating a group that aligns more closely with the full spectrum of urban demographics and socio-economic characteristics, while maintaining breadth in experiential perspectives.

In conclusion, the 40-participant panel represents a carefully balanced and broadly representative selection, providing meaningful diversity across demographics, socio-economic status, life experiences, and professional backgrounds. This design ensures that deliberations within the AntifragiCity Citizens' Forum are informed by multiple, varied perspectives, supporting inclusive and robust discussions that reflect the realities and priorities of the Bratislava population.

5.2.2 Pre-survey insights

This section summarizes the findings of the pre-participation survey conducted among participants of the Bratislava AntifragiCity Forum. The survey aimed to understand participants' current travel behaviors, motivations, attitudes toward sustainable urban transport, and their interest in contributing to urban mobility initiatives. A total of 34 valid responses were collected ($n = 34$), where the number of respondents is different, this is explicitly mentioned.

5.2.2.1 Citizens' participation in urban mobility

Respondents were asked whether they would like to be involved in shaping mobility initiatives in their city. The results indicate a strong overall interest in engagement (see Figure 4).

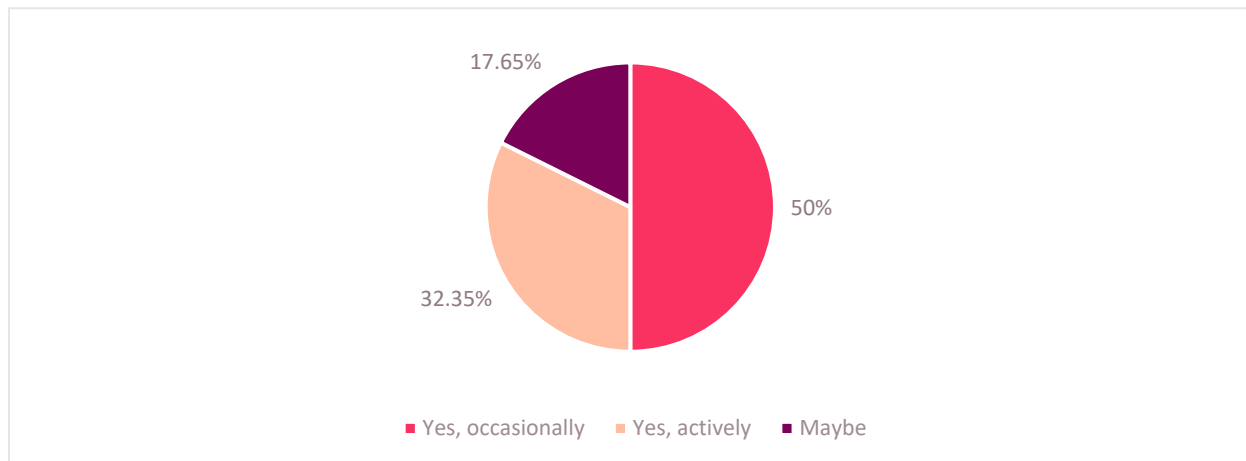


Figure 4 Willingness to shape mobility initiatives in city

Half of the respondents (50%) expressed a desire to participate occasionally, suggesting a willingness to contribute on a limited or flexible basis. A substantial share (32.35%) indicated that they would like to be involved actively, reflecting a highly motivated group eager to participate regularly or take a hands-on role in urban mobility initiatives. The remaining respondents (17.65%) answered maybe, indicating some openness to participation, though their engagement is uncertain or conditional. Overall, the data demonstrate that a large majority (82.35%) of respondents are at least somewhat interested in being involved in shaping mobility initiatives, either occasionally or actively.

The next question then asked participants how they would like to be involved in shaping mobility initiatives in their city. Four main forms of participation were offered: Participation in public consultations/citizens' forums, testing mobility-related applications, engaging with community or neighborhood mobility projects, and providing feedback via surveys.

	%
Sharing opinions in public consultations / citizens' forums	88.24%
Providing feedback via surveys / questionnaires	70.59%
Engaging in community-level mobility projects	58.82%
Testing mobility applications or tools	52.94%

Table 12 Participation preferences

The analysis reveals clear patterns in participation preferences. Public consultations and citizen forums emerged as the most popular channel, with the vast majority of respondents (88.24%) indicating interest in this mode of engagement. This underscores a strong preference for structured, visible channels where individuals can voice their opinions and contribute to decision-making processes.

Providing feedback through surveys or questionnaires was also highly favored (70.59%), reflecting a demand for flexible, low-effort opportunities to participate. This form of engagement is particularly relevant for respondents who may have time constraints or prefer asynchronous methods of contribution.

Engagement in community-level mobility projects was moderately popular (58.82%), suggesting that while a significant portion of respondents are motivated to take hands-on action in their neighborhoods, a smaller subset is willing to commit the time and effort required. Similarly, testing mobility applications or tools attracted interest from approximately half of respondents (52.94%), indicating that more technical or experimental forms of participation appeal primarily to digitally literate or innovation-oriented individuals.

A notable observation is that many respondents expressed interest in multiple forms of participation. Almost 60% of respondents (58.82%) selected three or four forms of participation, demonstrating a strong willingness to engage through multiple channels. Only 17.65% preferred a single form of

participation, indicating that a minority of respondents favor simpler, low-commitment options. This demonstrates a willingness to engage through diverse channels and suggests that cities could mobilize motivated individuals by providing varied opportunities for involvement.

	%
1 Mode	17.65%
2 Modes	23.53%
3 Modes	29.41%
4 Modes	29.41%

Table 13 Number of participation modes selected (*per participant*)

In summary, the findings indicate a strong overall willingness to contribute to urban mobility initiatives, with structured forums and surveys emerging as the most accessible and widely preferred forms of participation.

In this view, respondents were asked to indicate their agreement with three statements regarding public involvement in urban mobility planning: public participation is important for good transport decisions, citizens' everyday experiences are as important as expert knowledge, and involving diverse residents leads to fairer mobility outcomes.

	Strongly Agree	Somewhat Agree	Somewhat Disagree	Strongly Disagree
Public participation is essential	73.53%	17.65%	8.82%	0%
Citizen experience as valuable as expert opinion	55.88%	38.24%	5.88%	0%
Engagement of diverse residents leads	52.94%	32.35%	11.76%	2.94%

to fairer outcomes				
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Table 14 Opinions on citizens' participation

The majority of respondents strongly agree that public participation is necessary for effective transport policy (91.18%). Only a small minority (8.82%) expressed some disagreement, and no one strongly disagreed. This indicates widespread recognition of the importance of citizen input.

Even higher agreement (94.12%) was observed for the statement that everyday citizen experiences are as important as expert opinions, with a limited number of respondents disagreeing (5.88%). This suggests that respondents see practical knowledge and on-the-ground insights as crucial for designing policies that reflect real needs and conditions.

Most respondents (85.29%) agree that engaging a diverse range of residents leads to fairer transport outcomes. While slightly lower than the other two statements, this still reflects strong recognition that inclusivity and representation are critical for equity in decision-making. A small fraction (14.7%) expressed disagreement.

Across statements, the data suggest that respondents are motivated and supportive of participatory mechanisms.

5.2.2.2 Participants and urban mobility

In line with the questions on citizen participation and the topic of urban mobility, the pre-survey also measured three aspects of respondents' engagement with urban mobility: their general interest in urban mobility, how informed they feel about the city's mobility policies, and their confidence in contributing to urban mobility discussions (see Figure 5).

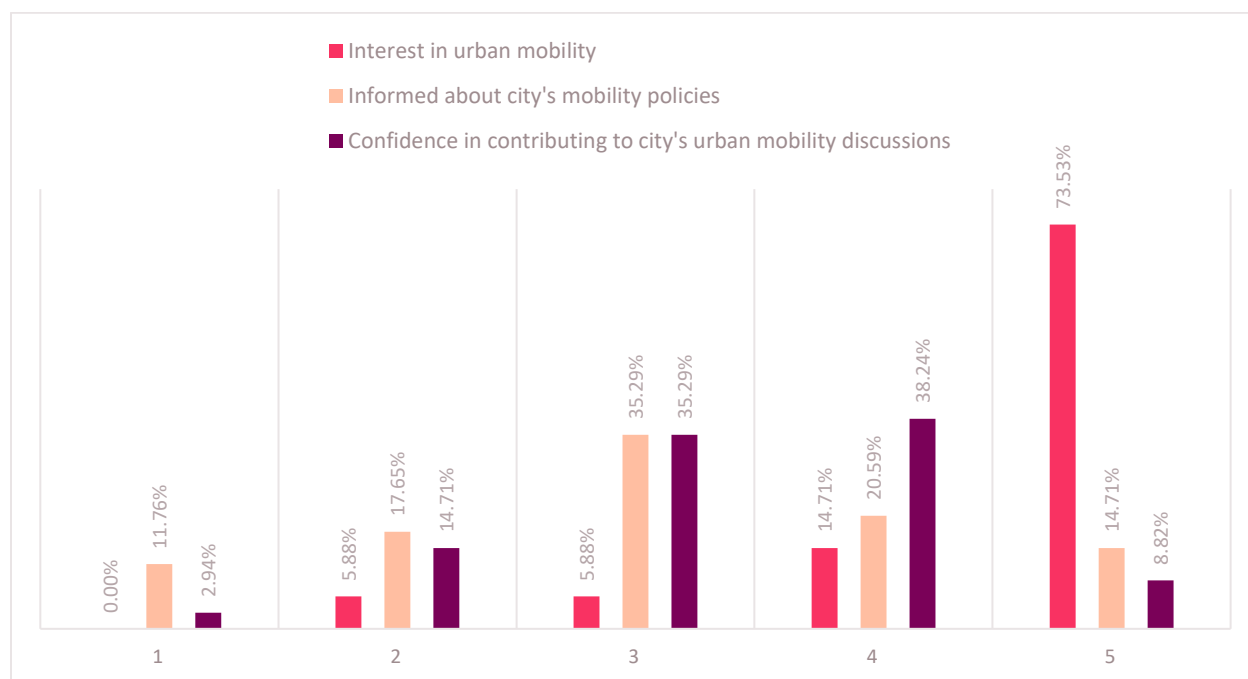


Figure 5 Knowledge urban mobility

Note: 1 = low interest, knowledge, or confidence; 5 = high interest, knowledge, or confidence

The results indicate that overall interest in urban mobility is very high among respondents. A large majority, 73.53%, rated their interest at the maximum level of 5, while 14.71% rated it 4. Only a small minority expressed lower interest, with 5.88% rating it 2 and 5.88% rating it 3. This suggests that respondents are generally highly engaged with urban mobility issues and see them as personally relevant.

In contrast, respondents are less informed about the city's mobility policies. Only 14.71% rated themselves as highly informed (5), and 20.59% rated themselves 4. The largest share, 35.29%, rated their knowledge as moderate (3), while smaller percentages rated themselves low (1 or 2: 11.76% and 17.65%, respectively). This indicates a gap between interest and policy awareness: although most respondents care deeply about mobility, many feel they lack detailed knowledge about the policies shaping the city's transport system.

Respondents' confidence in contributing to urban mobility discussions is moderate rather than high. Only 8.82% feel very confident (5), while 38.24% rated their confidence at 4. A further 35.29% reported moderate confidence (3), and smaller shares scored 1 (2.94%) or 2 (14.71%). This pattern suggests that while most respondents are willing and somewhat capable of engaging in discussions, they are hesitant or less certain about fully participating in decision-making, even when they are interested in and informed about urban mobility.

In summary, the data reveal an interesting dynamic: respondents are highly interested in urban mobility, but awareness of policies and confidence to engage in discussions are moderate rather than high. While most respondents feel at least somewhat capable of participating, very few feel fully confident.

5.2.2.3 Mobility habits and opinions

Participants were asked to identify their most-used mode of daily travel, selecting only one option (see Figure 6). Among the forum participants, public transportation is the most frequently used mode of transport, with 55.88% of respondents indicating that they use it for their daily travel. Private cars remain significant, with 29.41% of participants using them, while cycling and walking are less common, accounting for 8.82% and 5.88%, respectively.

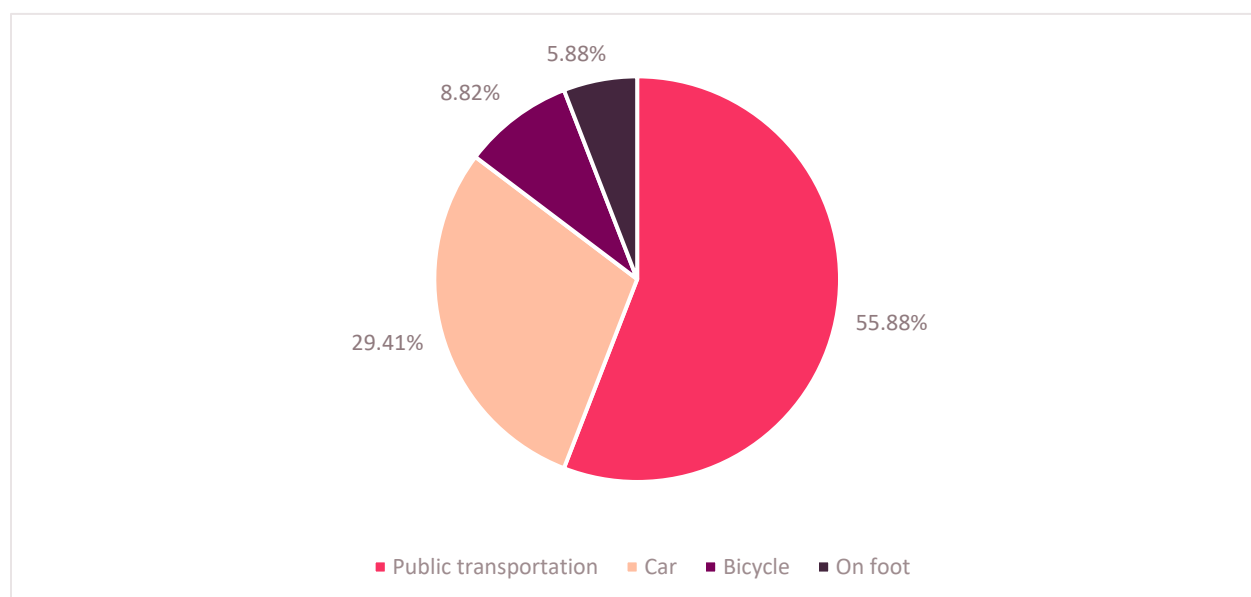


Figure 6 Most used daily transport mode

Respondents were asked to identify the main reasons for choosing their primary mode of transport, with the option to select up to three factors (see Figure 7). The results presented here reflect aggregated selections across all participants, meaning the percentages represent how frequently each factor was chosen overall. The aggregated results show that practical considerations clearly outweigh environmental or purely personal motivations, highlighting that everyday functionality is the dominant influence on travel behavior.

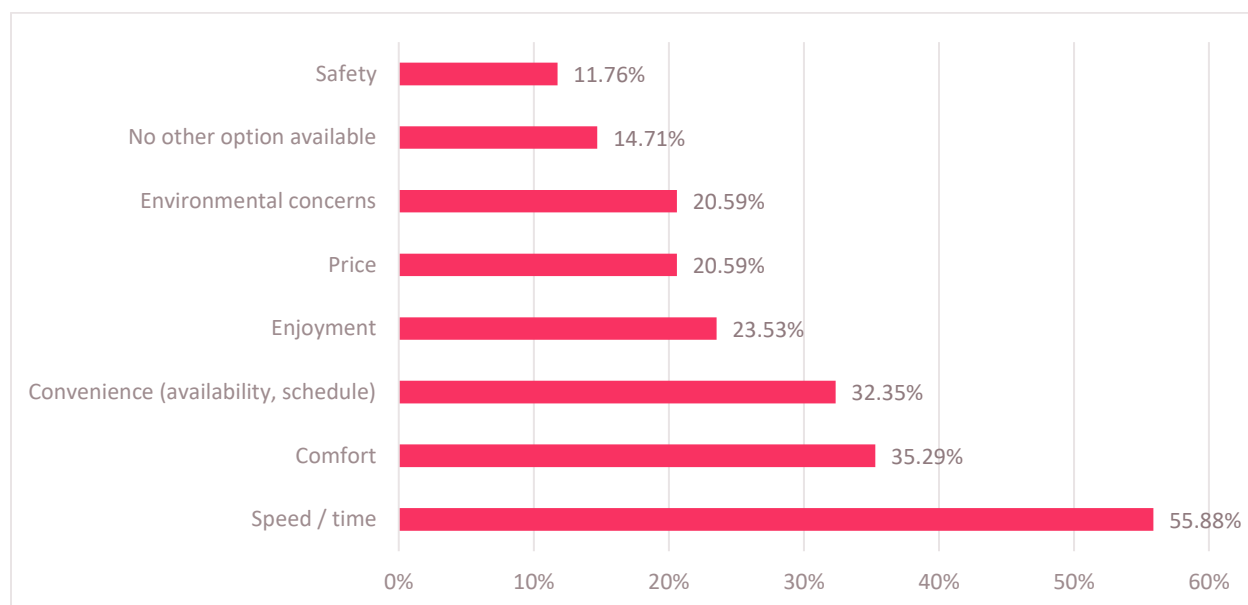


Figure 7 Main reasons for choosing daily mode of transport

The most frequently cited reason was speed or time efficiency, accounting for 55.88% of all selections. This indicates that travel time is the single most important determinant of modal choice. People tend to favor the option that allows them to reach their destination as quickly and reliably as possible, underscoring the central role of efficiency in daily mobility decisions.

Comfort (35.29%) and convenience, such as availability and schedule (32.35%), were the next most common factors. These findings suggest that beyond speed, the overall ease and quality of the travel experience strongly shape transport preferences. Respondents value modes that are physically comfortable and logistically simple to use, reinforcing the idea that reliability and user experience are critical to sustained modal use.

Pleasure or enjoyment was selected in 23.53% of responses, showing that for a notable share of participants, travel is not only a functional activity but also an experiential one. Cost and environmental concern were each selected by 20.59% of respondents. The equal weighting of these two factors suggests that financial considerations are just as influential as ecological values, but both remain secondary to time, comfort, and convenience in determining primary mode choice.

A smaller proportion of respondents indicated structural constraints or risk-related motivations. “No other option” was selected in 14.71% of responses, indicating that some individuals rely on their current mode due to limited alternatives rather than preference. Safety was selected by 11.76% of respondents, making it the least frequently chosen factor among those listed. While safety is a fundamental requirement for any transport system, these results suggest it may be perceived as a baseline expectation rather than a distinguishing reason for choosing a specific mode.

Overall, the findings demonstrate that transport mode choice is driven primarily by practical and experiential factors, especially time efficiency, comfort, and convenience. Environmental concern and cost play a meaningful but secondary role, while lack of alternatives influences a smaller yet important segment of users.

Respondents were also asked to identify factors that would encourage greater use of sustainable transport. The results indicate a clear preference for practical and functional improvements, suggesting that enhancements in infrastructure, connectivity, and service efficiency are the most influential drivers of increased adoption.

	%
Better connectivity / frequency	61.76%
Better infrastructure (e.g. bike lanes, covered bus stops, sidewalks)	58.82%
Time savings	44.12%
Improved safety	23.53%
Improved comfort	20.59%
Lower costs	17.65%
None	8.82%

Table 15 Factors encouraging greater sustainability

The most frequently cited factor is better connectivity and higher service frequency, selected by 61.76% of respondents. This suggests that many users are limited by gaps in network coverage or infrequent services, and that making sustainable transport more reliable and accessible would significantly encourage adoption.

Better infrastructure, such as bike lanes, sidewalks, and improved stops, was selected by 58.82% of respondents, highlighting the importance of safe, well-designed, and dedicated facilities. This indicates that physical infrastructure improvements are critical for encouraging walking, cycling, and public transport use, particularly when these modes must compete with the convenience of private vehicles.

Time savings is the third most commonly cited factor, with 44.12% of respondents selecting it. This aligns with broader findings across transport mode choice: efficiency is a primary driver of travel behavior. Sustainable modes that can reduce journey times, minimize transfers, or offer direct routes are therefore likely to see increased adoption.

Secondary factors, including improved safety (23.53%) and improved comfort (20.59%), were cited less frequently but remain important. Safety improvements, such as better lighting, segregated bike lanes, and traffic calming measures, can reduce perceived risk and encourage more cautious or hesitant users to shift to sustainable modes. Comfort, including seating, shelter, or smooth travel conditions, also plays a role in making these modes more attractive.

Lower costs were selected by 17.65% of respondents, indicating that financial incentives or affordability measures can support mode shift but are not the primary motivating factor for most users. Finally, 8.82% of respondents indicated that nothing would encourage them to use sustainable transport, suggesting a small but notable proportion of the population is resistant or has strong barriers to adoption.

Overall, the data suggest that functional and service-oriented improvements – especially connectivity, infrastructure, and time efficiency – are the most effective levers for increasing the use of sustainable transport. While factors such as safety, comfort, and cost contribute, they play a secondary role.

Regarding participants' willingness to make compromises to increase environmental sustainability, the survey results indicate a generally high level of willingness among respondents to accept compromises in order to improve environmental sustainability in their city, such as reducing parking availability, increasing green spaces, or lowering speed limits (see Figure 8). Responses were measured on a scale from 0 (not willing at all) to 10 (very willing), and the distribution is clearly weighted toward the higher end of the scale.

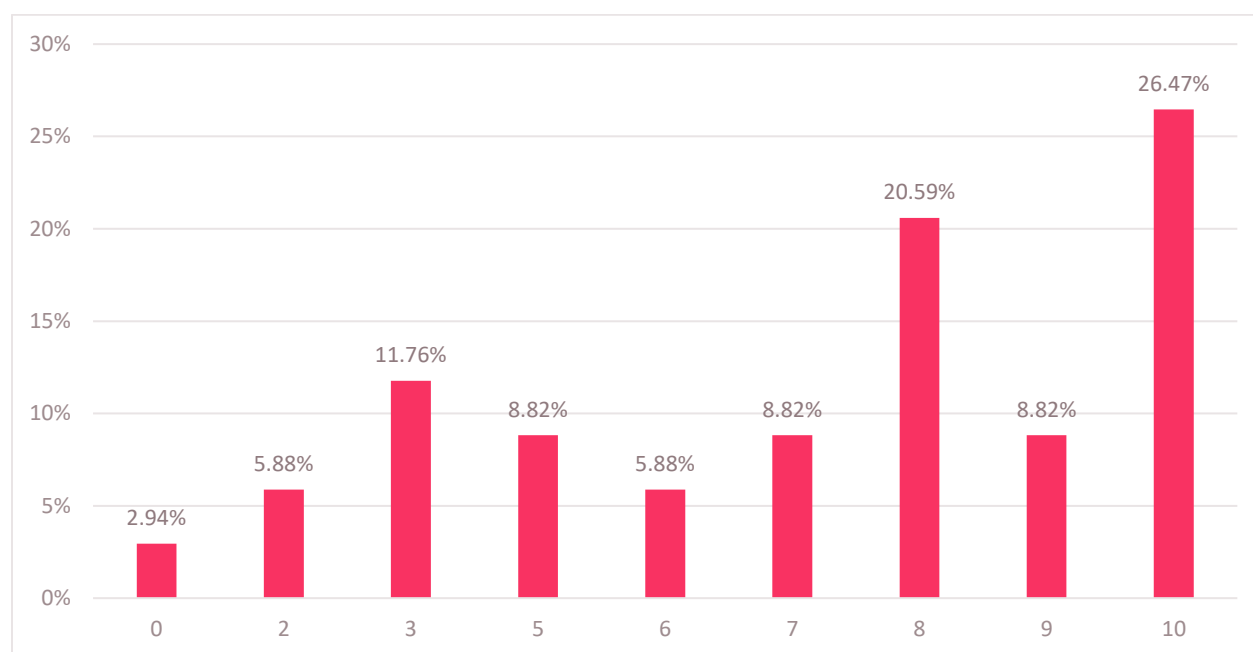


Figure 8 Willingness to accept trade-offs for environmental sustainability

Note: 0 = not at all, 10 = very willing

More than half of respondents (55.88%) selected values between 8 and 10, demonstrating strong support for sustainability-oriented trade-offs. The most frequently selected response was 10, chosen by 26.47% of participants, indicating that over one quarter of respondents are fully willing to accept such changes. Another 20.59% selected 8, reinforcing the conclusion that a substantial portion of the respondents is highly receptive to measures that may involve personal or practical compromises for environmental benefit.

An additional 23.52% of respondents expressed moderate willingness, selecting values between 5 and 7. This group appears open to sustainability measures but may have reservations about the extent, pace, or specific impacts of proposed changes. Their responses suggest that while they are not opposed in principle, their support may depend on how policies are implemented, communicated, and balanced with everyday needs such as accessibility and convenience.

In contrast, 20.58% of respondents indicated low willingness to accept compromises, selecting values between 0 and 3. However, only 2.94% chose 0, meaning that outright rejection of sustainability-related trade-offs is relatively rare. Most respondents in the lower range still selected values above zero, which may reflect hesitation or concern rather than firm opposition.

Overall, the findings demonstrate a readiness among participants to support urban sustainability measures even if they require adjustments to existing habits or infrastructure.

Lastly, respondents were asked to indicate which measures they believed would make their city a more pleasant place to live and improve mobility, both in everyday situations and during crises such as extreme weather events or emergencies. Multiple responses were permitted. The distribution of responses in the pre-survey shows a clear hierarchy of perceived priorities, with a strong emphasis on structural and spatial interventions.

	%
Better public transport	66.67%
Better active mobility infrastructure	53.33%
Greater share of greenery	53.33%
Digital traffic management	50.00%
Restrictions on cars in dense areas	43.33%
30 km/h speed zones	40.00%
Pedestrian zones	36.67%
Reduced parking supply	30.00%
Shared streets	23.33%
Congestion charging	13.33%
Shared mobility services	10.00%
None of the above	10.00%

Table 16 Measures perceived as necessary to be antifragile

Better public transport emerged as the most frequently selected individual measure, supported by 66.67% of respondents. This suggests that public transport is widely perceived as the backbone of urban mobility and a key contributor to resilience and everyday functionality. Measures related to active mobility and environmental quality followed closely: both better active mobility infrastructure and a greater share of greenery were selected by 53.33% of respondents. This indicates that

respondents associate improvements in walking and cycling conditions, as well as environmental enhancements, with broader notions of urban livability rather than mobility alone.

Digital traffic management was selected by exactly half of the respondents (50%), placing it among the more prominent measures but below core infrastructure-related interventions. This suggests that while technological solutions are recognized as important, they are not viewed as substitutes for physical changes to transport systems and public space. Measures aimed at restricting or calming car traffic received moderate levels of support. Restrictions on cars in dense areas (43.33%), 30 km/h speed zones (40%), and pedestrian zones (36.67%) indicate a substantial willingness among respondents to accept limitations on car use in favor of safer and more comfortable urban environments.

More interventionist demand-management measures, such as reduced parking supply (30%) and congestion charging (13.33%), were selected less frequently, suggesting greater ambivalence toward policies that directly impose costs or reduce convenience for drivers. Shared streets (23.33%) and shared mobility services (10%) were among the least frequently selected individual measures, possibly reflecting lower familiarity or more context-dependent acceptance. Notably, 10.00% of respondents selected “none of the above,” indicating that a small subset of participants either did not identify with the listed measures or had not yet formed clear preferences.

Overall, the data suggest that respondents prioritize integrated, people-centered urban mobility strategies. Preferences consistently favor visible, experiential interventions that reshape streets, improve public transport, and support walking and cycling, while technological and service-based solutions are viewed as secondary but supportive components.

5.2.3 Post-survey insights

This section synthesizes post-forum survey findings and qualitative feedback to evaluate the representativity, process design, expert contributions, facilitation quality, deliberative interactions, recommendation outcomes, participant takeaways, and attitudes toward mobility-related digital tools. A total of 32 valid responses were collected ($n = 32$). However, for some questions there are only 31 responses ($n = 31$), where this is the case, this is mentioned.

5.2.3.1 Evaluation of the process

A critical aspect of deliberative processes is the inclusion of diverse social perspectives. Accordingly, participants were asked whether they felt any groups or parts of society were missing from the process ($n = 31$). Almost half of respondents (48.39%) believed that certain perspectives were not sufficiently represented, while 16.13% felt no groups were missing and 35.48% were unsure.

	%
Yes	48.39%
No	16.13%
Don't know	35.48%

Table 17 Perceived representativity

The open-ended responses suggest that perceived gaps related primarily to specific mobility needs and user perspectives. Frequently mentioned were people with disabilities and reduced mobility, described for example as *“handicapped people who need a barrier-free city the most,” “disabled people with severely limited mobility,”* and *“immobile public transport users.”* One participant noted, *“People with disabilities. We tried to think of them too, but they would bring another perspective to the issue,”* highlighting an awareness that lived experience cannot be fully substituted through discussion alone. Participants also pointed to the absence of younger age groups, referring to *“children”* and *“students (children)”*, with one participant explaining why they believe children should have been include – although individuals younger than 18, sometimes 16, are generally not part of a citizens’ assembly: *“they travel a lot - school, clubs, public transport.”* In addition, several responses mentioned perspectives linked to car use and commuting patterns, including *“pedestrians,” “active motorists,”* and *“hardcore car enthusiasts.”* These responses suggest that while the forum successfully engaged a broad range of participants (see earlier section on representativity of the Bratislava AntifragiCity Forum),– perceived – societal representativity remains a challenge.

Furthermore, participants evaluated the structure, pacing, and duration of the meetings. An overwhelming majority of participants (90.63%) considered the duration and number of meetings adequate, while only a small minority (9.37%) felt the process was too short.

	%
Yes	90.63%
No, it was too short	9.37%

Table 18 Perceptions regarding timing

Qualitative feedback reinforces this positive assessment. Participants particularly appreciated the pacing of the meetings, which were spaced over time rather than held on consecutive days. This

structure was seen as beneficial for reflection and consolidation between sessions. As one participant noted, *“2 days are exactly right, it's good that they are not consecutive, but separated by a week.”* Other emphasized that the timeframe allowed for meaningful engagement with the topic, including sufficient opportunity to discuss *“the problem”* and to come up *“with steps that should change [things] for the better.”*

A small minority expressed a desire for additional time, suggesting that an extra session might reduce time pressure during solution development and ensure that no ideas were overlooked: *“I would add one more meeting so that we don't have to rush when formulating solutions and strategies and so that nothing is forgotten.”* Similarly, another participant claimed that they would have preferred more time *“to address topics individually, elaborate on them, [and] create structures and systems that could optimize the result.”*

This high level of satisfaction with the process design is mirrored in participants' willingness to remain engaged. When asked whether they would be open to participating in a longer process, 87.5% responded positively.

	%
Yes	87.5%
No	12.5%

Table 19 Willingness to participate in a longer process

This indicates not only a positive overall experience but also trust in the facilitation and a perception that participation was worthwhile. Together, these findings suggest that the forum succeeded in creating a participatory environment that participants valued and would be willing to re-enter, thereby laying a strong foundation for sustained engagement in future project activities.

Another crucial part of any deliberative process is the contributions from experts. Participants (n = 31) reported that expert input was generally clear, accessible, and useful. A large majority agreed that experts provided diverse viewpoints (83.87% agreement), strengthening the perceived balance and credibility of the forum. Clarity of communication was rated particularly highly: 96.77% of participants agreed that the information provided was clear and easy to understand. Similarly, over 90% considered the information useful, confirming that expert contributions were seen as relevant and valuable to the discussions.

At the same time, responses indicate a nuanced picture regarding the scope and depth of the information (see Figure 9). While most participants did not perceive the information as too specific (67.74% disagreed with this statement) or too technical (83.87% disagreed), a notable minority felt

that the content was too limited (35.48% agreement). This suggests that while the material was broadly accessible, some participants would have welcomed more in-depth or detailed input. Importantly, the volume of information was generally experienced as manageable: 87.10% of respondents disagreed that they felt overwhelmed, indicating that the pacing and structure of expert sessions were appropriate for most attendees.

	Strongly agree	Rather agree	Rather disagree	Strongly disagree
Experts provided diverse viewpoints	45.16%	38.71%	12.90%	3.22%
Information was clear and easy to understand	61.29%	35.48%	0%	3.22%
Information provided was useful	54.84%	35.48%	6.45%	3.23%
Information was too limited	12.90%	22.58%	45.16%	19.35%
Information was too specific	9.68%	22.58%	58.06%	9.68%
Information was too technical	9.68%	6.45%	61.29%	22.58%
Felt overwhelmed by the amount of information	6.45%	6.45%	45.16%	41.94%

Table 20 Perceptions on experts and information

Qualitative comments further illustrate this overall positive assessment. Expert presentations were described as “*general and concise*,” “*professional and informative*,” and effective in “*broadening my horizons*.” Participants also appreciated the effort to present complex topics in an accessible manner and to reflect the diversity of the audience: one participant described the expert presentation as “*nice and concisely processing demanding topics*”, another similarly appreciated them because they were “*professional and with an emphasis on the diversity of the participants*.”

A small minority, however, noted perceived bias in some presentations: “*I think that the people presented as experts were quite ‘activistically’ biased towards one side of the opinion spectrum*”, or a lack of a unified conceptual framework: “*The presentations lacked a unified concept*.” Other participants observed that the experts were constrained by time: “*They could have been even more specific, more examples, but I understand that there was time for that too – in addition to the discussion, to have personal consultations*,” while another remarked that the presentation were “*too general, [the experts] didn’t respond to criticism, not enough time*.” While these comments point to areas for potential refinement, they did not diminish the overall positive perception of session as clear, accessible, and relevant.

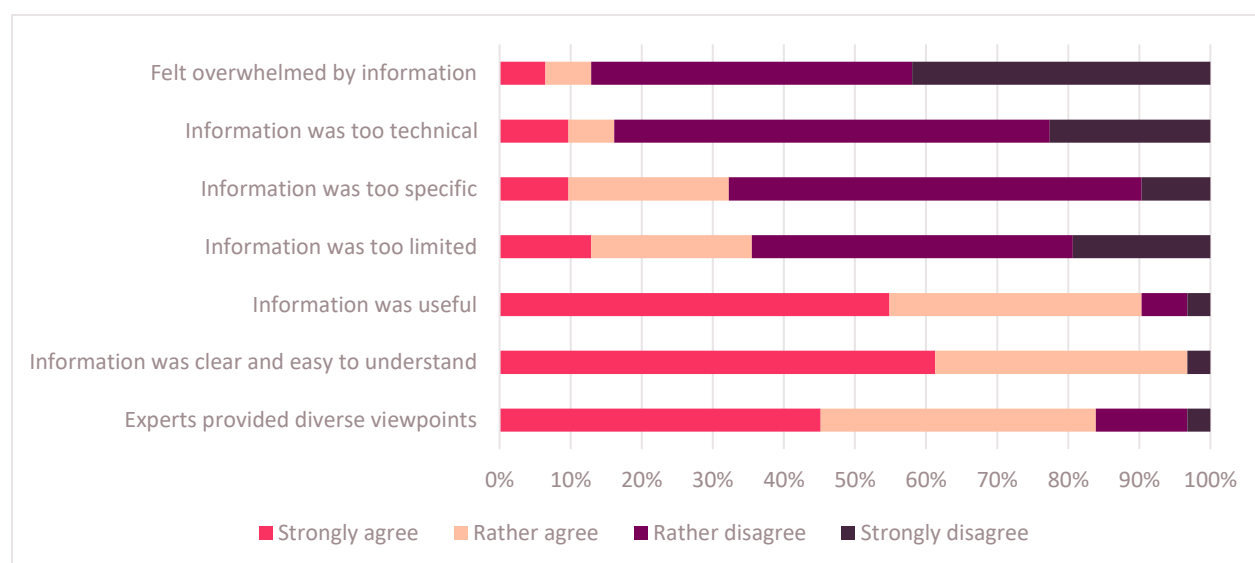


Figure 9 Perceptions of expert contributions

In addition to perceived representativity, duration, and expert input, facilitation plays a key role in shaping the deliberative environment. Participants overwhelmingly agreed that facilitators ensured fair and balanced discussions (100% agreement, with 83.87% strongly agreeing, n = 31).

%

Strongly agree	83.87%
Agree	16.13%
Disagree	0%
Strongly disagree	0%

Table 21 Perceptions on facilitation fairness

A similarly strong majority perceived facilitators as neutral and not favoring particular opinions (90.33% agreement, n = 31). Only a small minority reported concerns regarding potential bias.

	%
Strongly agree	80.65%
Agree	9.68%
Disagree	6.45%
Strongly disagree	3.23%

Table 22 Perceptions on facilitators' neutrality

These results indicate a high level of trust in the facilitation process and suggest that discussions were experienced as inclusive, well-managed, and respectful of diverse viewpoints.

5.2.3.2 Evaluation of deliberation

The evaluation also explored the quality of the deliberative interactions, aiming to assess both the diversity of viewpoints and the inclusivity of discussions. To begin, participants were asked how many people in their group held different views from their own. The majority of participants perceived a moderate diversity of perspectives: 56.25% reported that several participants held different views, 18.75% indicated about half, and 15.63% observed that most had different perspectives. Only a small minority felt there were no differing opinions (3.13%) or were unsure (6.25%). This distribution suggests that the forum successfully fostered a variety of viewpoints, creating space for meaningful debate while still allowing participants to find areas of common understanding.

	%
Most of them	15.63%
About half of them	18.75%
Several of them	56.25%
None	3.13%
Don't know	6.25%

Table 23 Perceptions on diversity

Participants were then asked about key elements of deliberative quality, including the ability to express opinions freely, equality of opportunity to speak, willingness to listen, respectfulness, and potential dominance in discussions (see Figure 10). Responses indicate strong support for inclusive and open dialogue. For example, 75% of participants strongly agreed that they were able to express their opinions freely, and 65.6% strongly agreed that all participants had equal opportunities to speak. Respectful deliberation was also widely reported, with 50% strongly agreeing that discussions occurred respectfully and 43.8% somewhat agreeing. This demonstrates that the forum created a safe space where participants felt their contributions were valued.

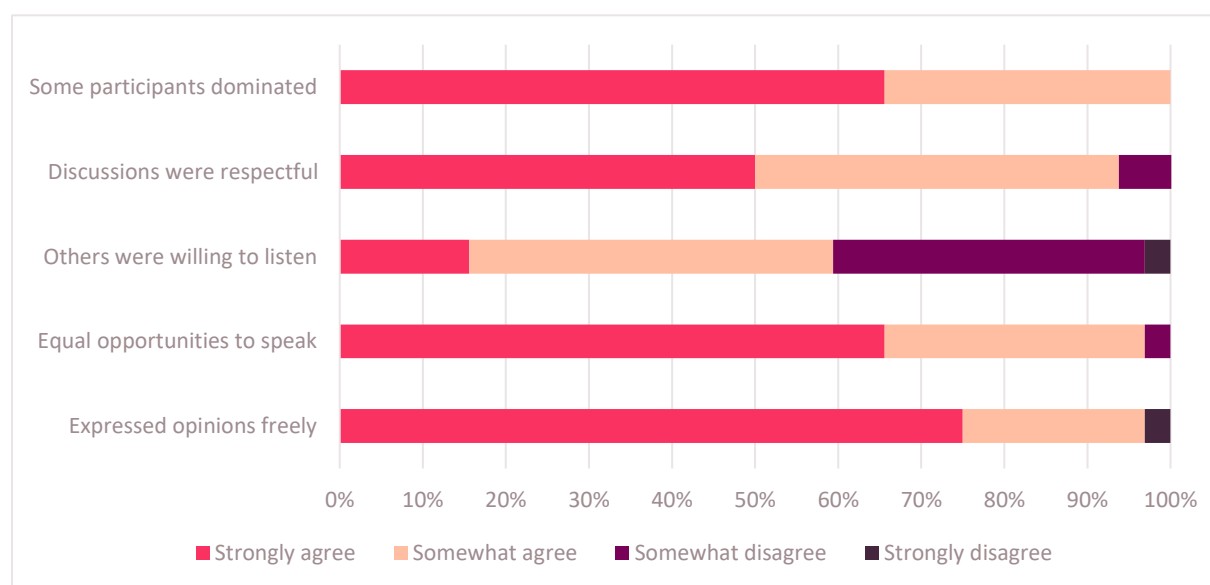
	Strongly agree	Somewhat agree	Somewhat disagree	Strongly disagree
Expressed opinions freely	75.0%	21.9%	0%	3.1%
Equal opportunities to speak	65.6%	31.3%	3.1%	0%
Others were willing to listen	15.6%	43.8%	37.5%	3.1%
Discussions were respectful	50.0%	43.8%	6.3%	0%

Some participants dominated	65.6%	34.4%	0%	0%
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Table 24 Perceptions on deliberations

Regarding attentiveness and active listening, responses were slightly more varied. While 15.6% strongly agreed that others were willing to listen, a larger proportion (43.8%) somewhat agreed, and 37.5% somewhat disagreed. This suggests that, although participants generally perceived an openness to diverse opinions, there were occasional challenges in fully hearing and integrating all voices, which is not uncommon in participatory processes with heterogeneous groups. Interestingly, as in every deliberative process, concerns about dominance were uniform. A majority (65.6%) strongly agreed that some participants dominated discussions, and 34.4% somewhat agreed.

Only very few participants reported strong disagreement with these statements, reflecting a largely shared perception of fair engagement despite natural variations in individual assertiveness. Taken together, these findings indicate that deliberations were broadly inclusive, respectful, and characterized by substantive diversity of viewpoints. Participants generally felt empowered to express their perspectives and engage in discussion, while the presence of dominant voices was acknowledged but did not seem to compromise the overall quality of dialogue. The results suggest that the forum successfully balanced freedom of expression with structured interaction, enabling meaningful, deliberative exchange among participants with diverse backgrounds and mobility experiences.

**Figure 10** Perceived deliberative quality

5.2.3.3 Evaluation of recommendations

The survey also aimed to understand participants' perceptions regarding the recommendations. In this sense, participants were asked to evaluate the extent to which their contributions were reflected in the final recommendations. The majority of respondents felt that their input had been meaningfully incorporated: 74.19% strongly agreed and 16.13% rather agreed that their contributions were considered in the outcomes. Only a small minority expressed disagreement (6.45% rather disagree, 3.23% strongly disagree). These results indicate that participants generally perceived the process as responsive and that their voices were taken seriously, reinforcing the sense of ownership and legitimacy of the recommendations.

	%
Strongly agree	74.19%
Agree	16.13%
Disagree	6.45%
Strongly disagree	3.23%

Table 25 Agreement with overall recommendations

To further explore participants' alignment with the outcomes, they were asked to consider the recommendations from the perspective of a decision-maker, assessing how many they would adopt. Responses were distributed across the spectrum: 6.45% reported they would adopt all recommendations, 29.03% would adopt most, 38.71% would adopt about half, and 25.81% would adopt some. While few participants would adopt every recommendation, the majority were willing to endorse at least half of them. This suggests that the recommendations were largely seen as relevant and actionable, though naturally not all proposals aligned perfectly with individual priorities or perspectives.

	%
All	6.45%
Most of them	29.03%

About half of them	38.71%
Some of them	25.81%
A few	0%
None	0%

Table 26 Perceptions on decision-makers' uptake of recommendations

Taken together, these findings highlight two important points. First, participants felt that the process was participatory and that their contributions influenced the outcomes, supporting the credibility of the recommendations. Second, while there was strong general agreement with the final set of proposals, the distribution of responses indicates that the process successfully balanced diverse viewpoints, producing recommendations that are broadly acceptable without necessarily achieving unanimous endorsement. This pattern is consistent with a deliberative process involving heterogeneous stakeholders, reflecting both inclusivity and the practical realities of negotiating shared outcomes.

5.2.3.4 Impact on participants

Beyond the evaluation of the process, quality of the deliberations, and recommendations, the post-survey also explored the perceived impact of participation on individuals. In this sense, participants assessed how useful they found the forum in three key areas: (i) learning new perspectives, (ii) reflecting on their own mobility habits, and (iii) feeling empowered to contribute (see Figure 11).

	Very useful	Somewhat useful	Not useful
Reflecting on personal mobility habits	40.63%	37.50%	21.88%
Learning new perspectives	90.63%	9.37%	0%
Feeling empowered to contribute	62.50%	31.25%	6.25%

Table 27 Perceived usefulness of the process

Encouragingly, process encouraged personal reflection. Nearly four in five participants (78.13%) reported that participation helped them reflect on their own mobility habits. The forum proved particularly effective in broadening participants' perspective. All respondents (100%) found the process at least somewhat useful for learning new viewpoints, and over 90% rated it as very useful in this regard. A similarly strong result was observed for empowerment: 93.75% of participants reported that the forum helped them feel able to contribute. These findings indicate that the process not only informed participants but also strengthened their sense of agency within discussions on urban mobility.

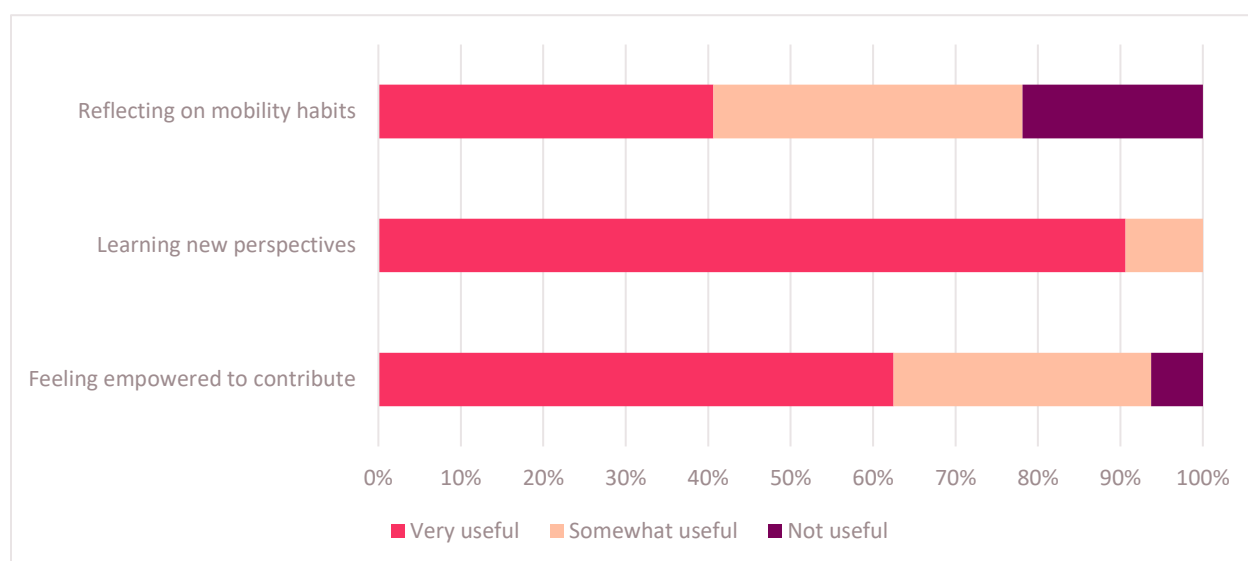


Figure 11 Impact on participants

More precisely, 75.01% indicated that they gained new arguments and perspectives.

	%
Strongly agree	34.38%
Agree	40.63%
Disagree	25.00%
Strongly disagree	0%

Table 28 Perceptions on gaining new arguments and perspectives

Beyond learning new perspectives, the quality of the forum is underscored by responses concerning openness to different options and perspectives (n = 31). Among participants, 93.55% agreed or strongly agreed that the process helped them consider alternative viewpoints.

	%
Strongly agree	54.84%
Agree	38.71%
Disagree	6.45%
Strongly disagree	0%

Table 29 Perceptions on considering alternative viewpoints

This is closely linked to reported learning in terms of dialogue and mutual understanding: 96.88% of respondents stated that they developed a better understanding of others' opinions.

	%
Strongly agree	46.88%
Agree	50.00%
Disagree	3.13%
Strongly disagree	0%

Table 30 Perceptions on understanding other's opinions

These results highlight the forum's strong contribution to deliberative capacity, demonstrating that participants not only absorbed information but also engaged constructively with diverse viewpoints.

In addition to these general effects, participants reported increased understanding across several thematic dimensions of urban mobility (see Figure 12).

	Strongly agree	Agree	Disagree	Strongly disagree
Urban mobility	21.88%	59.38%	18.75%	0%
Accessibility & inclusivity	15.63%	56.25%	28.13%	0%
Crisis preparedness	25.00%	46.88%	28.13%	0%
Diversity & redundancy	21.88%	46.88%	31.25%	0%
Sustainable & low-carbon transport	12.50%	37.50%	46.88%	3.13%

Table 31 Perceived increase in knowledge

A large majority indicated improved overall understanding of urban mobility (81.26%). Learning gains were also substantial in the areas of accessibility and inclusivity (71.88%), crisis preparedness (71.88%) and system diversity and redundancy (68.76%). Perceived learning related to sustainable and low-carbon transport was somewhat more mixed, though still positive overall, with half of participants reporting improved understanding. This pattern may suggest that participants entered the process with comparatively higher prior knowledge in this field, or that discussions placed greater emphasis on social and systemic resilience aspects of mobility.

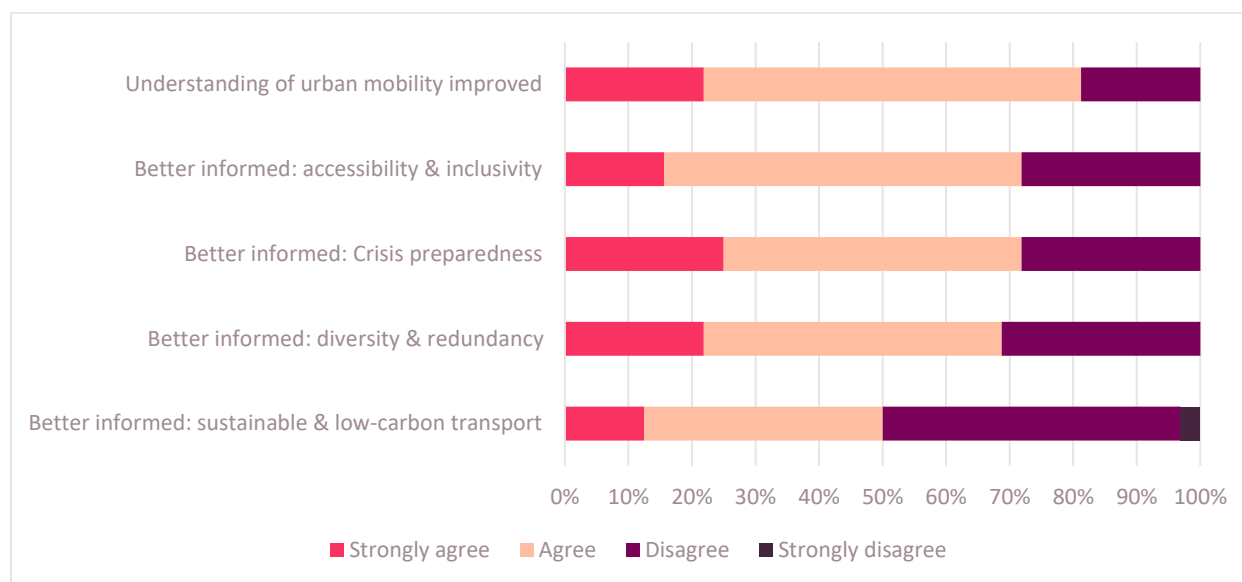


Figure 12 Participants' learning gains

To place these findings, participants were asked which elements of the process had the greatest influence on them. Responses were distributed relatively evenly among plenary sessions (34.38%), small working group discussions (34.38%), and expert presentations (31.25%).

	%
Plenary sessions	34.38%
Small working group sessions	34.38%
Expert presentations	31.25%

Table 32 Perceptions on various elements of process

This balanced distribution indicates that the process design functioned effectively as a combination of complementary formats. Notably, interactive formats – small group and plenary discussions – together accounted for nearly 70% of responses, underlining the importance of dialogical and participatory exchange in shaping participants' experiences and learning outcomes.

Beyond learning gains *per se*, the post-forum survey asked participants about the changes they expected to make in their own mobility behaviors following the discussions. The most frequently cited action, reported by 31.11% of respondents, is advocating for improved transport. This suggests that participants see their role not only in modifying personal habits but also in influencing broader

transport policies and contributing to systemic improvements. Advocacy emerged as a stronger motivation than direct behavior change, highlighting the forum's success in fostering a sense of agency and engagement with urban mobility issues.

Following advocacy, 22.22% of participants indicated that they do not expect any change in their personal mobility. This proportion shows that a subset of participants may view the forum primarily as an informative or reflective experience, rather than as a trigger for immediate action. It also suggests that some structural or personal constraints may limit their ability to alter travel behavior.

Behavioral intentions toward more sustainable mobility modes were moderately frequent. Using public transport more was cited by 17.77% of participants, while adopting more active forms of transport, such as walking or cycling, accounted for 15.55% of responses. These results demonstrate that the forum encouraged reflection on practical, everyday mobility choices and a willingness among participants to shift toward environmentally friendly and health-supportive travel modes. In contrast, trialing shared mobility services and reducing car use were less common intentions, with only 8.89% and 4.44% of responses, respectively. This pattern may indicate perceived barriers related to convenience, availability, or feasibility within the current urban context.

	%
I will advocate for improved transport	31.11%
I do not expect any change	22.22%
I will use public transport more	17.77%
I will use more active forms of transport (walking/cycling)	15.55%
I will try shared mobility services	8.89%
I will reduce car use	4.44%

Table 33 Perceived behavioral changes after participation

Taken together, the data reveal a dual focus among participants: a strong orientation toward policy advocacy and civic engagement, accompanied by a moderate commitment to adopting sustainable mobility behaviors. The relative prominence of advocacy and the comparatively lower rates of direct car reduction or shared mobility use suggest that while the forum effectively stimulated awareness and reflection, further interventions – such as infrastructure improvements, incentives, or targeted

awareness campaigns – may be needed to translate participants’ insights into broader behavioral change. This pattern is consistent with other findings from the forum, which highlight that participants valued learning, deliberation, and engagement, while intentions for personal mobility changes, though present, were secondary to systemic involvement.

5.2.3.5 Open-ended feedback

To also allow participants a say in the survey – not just in the process, they were invited to share their main takeaways from the forum and suggest one action they would like to see the city of Bratislava undertake next. Analysis of these open-ended responses reveals recurring themes related to learning, reflection, and aspirations for urban mobility improvements.

A primary insight emphasized the value of deliberation and dialogue. Several participants highlighted that *“meeting and discussing is important”* and that discussions could be conducted largely calmly, even when minor disagreements occurred. This reflects participants’ appreciation for the forum as a safe and constructive space for exchanging ideas. In terms of actionable follow-up, participants consistently called for more frequent citizen engagement and transparent communication, including informing the public about the forum outcomes and using similar meetings for broader city development topics, such as safety, ecology, and urban planning.

Another key theme concerned gaining diverse perspectives on transport. Many participants noted that the forum helped them see the city’s mobility system from viewpoints other than their own, including those of pedestrians, cyclists, and public transport users. One participant reflected, *“I gained perspectives from people who view transport differently.”* Participants suggested that this insight should be translated into practical measures, such as systematizing bike routes, improving public transport education, and giving greater attention to non-motorized mobility, emphasizing that city planning should balance the needs of all users.

A third theme related to recognizing shared challenges and strategic priorities. Participants observed a general consensus on key urban mobility problems, while also reflecting on broader urban quality: *“That a quality city means living well overall; transport is not the goal but the means; the point is to make it a good place to live, not just to have the fastest transport.”* Correspondingly, suggested next steps focused on tangible infrastructure improvements – such as building tram or railway tracks, safe bike routes, and parking management – but also on strategic implementation, including turning existing planning documents and discussions into actionable project.

A final theme highlighted civility, respect, and collective learning. Participants valued the respectful atmosphere of discussions and the sense of agreement on many points. They saw this as an opportunity for the city to reinforce participatory governance, suggesting high-level organization of future forums and continuous dialogue with citizens to ensure that policy reflects lived experiences.

In sum, the qualitative feedback demonstrates that the forum achieved both cognitive and practical outcomes. Cognitively, participants expanded their perspectives on mobility and urban living; practically, they articulated specific priorities and steps for city action. These insights indicate a strong link between deliberative participation and actionable recommendations, highlighting the forum's dual role as a space for reflection and a driver of concrete urban improvements.

Main insight	Next steps for the city of Bratislava
Meeting and discussing is important	Different perspectives on transport — pedestrians and cyclists. Systematize bike routes separated from roads and sidewalks.
Discussion can be conducted 98% calmly, the remaining 2% are not disruptive and can also be constructive	Inform that discussion with citizens is possible, and also perhaps point out that experts were not so far off from public opinions
Tolerance	Continue building already planned projects with quality architecture
Greater perspective	Construction of tram tracks, construction of railway tracks, double-tracking
More or less we all perceive the same problems in transport	Educate people about public transport
That a quality city means living well overall; transport is not the goal but the means; the point is to make it a good place to live, not just to have the fastest transport	Build safe bike routes
Decency and respect	Meaningful, high-level organization. Continue these discussions
I gained perspectives from people who view transport differently	Inform the public about this project, including the forum results, and use such meetings more often for city development in other areas such as safety, ecology, urban development...
Participants focused more on vehicles and less on pedestrians as transport users	Continue managing and designing city spaces as before. Greater emphasis on following road

	rules, limiting speeds in the city, on the riverside
Interesting perspective and agreement on most opinions	More frequent meetings like this
The forum revealed many interesting inputs, which showed me other perspectives on transport and movement in the city	It would be great to see, in the near future, the beginning of implementation of the strategic document into reality
Stimulating outputs, people were able to agree, opinions were intertwined	Improve public transport coverage and reinstate the Jurigovo námestie stop along with the necessary infrastructure.
That I see a vision that maybe it will be great one day, and I will be able to stay here and live.	Repair sidewalks, expand trams, resolve transfer issues, and make public transport cheaper.
Educational and informative	Involve citizens in transport planning
Beneficial! In terms of communication, awareness and feedback of mutual enrichment: “vox populi, vox dei”	Increase monitoring of cyclists in traffic
I believe that our opinions will not be lost	

Table 34 Open-ended feedback

5.2.3.6 Mobile application

In line with the broader purpose of the AntifragiCity project, the post-survey also explored participants’ perceptions of the potential role of mobile applications in urban mobility, particularly in relation to crisis response, everyday transport management, and inclusivity. Overall, responses suggest a strong recognition of the potential value of digital tools, alongside a clear awareness of accessibility challenges.

When asked whether a mobility app could help the city respond more effectively during crises or emergencies, such as heatwaves or flooding, 50% of participants strongly agreed and 31.25% agreed. Only a small minority disagreed (6.25%) or strongly disagreed (12.5%). Similarly, a majority of participants (62.5% strongly agree, 21.88% agree) endorsed the idea that residents should be able to share information through a mobility app to help improve transport. These findings indicate that

participants see mobile applications as potentially valuable for enhancing responsiveness, coordination, and citizen engagement in the urban mobility system.

At the same time, participants emphasized the importance of inclusive access. When asked whether the city should provide non-digital ways to receive information and give feedback, 59.38% strongly agreed and 28.13% agreed. This underscores that while digital tools are valued, complementary non-digital channels are essential to ensure participation by all residents, including those who may be technologically disadvantaged.

	Strongly agree	Agree	Disagree	Strongly disagree
A mobility app could help the city respond better during crises or emergencies (such as heatwaves or flooding)	50%	31.25%	6.25%	12.50%
Residents should be able to share information through a mobility app to help improve transport	62.50%	21.88%	3.13%	12.50%
The city should also provide non-digital ways to receive information and give feedback, so everyone can benefit	59.38%	28.13%	3.13%	9.38%

Table 35 Perceptions on a mobile application

Participants were also asked about their personal likelihood of using specific app functionalities. Real-time public transport and traffic information emerged as the most highly valued feature, with 68.75% very likely and 18.75% likely to use it. Similarly, a feedback/reporting function was seen as highly attractive (71.88% very likely, 18.75% likely). Route suggestions during disruptions or emergencies were also considered relevant (62.5% very likely, 25% likely). Features supporting sustainable travel choices and accessibility information were perceived as useful but showed somewhat lower adoption likelihood (40.63% and 37.5% very likely, respectively), suggesting that while participants appreciate these functions, they may prioritize immediate operational information and participatory functionalities over long-term sustainability or accessibility features.

	Very likely	Likely	Unlikely	Very unlikely
Real-time public transport and traffic information	68.75%	18.75%	6.25%	6.25%
Route suggesting during disruptions or emergencies	62.50%	25%	6.25%	6.25%
Travel suggestions that support more sustainable choices	40.63%	28.13%	9.38%	21.88%
Accessibility information (for example, step-free routes)	37.50%	34.38%	9.38%	18.75%
A way to give feedback or report problems	71.88%	18.75%	0%	9.38%

Table 36 Uses for mobile application

Qualitative responses identified several potential barriers to app adoption. These include limited internet access, particularly for socio-economically disadvantaged groups, seniors, or individuals with health or mobility limitations, as well as issues of technological literacy. One participant highlighted that *“pensioners will have a problem with this; they do not use a phone and do not have access to technology,”* while others noted broader digital divides, such as *“inaccessibility to the Internet, age, health disadvantage, [and] education for seniors”* and *“access to the Internet, or general use of modern technologies.”* Another participant underscored that *“network accessibility... for minorities who live in generational poverty... for them other alternatives would legitimately be found,”* while others simply referred to *“internet availability,”* with one asking what if there is no *“Wi-Fi connection.”* Participants also expressed concerns about fragmentation and single-purpose apps: if the app focuses too narrowly, it risks being underutilized, and ideally it should integrate multiple aspects of urban life to encourage sustained use, though *“simplicity”* should be at its core. One participant noted the risk of *“yet another application,”* and emphasized that if the app were single-purpose, it *“would have to include several aspects of life in Bratislava”* to be meaningful and widely adopted. Another similarly remarked that they are already *“overwhelmed by applications,”* and that for it to be used it should be *“up-to-date and beneficial.”* Another respondent asked remarked on the

perceived usefulness of such an application, believing it is *“useless, since there are few crisis situations and Bratislava is not that big.”* In line with the benefit and usefulness of the application, one participant mentioned that it should be available in various languages *“for foreigners.”*

Taken together, these findings indicate a strong perceived value of a city mobility app as a tool for both crisis management and citizen engagement, particularly for real-time information and feedback. At the same time, participants emphasized the necessity of designing such tools in an inclusive and integrated manner, with attention to digital accessibility and complementary non-digital communication channels. This aligns with broader themes from the forum regarding inclusivity, representativity, and practical implementation, suggesting that technology can support, but not replace, the participatory and equitable urban mobility strategies discussed throughout the process.

5.2.4 Comparative insights

30 participants (n = 30) completed pre- and post-surveys assessing their attitudes toward urban mobility policies, civic participation, and their perceived role in transport decision-making. Both surveys included identical questions, allowing assessment of how participants’ views shifted after the process, with particular attention to:

- Attitudes toward participatory and inclusive decision-making
- Willingness to engage in future mobility initiatives
- Willingness to accept sustainability-related trade-offs
- Interest, knowledge, and confidence regarding mobility issues
- Support for specific mobility measures

5.2.4.1 Citizens’ participation in urban mobility

In both surveys, participants rated agreement (Strongly disagree – Strongly agree) with three statements: public participation is important for good transport decisions, citizens’ everyday experiences are as important as expert knowledge, and involving diverse residents leads to fairer mobility outcomes.

	More Positive	Less Positive	No Change
Public participation is important	10%	13.33%	76.67%

Citizens' experiences = expert knowledge	20%	13.33%	66.67%
Diversity leads to fairer outcomes	13.33%	20%	66.67%

Table 37 Changes in perceptions regarding citizens' participation

Support for participatory and inclusive governance was already high. Still, a meaningful minority strengthened their agreement, particularly regarding the importance of public participation. This suggests the process reinforced the perceived legitimacy of citizen involvement in mobility decision-making.

Comparing participants' willingness to be involved in mobility initiatives, shows that one-fifth of participants moved toward more active involvement (from "no" or "don't know" to "yes, participate occasionally" or from "yes, participate occasionally" to "yes, participate actively"), indicating that the process translated attitudes into a stronger declared willingness to participate in real-world activities.

	%
More active	20%
Less active	6.67%
No change	73.33%

Table 38 Direction of change

Comparing pre- and post-process responses also reveals shifts in how participants imagine their role in urban mobility governance. On the aggregate level, three main trends are observable.

I. Continued strong interest in consultative participation

The option 'Participation in public consultations/citizens' forums' remains one of the most consistently selected options both before and after the process. However, post-survey responses show this option more often appearing as part of a broader participation portfolio, rather than as the sole form of engagement. In other words, while participants continue to value deliberative and

discussion-based spaces, they increasingly see them as one component of involvement rather than the only avenue.

II. Growth in multi-modal participation preferences

A clear pattern across the dataset is the expansion from one or two participation modes to three or four. Many participants added additional ways of contributing post-survey, most frequently:

- Testing mobility-related applications
- Engagement in community or neighborhood mobility projects

Both options show notable post-survey uptake, including among participants who previously selected only consultations or surveys. This shows an interest toward more hands-on participation, where citizens contribute as testers, co-creators, and local actors.

III. Stable but more strategically positioned role of surveys

‘Providing feedback via surveys’ are common both pre and post. However, post-survey it more often appears alongside other options, rather than as the only selected mode. Surveys thus remain seen as a legitimate form of engagement, but increasingly as supplementary rather than primary participation. This suggests a more differentiated understanding of participation depth.

5.2.4.2 Participants and urban mobility

In line with the questions on citizen participation and the topic of urban mobility, the surveys also measured three aspects of respondents’ engagement with urban mobility: their general interest in urban mobility, how informed they feel about the city’s mobility policies, and their confidence in contributing to urban mobility discussions.

Interest levels were already high at baseline, creating a ceiling effect. Still, nearly one-quarter reported increased interest, while very few reported declines (see Figure 13). The process therefore appears to have maintained or modestly strengthened engagement with mobility topics.

	%
Increased	23.33%
Decreased	3.33%

No change	73.33%
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Table 39 Direction of change

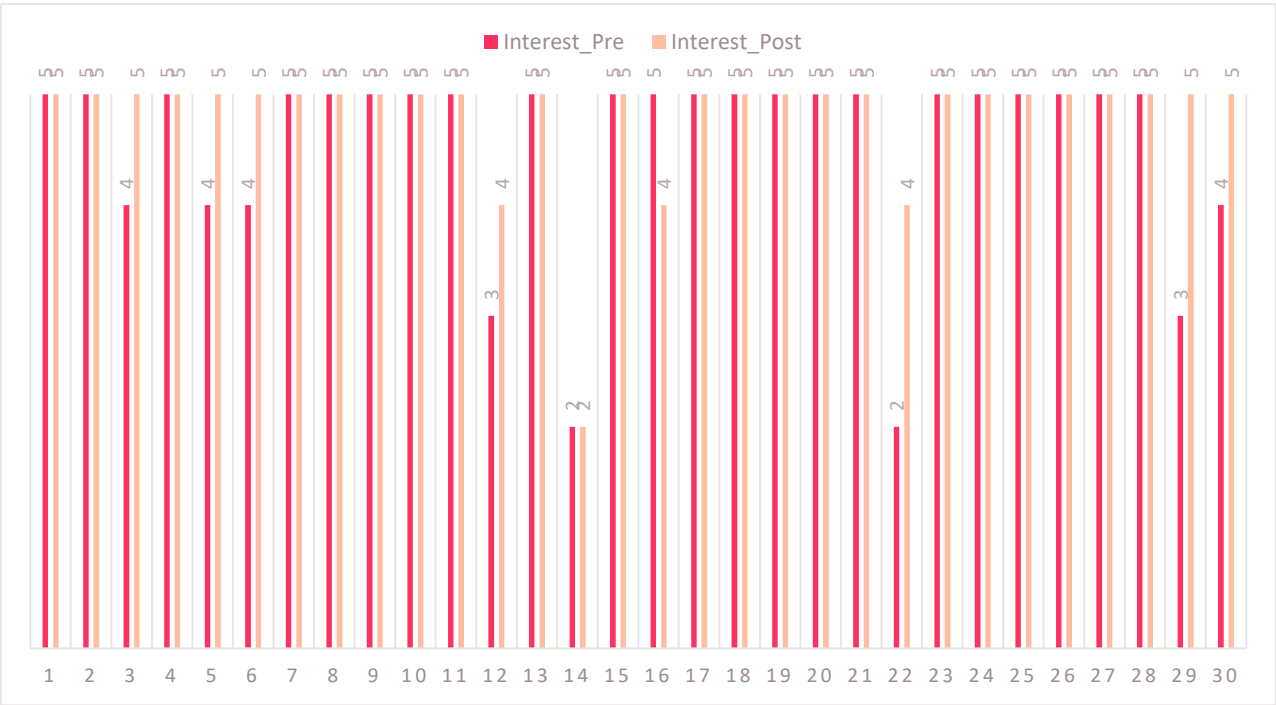


Figure 13 Comparison of participants' interest in urban mobility

Notably, more participants felt less informed after the process than more informed (see Figure 14). This pattern might be about the “complexity effect”: as individuals are exposed to more nuanced discussions, they may reassess and lower their self-perceived knowledge. Rather than indicating failure, this likely reflect greater awareness of the complexity of urban mobility systems.

	%
Increased	16.67%
Decreased	43.33%
No change	40%

Table 40 Direction of change

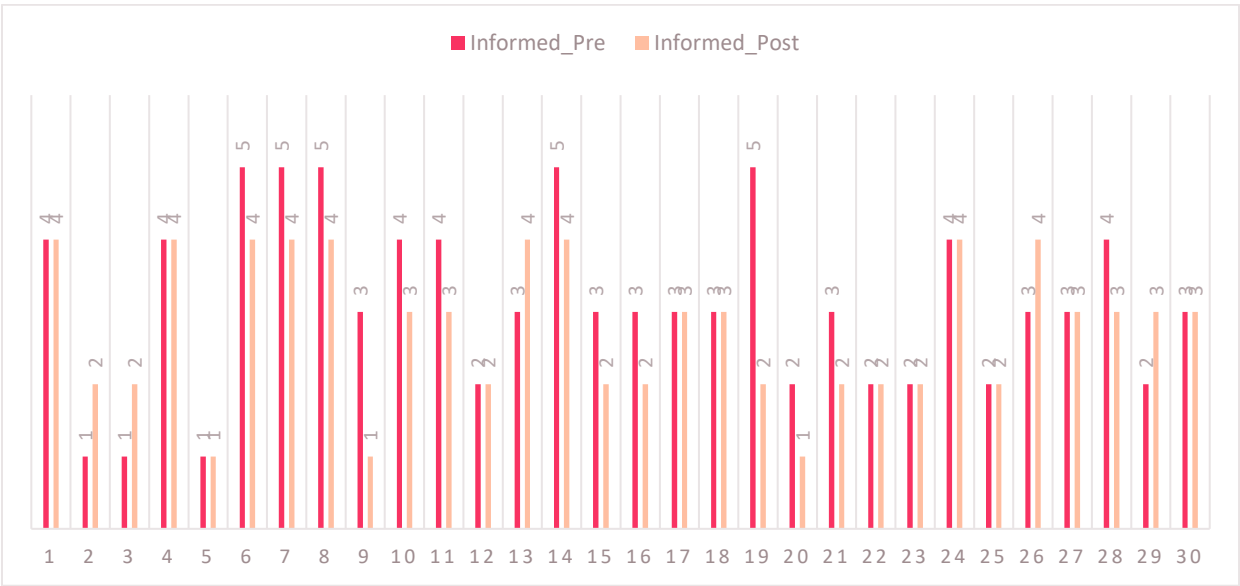


Figure 14 Comparison of participants' perceptions of feeling informed

Despite some decline in perceived knowledge, a significant number of participants felt more capable of contributing to mobility-related discussions or initiatives (see Figure 15). This suggests the process strengthened participants’ sense of civic efficacy, even where technical confidence did not increase.

	%
Increased	40%
Decreased	13.33%
No change	46.67%

Table 41 Direction of change

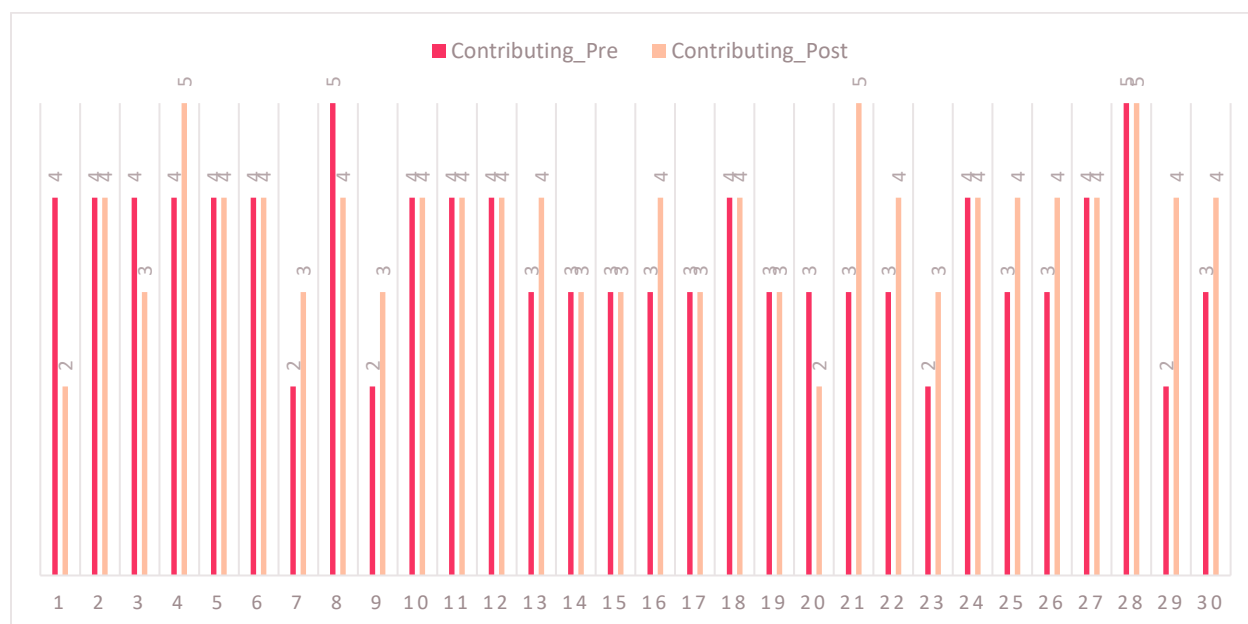


Figure 15 Comparison of participants' willingness to contribute to urban mobility policies

5.2.4.3 Mobility habits and opinions

In both surveys, participants rated their willingness to accept trade-offs related to sustainable mobility measures (scale 0–10) (see Figure 16).

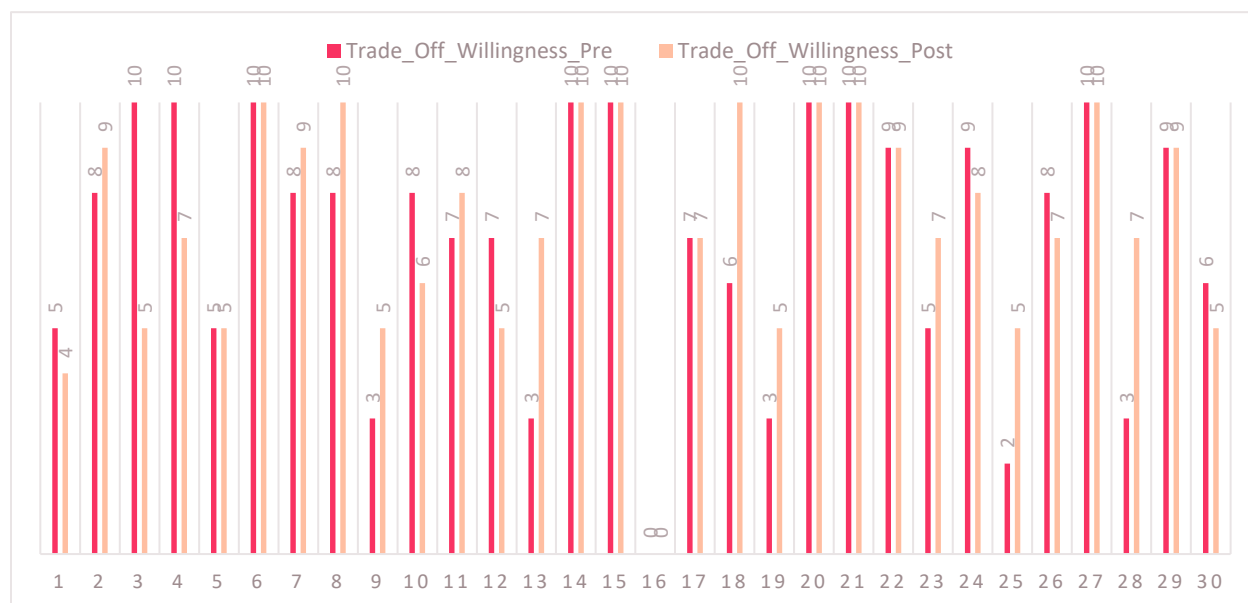


Figure 16 Comparison of participants' willingness to accept trade-offs

	%
Increased	40%
Decreased	26.67%
No change	33.33%

Table 42 Direction of change

A substantial share of participants became more willing to make compromises or accept restrictive mobility measures. While over a quarter reduced their willingness, the dominant pattern is a shift toward greater acceptance of trade-offs, suggesting increased openness to systemic or regulatory interventions (e.g., car restrictions, space reallocation).

In a related question, participants were also asked to identify factors that would encourage greater use of sustainable transport.

	Pre (%)	Post (%)	Δ (Post–Pre)
Better public transport	66.67%	76.67%	+10%
Better active mobility infrastructure	53.33%	50.00%	–3.33%
Greater share of greenery	53.33%	50.00%	–3.33%
Digital traffic management	50.00%	56.67%	+6.67%
Restrictions on cars in dense areas	43.33%	43.33%	0

30 km/h speed zones	40.00%	33.33%	-6.67%
Pedestrian zones	36.67%	43.33%	+6.67%
Reduced parking supply	30.00%	20.00%	-10%
Shared streets	23.33%	36.67%	+13.34
Congestion charging	13.33%	23.33%	+10%
Shared mobility services	10.00%	16.67%	+6.67%
None of the above	10.00%	0.00%	-10%

Table 43 Measures perceived as necessary to be antifragile

The comparison of pre- and post-survey responses reveals a clear reconfiguration of preferences rather than a wholesale replacement of priorities. Interestingly, the average breadth of support increased, with respondents selecting a larger number of measures in the post-survey than in the pre-survey. This indicates that participants moved away from single-solution preferences and toward combinations of interventions addressing infrastructure, governance, technology, and urban space simultaneously.

Better public transport was already the most strongly supported measure in the pre-survey, selected by 66.67% of respondents, and its support increased further to 76.67% in the post-survey. This confirms its role as the central pillar of perceived mobility improvement both before and after the intervention. In contrast, support for better active mobility infrastructure and a greater share of greenery remained high but showed slight declines of 3.33 percentage points each, suggesting that these measures were already widely accepted and did not experience further growth post-intervention.

The most notable increases occurred among measures associated with regulation, spatial reallocation, and system management. Support for shared streets rose from 23.33% to 36.67%, representing the largest absolute increase (+13.34 percentage points). Pedestrian zones also gained support, increasing from 36.67% to 43.33%, while congestion charging rose from 13.33% to 23.33%. These changes indicate growing acceptance of measures that actively restructure street use and manage demand rather than solely expanding infrastructure capacity.

Digital traffic management also increased in support, from 50% to 56.67%, making it one of the most frequently selected measures in the post-survey. This rise suggests greater recognition of coordination, real-time control, and adaptive management as integral components of urban mobility systems. Shared mobility services, while still selected by a smaller proportion of respondents overall, increased from 10% to 16.67%, indicating modest but consistent growth in interest in flexible, service-based mobility options.

By contrast, support for some car-related regulatory measures remained stable or declined slightly. Restrictions on cars in dense areas showed no net change, remaining at 43.33% in both surveys. Support for 30 km/h speed zones decreased from 40% to 33.33%, and reduced parking supply declined from 30% to 20%, suggesting more ambivalent attitudes toward certain restrictive measures when considered in isolation.

Finally, the disappearance of the “none of the above” option, which fell from 10% in the pre-survey to 0% in the post-survey, indicates a shift away from non-selection toward active endorsement of at least one measure. Taken together, the table suggests that post-survey responses did not simply favor more measures overall, but rather reweighted support toward integrative, regulatory, and system-oriented interventions while maintaining strong backing for core public transport provision.

5.2.4.4 Synthesis

The pre-post comparison results suggest that the process did not simply make participants more supportive of mobility or sustainability policies in a general sense. Instead, it contributed to deeper shifts in how participants understand urban mobility challenges, the nature of policy change, and their own role within governance processes.

Participants demonstrated a more realistic acceptance of sustainability trade-offs, showing greater willingness to support measures with tangible personal or spatial implications. Policy preferences broadened and became more integrated, with increasing support for coordinated packages combining regulation, infrastructure, technology, and environmental design.

Changes were also evident in participants’ civic self-perception. Confidence in contributing to mobility discussions increased for many, alongside a greater stated willingness to engage. This

growth in civic efficacy occurred even where perceived knowledge declined, suggesting increased awareness of complexity paired with stronger feelings of legitimacy as contributors.

Support for participatory governance remained consistently strong. Participants continued to value public involvement, lived experience, and diversity in decision-making, reinforcing the legitimacy of inclusive approaches to mobility policy.

Finally, preferences for how to participate became more differentiated and multi-modal. Participants increasingly envisioned themselves not only as respondents, but also as testers, collaborators, and community-level actors.

Taken together, these developments point toward more reflective, engaged, and system-aware perspectives on urban mobility and sustainability. Participants appear better equipped to understand policy trade-offs and to situate themselves as active contributors within broader governance processes.

5.2.5 Recommendations

The citizens' forum in Bratislava produced a structured and prioritized set of recommendations addressing urban mobility resilience, safety, governance, and communication, reflecting collective preferences and perceived trade-offs.

Across all thematic areas, participants consistently emphasized predictability, safety, coordination, and trust as key enablers of antifragile urban mobility systems. The recommendations indicate a public preference for mobility regimes that are pre-planned, clearly communicated, and centered on human safety and accessibility rather than short-term operational expediency.

1. Compliance with a binding mobility plan

Participants emphasized the importance of adhering to a long-term, binding mobility plan, particularly in times of disruption. The discussion reflected a strong expectation that emergency responses should not override established principles of accessibility, safety, and public-space quality, but rather operate within them.

Key recommendations included:

- Systematic mapping of transport habits and the use of data in long-term mobility strategies
- Ensuring that public space prioritizes people, with no exceptions that reduce safety or accessibility (e.g. reduced bus station capacity or unsafe tram stop slopes)
- Greater public involvement in the development of strategic mobility documents

Taken together, these recommendations point to a preference for predictable, rule-based planning that limits ad hoc decision-making. Citizens associate clearly defined plans and data-driven approaches with fairness, accountability, and continuity, especially when difficult trade-offs must be made.

2. Increasing safety in transport

Safety emerged as a dominant and non-negotiable priority, encompassing physical design, operational practices, and psychological perceptions of security across the transport system.

Recommended measures included:

- Safer stations, underpasses, stops, public transport vehicles, and crossings
- High-quality lighting of stops, sidewalks, and crossings, eliminating dark or unsafe spaces
- Clean and well-maintained public spaces to enhance perceived safety
- CCTV systems that are actively monitored and linked to police operations
- Increased police presence at critical locations and times
- Emergency buttons in public transport and stations, and incident reporting via mobile applications
- Durable, non-slip horizontal road markings
- Adequate green-light duration for pedestrians
- Crosswalks across large intersections and raised crossings (approx. 30 cm)
- Physical separation of corridors for pedestrians, cyclists, and motorized traffic

The emphasis on visibility, cleanliness, and monitoring indicates that perceived safety is as influential as objective safety in shaping mobility behavior. Participants clearly linked safety to confidence in using the transport system, particularly under stressful or uncertain conditions.

3. Greater communication with the public

Participants stressed the need for clear, consistent, and understandable communication, especially during service disruptions or periods of change.

Key recommendations included:

- Clear explanation of transport plans and measures
- Establishment of a single, unified communication channel
- Educational campaigns focused on sustainable mobility
- Information boards in public transport vehicles
- Direct, on-site communication during service disruptions or closures

Communication was framed not merely as information delivery, but as a way to explain intent, reduce confusion, and manage expectations. Citizens expressed a clear expectation that authorities should actively justify decisions and provide context, particularly when measures affect daily routines.

4. Centralized, multi-channel transport information

Participants called for a single, comprehensive source of real-time transport information, made accessible through multiple platforms to reach diverse user groups.

Recommended features included:

- Real-time location data for public transport vehicles
- Multilingual information availability
- Options for users to report incidents or disruptions
- Live traffic camera feeds
- Map-based visualization of traffic and transport conditions
- Inclusion of information on fire brigade interventions, planned closures, and emergency shutdowns
- Inspiration from best practices such as Prague's "Metropola všem" platform

Preferred communication channels included:

- Online applications
- Social media
- Telephone hotlines
- Integration with navigation systems and Google Maps
- Websites
- Information boards in public transport and at stops

These recommendations reflect a desire for clarity, redundancy, and inclusiveness, acknowledging that different users rely on different channels and that no single platform is sufficient in all situations.

5. Rapid and understandable solutions during crisis situations

This category focused on concrete operational expectations during disruptions.

Citizens recommended:

- Immediate information via boards in vehicles and at stops
- Rapid deployment and sufficient capacity of replacement services
- Clear guidance on how to reach replacement connections, including walking routes if necessary
- Communication of alternative routes

- Active passenger information by dispatchers and drivers

The emphasis here is on speed, legibility, and human presence. Participants highlighted that timely and understandable responses can significantly reduce stress and uncertainty, shaping whether passengers adapt constructively or disengage from the system.

6. Improved planning and coordination of traffic restrictions

Participants identified fragmented planning and uncoordinated restrictions as a recurring problem.

Recommendations included:

- Synchronizing multiple traffic restrictions to avoid cumulative impacts
- Clearly marked and well-planned detour routes
- Use of dynamic traffic management systems
- Improved coordination between state, city, and municipal district authorities

These points underscore the need for system-wide coordination, particularly when multiple actors and projects affect mobility simultaneously. Poor coordination was perceived as amplifying disruption beyond what is technically necessary.

7. Improved interchanges and park-and-ride systems

Interchanges and park-and-ride (P+R) facilities were discussed as strategic elements of the mobility system, supporting both everyday travel and periods of disruption.

Key recommendations included:

- Analysis of existing P+R systems to identify strengths and weaknesses
- Targeted communication with commuters from surrounding municipalities
- Physical wayfinding and signage within the city
- Strong integration with public transport, including safe pedestrian access
- Fare integration, where parking includes a public transport ticket
- Expansion of parking policy (PAAS) citywide and discouragement of car entry into the city
- Improved interchanges within urban areas and better integration with regional transport
- Inclusion of amenities such as toilets, services, bike sharing, and bike parking
- A comprehensive P+R system covering the whole city
- Free P+R at the outskirts, with increasing fees closer to the city center
- Timetable coordination to ensure smooth transfers

These recommendations suggest that users are willing to adjust travel behavior when alternatives are clear, reliable, and reasonably priced, and when interchanges are designed as functional spaces rather than mere transfer points.

8. Safe infrastructure for pedestrians

Pedestrian infrastructure was highlighted as a fundamental element of a well-functioning mobility system.

Recommended measures included:

- Clearly marked routes for cyclists (to reduce conflicts with pedestrians)
- Higher-quality sidewalks and barrier-free access
- Adequate lighting
- Traffic calming in exposed areas, including speed radars
- Signalized and raised pedestrian crossings
- Improved acoustic signals
- Direct pedestrian routes prioritized through city blocks
- Safe refuge islands

Participants emphasized that walking must remain safe, intuitive, and continuous, particularly when other modes are unavailable or disrupted. Pedestrian infrastructure was treated as a basic condition for accessibility rather than a supplementary feature.

9. Safe and connected cycling infrastructure

Participants identified cycling infrastructure as an important component of the urban mobility system, particularly for short- and medium-distance travel.

Key recommendations included:

- Fully connected cycling networks
- Uniform cycle path signage across Bratislava
- Practical, intuitive, and comfortable routes
- Priority for cyclists at intersections with cars
- Expanded bicycle parking
- Safe surfaces and high-quality maintenance

Cycling was discussed in terms of usability and continuity, with attention to design quality and network coherence rather than isolated segments.

10. Creation of an apolitical expert platform

Participants proposed the establishment of an independent expert body to support strategic mobility decisions.

Recommended functions included:

- Improved coordination between the city, municipal districts, state authorities, rail and road operators, and the Bratislava region
- Strategic decisions on the backbone transport system
- Efficient and transparent financial management

This recommendation reflects a desire to separate long-term mobility planning from short-term political cycles, and to base key decisions on expertise and cross-institutional coordination.

11. Expansion of tram networks

Participants discussed tram networks as a central element of the urban transport system, particularly for high-capacity movement.

Citizens highlighted:

- Trams as the fastest and highest-capacity mode
- Reduced conflicts with other transport modes through segregation from cars
- Ability to transport large numbers of people during disruptions
- Extended operating hours and improved timetables
- Extension and interconnection of radial lines
- Barrier-free tram stops

Trams were consistently framed as structuring infrastructure, shaping travel patterns and supporting reliable movement across the city.

12. Making public transport more attractive

Participants also identified softer measures aimed at improving the everyday appeal of public transport.

Key recommendations included:

- Emphasizing emissions reduction and environmental benefits
- Ensuring financial accessibility, with public transport cheaper than car use
- Highlighting convenience (no parking search, direct access to destinations)

- Development of a “transport calculator” to support informed mode choice
- Free public transport for children under 15
- Improved travel comfort (heating, air conditioning)

These measures focus on habit formation and user experience, reinforcing public transport as a normal, attractive choice rather than an option of last resort.

5.2.5.1 Prioritization / triage of citizens’ recommendations

To support the translation of citizens’ inputs into actionable guidance, the recommendations generated during the Bratislava citizens’ forum were prioritized based on the voting scores assigned by participants. Each proposal received positive (“plus”) and, in some cases, negative (“minus”) points, reflecting collective judgments about importance, urgency, and perceived trade-offs. Proposals without an explicit score were interpreted as neutral, indicating neither strong endorsement nor opposition.

The resulting prioritization does not represent a prescriptive implementation sequence, but rather a collective assessment of relative importance as expressed by participants. It highlights which interventions were perceived as foundational to the functioning of the urban mobility system, which were seen as important enabling measures, and which were considered situational or supportive. This triage provides insight into how citizens differentiate between structural investments, governance mechanisms, safety measures, and operational responses when considering urban mobility challenges and potential crisis conditions.

Tier 1 - highest priority: very strong collective support, perceived as foundational and urgent

- Expansion of tram networks (+25) → Seen as the backbone of the urban mobility system, essential for capacity, reliability, and continuity under stress
- Creation of an apolitical expert platform (+16 / -1) → Viewed as critical for coordination, strategic decision-making, and depoliticizing mobility governance
- Safe and connected cycling infrastructure (+16 / -3) → High priority despite minor concerns, indicating strong support with acknowledged trade-offs around space and implementation

Tier 2 – high priority: strong support, considered necessary conditions for a functioning system

- Increasing safety in transport (+14) → Encompasses stations, vehicles, crossings, lighting, monitoring, and enforcement
- Improved interchanges and park-and-ride systems (+13) → Framed as strategic infrastructure for managing demand and enabling modal shifts

- Safe infrastructure for pedestrians (+13) → Treated as a baseline requirement rather than an optional improvement

Tier 3 - medium-high priority: important enabling measures that support system performance and acceptance

- Compliance with a binding mobility plan (+12) → Emphasizes predictability, consistency, and rule-based decision-making
- Greater communication with the public (+11) → Focused on explanation, transparency, and trust-building

Tier 4 – medium priority: supportive measures that improve usability and adaptability

- Centralized, multi-channel transport information (+8) → Valued for real-time awareness, redundancy, and accessibility across user groups
- Making public transport more attractive (+6 / -1) → Generally supported, with minor concerns about feasibility or cost

Tier 5 – low-neutral priority: Recognized as important, but not strongly emphasized in voting

- Rapid and understandable solutions during crisis situations (+1) → Operationally critical but likely seen as implicit or overlapping with other categories
- Improved planning and coordination of traffic restrictions (0) → Viewed as necessary but possibly perceived as a background administrative function

The prioritization reveals a clear preference among participants for structural and system-defining interventions over short-term or purely operational measures. Proposals related to core infrastructure and long-term system capacity – most notably tram network expansion, cycling infrastructure, and strategic governance mechanisms – received the highest levels of support. This suggests that citizens perceive durable, capital-intensive investments as more consequential for addressing mobility challenges than incremental or reactive adjustments.

Safety-related measures consistently ranked among the highest priorities, indicating that participants view safety as a foundational condition for mobility, rather than an additional benefit. The strong emphasis on station safety, lighting, pedestrian crossings, and the separation of transport modes reflects an understanding that unsafe environments undermine the usability of all other mobility interventions. Participants appear to associate safety directly with system reliability and personal confidence in using public space.

Governance and coordination emerged as significant priorities when framed as system-level enablers rather than administrative processes. The strong support for an apolitical expert platform and compliance with a binding mobility plan indicates that participants recognize fragmented

decision-making and institutional inconsistency as key sources of inefficiency and disruption. Citizens appear to value clear roles, long-term direction, and expert-led coordination as essential for coherent mobility outcomes.

Interventions related to interchanges and park-and-ride systems occupied a high-priority position, reflecting an appreciation for network integration and accessibility across scales. Participants viewed these facilities not only as convenience measures but as strategic tools for managing demand, enabling modal shifts, and balancing urban and regional mobility needs. Their prioritization suggests sensitivity to the everyday experience of transfers and the importance of seamless connections.

Communication- and information-oriented recommendations formed a distinct middle tier in the prioritization. While clearly valued, they were generally perceived as enabling or supportive functions rather than primary drivers of system performance. The emphasis on centralized, multi-channel information indicates that participants expect information systems to complement physical infrastructure and operational capacity, rather than substitute for them.

Operational measures focused on crisis response – such as rapid deployment of replacement services and immediate passenger guidance – received comparatively fewer bonus points. This likely reflects the perception that such measures are baseline expectations of transport operators rather than discretionary policy choices. Their lower ranking should therefore not be interpreted as a lack of importance, but rather as an indication that participants considered them implicit components of competent system management.

Overall, the triage demonstrates that participants differentiate clearly between foundational interventions, enabling measures, and situational responses. The voting results suggest a public logic that prioritizes long-term capacity, safety, and governance clarity as prerequisites for effective day-to-day operations and credible crisis response.

5.2.5.2 Synthesis for antifragile mobility planning

The citizens' forum in Bratislava provides a coherent picture of how different types of mobility interventions are perceived in terms of social acceptability, priority, and legitimacy. The recommendations and their associated voting scores indicate that public acceptance is not driven by single measures in isolation, but by how interventions fit into a consistent, understandable, and fair mobility system.

Across the recommendations, participants demonstrated strong acceptance of structural and transformative interventions, particularly when these were framed as improving system-wide capacity, safety, and reliability. Investments in tram networks, cycling infrastructure, pedestrian safety, and interchanges were not perceived as disruptive or controversial, but rather as necessary components of a well-functioning urban mobility system. This suggests that citizens are willing to

support significant spatial reallocation and long-term investment when benefits are clear, tangible, and broadly shared.

Safety emerged as a central determinant of social acceptability. Measures addressing lighting, crossings, surveillance, separation of modes, and the quality of public space were consistently prioritized, indicating that interventions are only acceptable if they visibly reduce both physical risk and perceived vulnerability. In this sense, safety functions as a threshold condition: without it, other mobility interventions, regardless of environmental or efficiency benefits, risk rejection or underuse.

Governance-related recommendations reveal that social acceptability is closely linked to trust in decision-making processes. The strong support for a binding mobility plan and an apolitical expert platform indicates that participants value predictability, expertise, and institutional coherence. Interventions are more readily accepted when they are seen as part of a long-term strategy, guided by data and professional competence, rather than as reactive or politically driven decisions.

Communication and information provision were consistently identified as critical enablers of acceptance, particularly during periods of disruption. Citizens expect timely, clear, and multi-channel communication that explains not only what is happening, but why. The emphasis on centralized information systems and direct communication reflects an understanding that uncertainty and confusion can undermine compliance even with otherwise well-designed interventions.

The voting results also reveal an important distinction between strategic choices and baseline expectations. Operational crisis-response measures, such as replacement services and on-site passenger guidance, received fewer bonus points not because they were considered unimportant, but because participants viewed them as inherent responsibilities of transport authorities. Their lower prioritization underscores that social acceptability depends not only on what is implemented, but on whether it is perceived as a discretionary intervention or a minimum standard of competence.

Taken together, the Bratislava citizens' forum suggests that socially acceptable mobility interventions share several key characteristics: they enhance safety, strengthen system capacity, are embedded in coherent governance frameworks, and are accompanied by transparent communication. The findings indicate that citizens are prepared to accept prioritization, regulation, and long-term transformation of the mobility system, provided these measures are implemented consistently, visibly improve everyday experience, and are perceived as fair and legitimate.

5.2.6 Social acceptability of mobility interventions: Insights from Bratislava

The findings from the Bratislava AntifragiCity Citizens' Forum – combining pre- and post-surveys, and recommendations formulated and voted upon by the participants – provide in-depth insights into the social acceptability of mobility interventions, with particular focus on crisis-resilient urban mobility. Through structured deliberation, participants evaluated a broad spectrum of proposals –

from infrastructure upgrades and regulatory measures to digital tools and governance mechanisms – considering both everyday functionality and extreme-event scenarios.

The process reveals social acceptance is strongly shaped by perceived reliability, safety, fairness, and transparency. Interventions gain broad support when they enhance predictability, reduce uncertainty, and are embedded within a coherent system of rules and procedures. Participants repeatedly emphasize that legitimacy derives from clear planning frameworks, expert coordination, and evidence-based decision-making, particularly during crises where ad hoc measures could undermine trust.

Infrastructure and system robustness form central criteria for acceptability. Citizens consistently prioritize high-capacity public transport, continuous pedestrian networks, connected cycling routes, and well-integrated interchanges. These measures are perceived as foundational, broadly beneficial, and resilient under stress. Acceptance is highest when infrastructure delivers visible, tangible improvements in safety, accessibility, and continuity of service.

Safety and predictability dominate discussions across all modes. Both objective and perceived safety influence willingness to use transport services during crises. Measures such as well-lit and maintained public spaces, monitored stations, physical separation of transport modes, and signalized crossings are considered essential prerequisites for any socially acceptable intervention. Similarly, adherence to binding mobility plans and the establishment of apolitical, expert-led coordination bodies are viewed as critical for maintaining trust, legitimacy, and compliance.

Communication and information provision emerge as essential enablers of acceptability. Participants emphasize the importance of multi-channel, inclusive, and real-time information that explains the rationale for interventions, guides behavior during disruptions, and provides clarity on alternative routes or services. Interventions that involve trade-offs – such as reallocating street space, restricting car access, or prioritizing certain modes – gain acceptance only when paired with clear communication and predictable rules.

Participation in the process itself strengthens openness to integrated and regulatory measures. Compared with pre-workshop insights, participants show increased willingness to accept interventions that combine infrastructure, governance, and operational adjustments, provided they are transparent, fair, and coherent. This demonstrates that social acceptability is not static but can grow through informed, participatory processes.

In summary, socially acceptable mobility interventions combine:

- Robust infrastructure ensuring safe, reliable, and continuous mobility across modes
- Transparent governance and rule-based planning, including expert coordination and binding mobility plans

- Effective, multi-channel communication to reduce uncertainty and build trust
- Adaptive operational measures capable of maintaining usability during disruptions
- Equitable trade-offs, where restrictions or prioritizations are clearly justified and consistently applied

These insights provide a comprehensive social baseline, reveal citizen priorities and behavioral expectations under both routine and crisis conditions, and inform social response models that illustrate when and why citizens accept different interventions. They confirm that public engagement through citizens' forums actively increases acceptability of both technical and regulatory measures, supporting the objectives of the AntifragiCity project to develop resilient, inclusive, and socially endorsed urban mobility strategies.

6 Larissa

This section of the report presents the findings from the Larissa AntifragiCity Forum. The objectives of this evaluation encompasses two main aspects:

- Learning about the Larissa Forum proceedings: The evaluation assesses its success as a deliberative process, gaining valuable insights into its strengths and areas for improvement.
- Understanding the Larissa Forums' impact: The evaluation seeks to delve into its impact on its members, as well as delve into the recommendations.

6.1 Introduction

The Larissa AntifragiCity Forum was held over two days: Friday 6 February 2026, and Saturday 7 February 2026, with participant sessions running from 10:00 to 15:00 on both days.

The recruitment strategy was carefully designed to maximize diversity and ensure that the LAF membership reflected, as closely as possible, the demographic and attitudinal composition of Larissa's population. The process was led by LISER in close collaboration with the Municipality of Larissa and IED, leveraging their institutional networks and established best practices in citizen engagement.

A public call for participation was launched in January 2026 through multiple channels, including the Municipality of Larissa, IED, and the AntifragiCity project's communication platforms, spanning both social and mainstream media. Despite this extensive outreach, 26 individuals registered to participate in the Forum. Given the limited number of pre-registrations, a strategic decision was made to extend invitations to all registered individuals and additionally allow participation from

individuals who had not pre-registered. This adaptive approach aimed to maximize turnout while maintaining an inclusive and open deliberative environment.

The LAF welcomed 41 participants over two days. However, most participants attended only for a part of the first day, with only 5 participants returning on the second day. Following participants, this drop-out can be explained by the overall duration (10 hours across two days), which was perceived as too demanding for ‘volunteers’. A second stated reason was that the second day fell on a Saturday, a non-working day in Greece. This day proved unsuitable for interested individuals, significantly reducing participation on the second day. Despite this constraint, the Larissa AntifragiCity Forum effectively met its objectives: allowing meaningful participant-expert interaction, hearing from citizens regarding urban mobility, and the production of high-quality recommendations.

6.2 Evaluation

6.2.1 Representativity

Given that no sortition occurred for the Larissa AntifragiCity Forum, this section first briefly summarizes the diversity and representativity of the participants.

From the 24 respondents to the pre-survey, 16 were female and 8 were male.

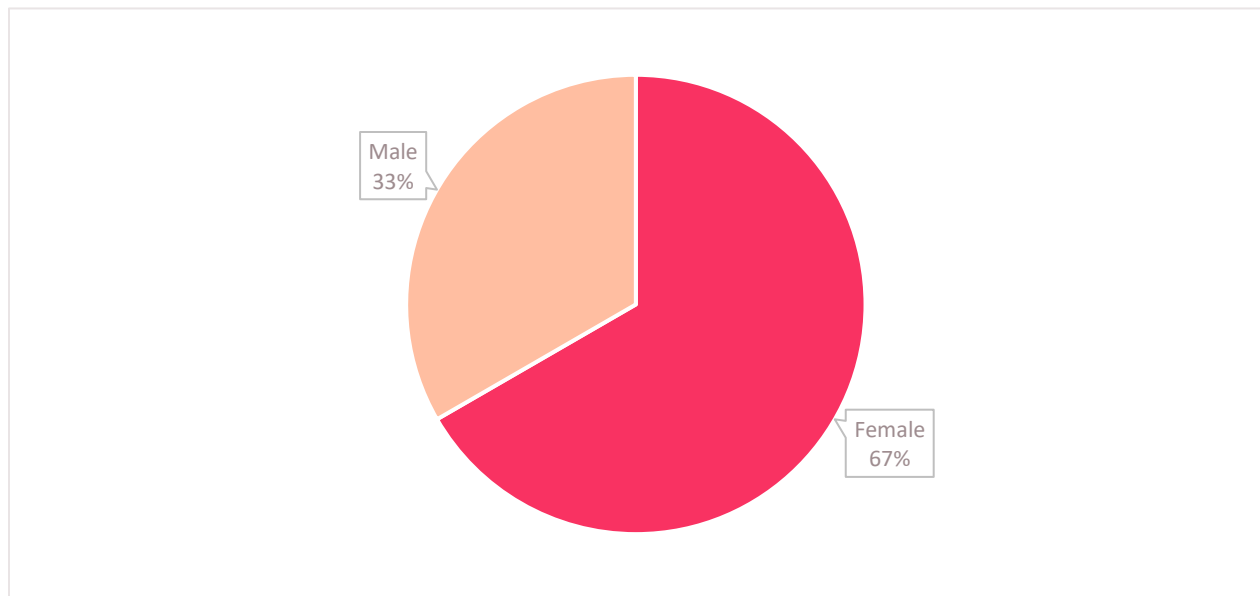


Figure 17 Participants by gender

When it comes to age, participants were between 18 and 74 years old.

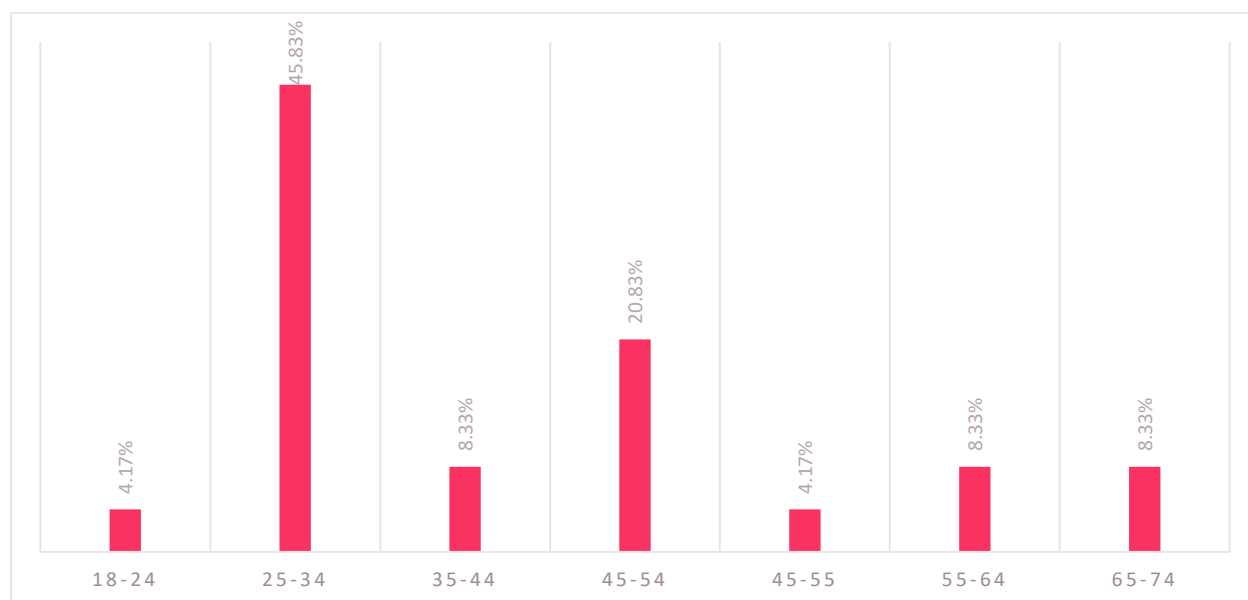


Figure 18 Participants by age groups

As typical in deliberative processes, most participants are highly educated. The vast majority of participants has at least a bachelor's degree, with a small minority of participants 'only' having a school-leaving degree.

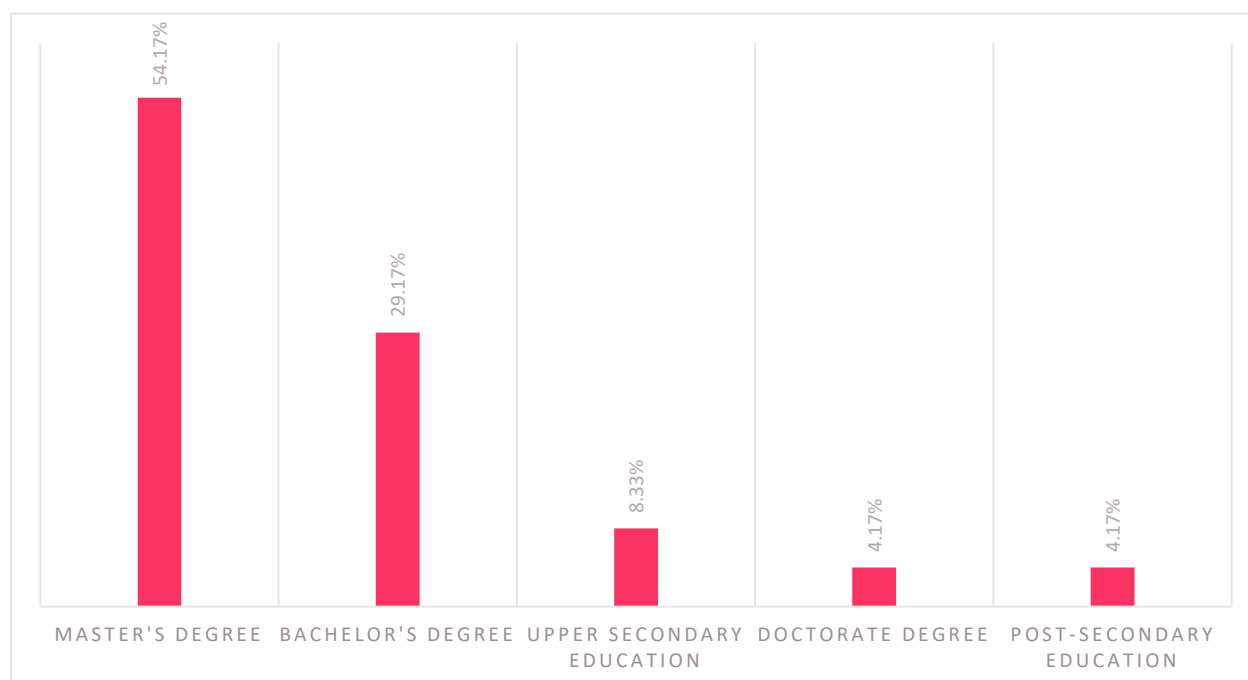


Figure 19 Participants by educational level

Considering participants' employment status, the majority are in paid employment ($n = 21$), while two participants are still studying and one participant is retired.

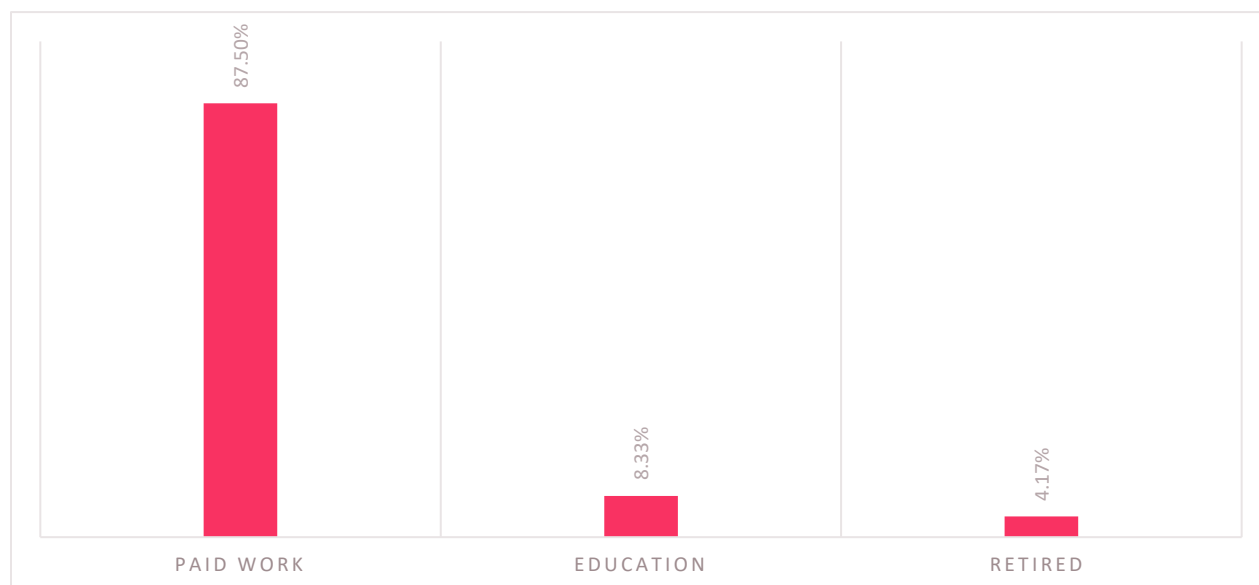


Figure 20 Participants by employment status

6.2.2 Pre-survey insights

This section summarizes the findings of the pre-participation survey conducted among participants of the Larissa AntifragiCity Forum. The survey aimed to understand participants' current travel behaviors, motivations, attitudes toward sustainable urban transport, and their interest in contributing to urban mobility initiatives. A total of 24 valid responses were collected ($n = 24$). Were this number of responses is different, this is acknowledged.

6.2.2.1 Citizens' participation in urban mobility

Respondents were asked whether they would like to be involved in shaping mobility initiatives in their city (see Figure 21).

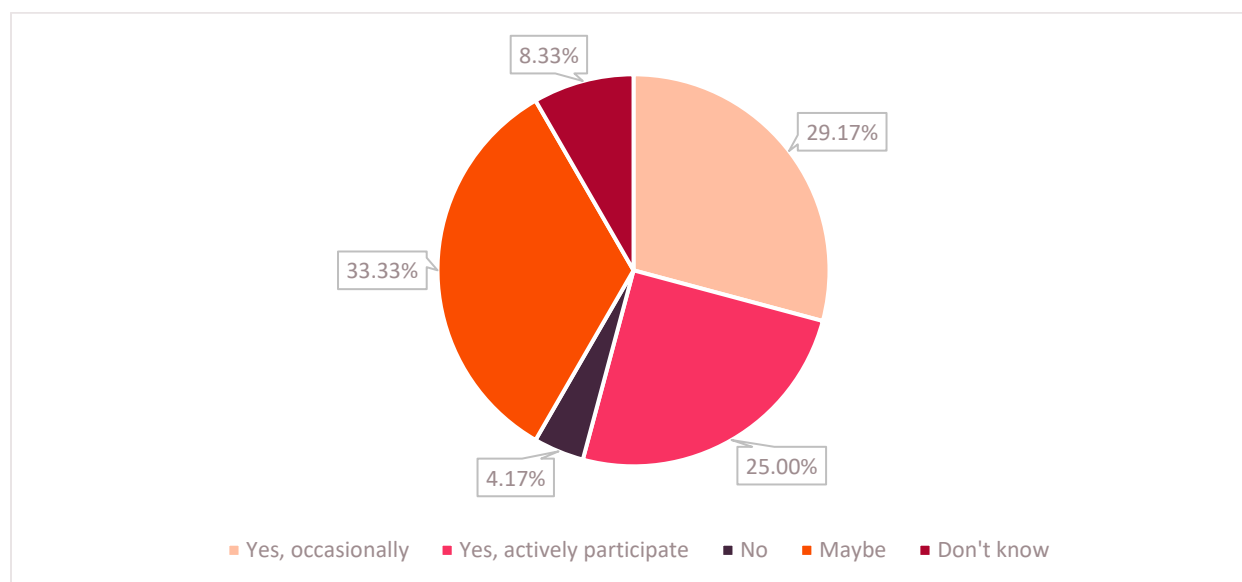


Figure 21 Willingness to shape mobility initiatives in city

A substantial share of respondents, 33.33%, answered “Maybe,” indicating some openness to involvement, though their participation would be conditional or limited. A further 8.33% responded “Don’t know”. 29.17% indicated that they would like to participate occasionally, suggesting a willingness to contribute on a flexible or periodic basis. A slightly lower share, 25%, expressed a desire to participate actively, reflecting a highly motivated group ready to engage regularly or take a hands-on role in mobility initiatives. Only a small minority, 4.17%, stated that they would not be interested in participating at all.

Overall, the data indicate that a slight majority (54.17%) of respondents are at least somewhat open to involvement, whether occasionally or actively. This suggests significant potential for public engagement and participatory programs, especially if opportunities are structured to accommodate both casual contributors and those willing to take a more active role.

To better understand participants’ willingness to participate, the next question asked participants how they would like to be involved in shaping mobility initiatives in their city. Four main forms of participation were offered: Participation in public consultations/citizens’ forums, testing mobility-related applications, engaging with community or neighborhood mobility projects, and providing feedback via surveys.

	%
Testing mobility-related applications	58.33%
Participation in public consultations / citizens’ forums	45.83%
Engaging with community or neighborhood mobility projects	37.50%

Providing feedback via surveys

37.50%

Table 44 Participation preferences

The results indicate diverse but clearly structured participation preferences. Testing mobility-related applications emerges as the most popular form of engagement, selected by 58.33% of respondents. This suggests a strong interest in innovative, practical, and digitally oriented modes of participation. Many participants appear willing to engage with experimental or trial-based initiatives, particularly those that allow them to interact directly with new tools or technologies related to mobility.

Participation in public consultations or citizens' forums is the second most preferred option, cited by 45.83% of respondents. This highlights the importance of structured, formal channels for civic engagement. Such settings offer visibility, opportunities for dialogue, and a sense of direct involvement in decision-making processes, which remain appealing to a substantial share of participants.

Engagement in community or neighborhood mobility projects is selected by 37.50% of respondents. This indicates moderate interest in hands-on, localized forms of participation that focus on tangible improvements in one's immediate environment. While these initiatives may foster a strong sense of ownership and community involvement, they also typically require greater time and personal commitment, which may limit participation to a smaller group.

Providing feedback via surveys is also selected by 37.50% of respondents, reflecting the value of flexible and low-effort participation options. Surveys allow individuals to contribute their opinions without the need for sustained engagement, travel, or coordination, making them accessible to a wide range of participants.

A notable observation is the variation in the number of participation modes selected per respondent. The distribution is as follows:

	%
0 Mode	20.83%
1 Mode	29.17%
2 Modes	20.83%
3 Modes	8.33%
4 Modes	20.83%

Table 45 Number of participation modes selected (*per participant*)

In total, 79.16% of respondents expressed interest in at least one form of participation, with almost half (49.99%) willing to engage through two or more channels. Overall, the findings demonstrate a strong willingness among respondents to engage with urban mobility initiatives, with preferences distributed across a variety of participation formats.

Finally, respondents were asked to indicate their agreement with three statements regarding public involvement in urban mobility planning: that public participation is essential for good transport decisions, that citizens' everyday experiences are as important as expert knowledge, and that involving diverse residents leads to fairer mobility outcomes. The distribution of responses is summarized below:

	Strongly Agree	Somewhat Agree	Somewhat Disagree	Strongly Disagree
Public participation is essential	58.33%	37.50%	4.17%	0%
Citizen experience as valuable as expert opinion	50%	37.50%	12.50%	0%
Engagement of diverse residents leads to fairer outcomes	54.17%	37.50%	8.33%	0%

Table 46 Opinions on citizens' participation

The majority of respondents strongly agree that public participation is essential for effective transport policy (58.33%), while a substantial portion (37.50%) somewhat agree. Only a small minority (4.17%) somewhat disagreed, and no respondents strongly disagreed, reflecting broad recognition of the importance of citizen input in shaping urban mobility decisions.

A similarly high level of support is observed for the statement that everyday citizen experiences are as important as expert opinions, with 50.00% strongly agreeing and 37.50% somewhat agreeing. A smaller group (12.50%) somewhat disagreed, indicating that while respondents overwhelmingly value practical, on-the-ground knowledge, a minority may still prioritize technical expertise in decision-making.

Strong agreement is also evident for the statement that engaging a diverse range of residents leads to fairer mobility outcomes, with 54.17% strongly agreeing and 37.50% somewhat agreeing. Only 8.33% somewhat disagreed, and no respondents strongly disagreed.

Overall, the findings indicate that respondents are motivated and supportive of participatory mechanisms in urban mobility planning.

6.2.2.2 Participants and urban mobility

In line with the questions on participation and urban mobility, the survey also measured three aspects of respondents' engagement with urban mobility: their general interest in urban mobility,

how informed they feel about the city's mobility policies, and their confidence in contributing to urban mobility discussions (see Figure 22).

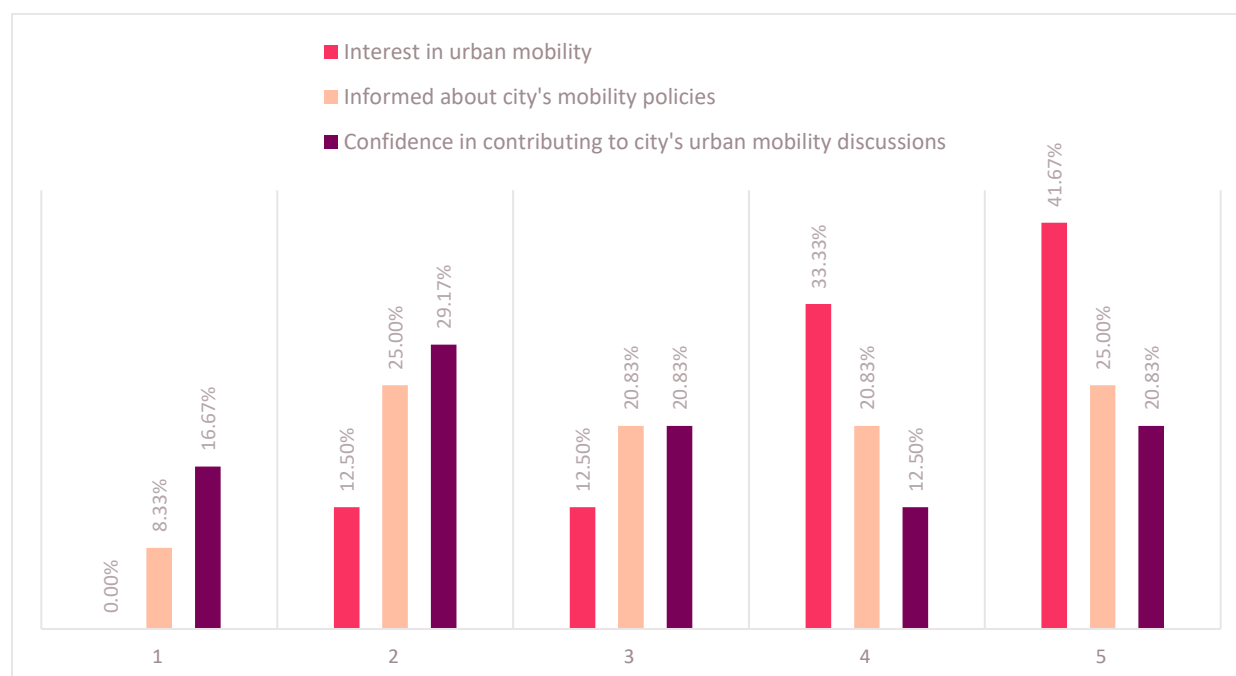


Figure 22 Knowledge urban mobility

Note: 1 = low interest, knowledge, or confidence; 5 = high interest, knowledge, or confidence

The results indicate a high overall level of interest in urban mobility among respondents. A large majority rated their interest at the upper end of the scale, with 41.67% selecting the maximum value of 5 and a further 33.33% selecting 4. Moderate interest (3) was reported by 12.50% of respondents, while an additional 12.50% selected 2. Notably, no respondents rated their interest at the lowest level of 1. Overall, this distribution suggests that urban mobility is perceived as a relevant and engaging topic by most participants, although a small minority express only limited interest.

In contrast, respondents report lower levels of awareness regarding the city's mobility policies. One quarter of participants (25%) rated themselves as highly informed (5), and an equal share selected the lowest levels of awareness (2). A further 20.83% rated their knowledge as 4, while another 20.83% selected 3. Only 8.33% reported very low awareness (1). This more even distribution across the scale indicates a clear gap between general interest in urban mobility and familiarity with the specific policies shaping the city's transport system.

Respondents' confidence in contributing to discussions on urban mobility is similarly moderate. While 20.83% feel very confident (5) and 12.50% rate their confidence at 4, the largest share of respondents selected lower to moderate levels: 29.17% chose 2 and 20.83% selected 3. A further

16.67% reported very low confidence (1). This pattern suggests that although some respondents feel capable of engaging in discussions, a considerable proportion remain hesitant or uncertain about their ability to contribute meaningfully.

Overall, the data reveal a notable imbalance between interest, knowledge, and confidence. While respondents demonstrate strong interest in urban mobility, awareness of policy frameworks and confidence to engage in discussions are comparatively weaker.

6.2.2.3 Mobility habits and opinions

Participants were asked to identify their most-used mode of daily travel, selecting only one option (see Figure 23). Among the forum participants, private car is the most frequently used mode of transport, with 41.67% of respondents indicating that they use it for their daily travel. Walking is also significant, being used by 37.5% of participants. Public transport is used by 16.67% of participants, while cycling is less common, accounting for 4.17%. This distribution suggests that while private car is the dominant mode, there is a substantial reliance on walking.

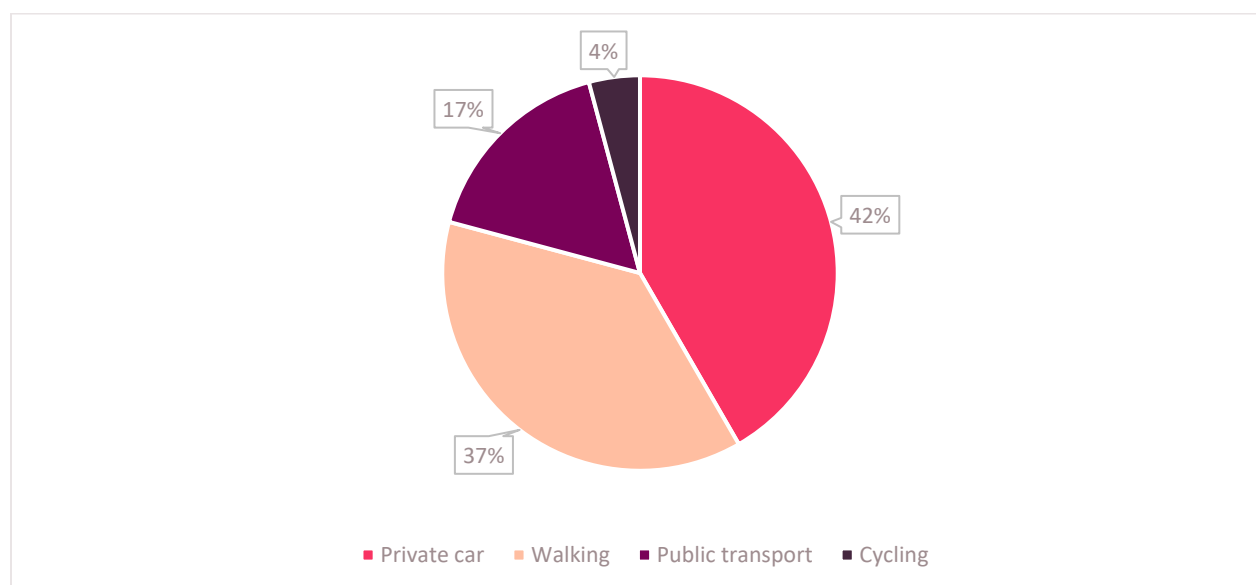


Figure 23 Most used daily transport mode

Respondents were asked to identify the main reasons for choosing their primary mode of transport, with the option to select up to three factors (see Figure 24). The results presented here reflect aggregated selections across all participants, meaning the percentages represent how frequently each factor was chosen overall. The aggregated results show that practical considerations clearly outweigh environmental or purely personal motivations, highlighting that everyday functionality is the dominant influence on travel behavior.

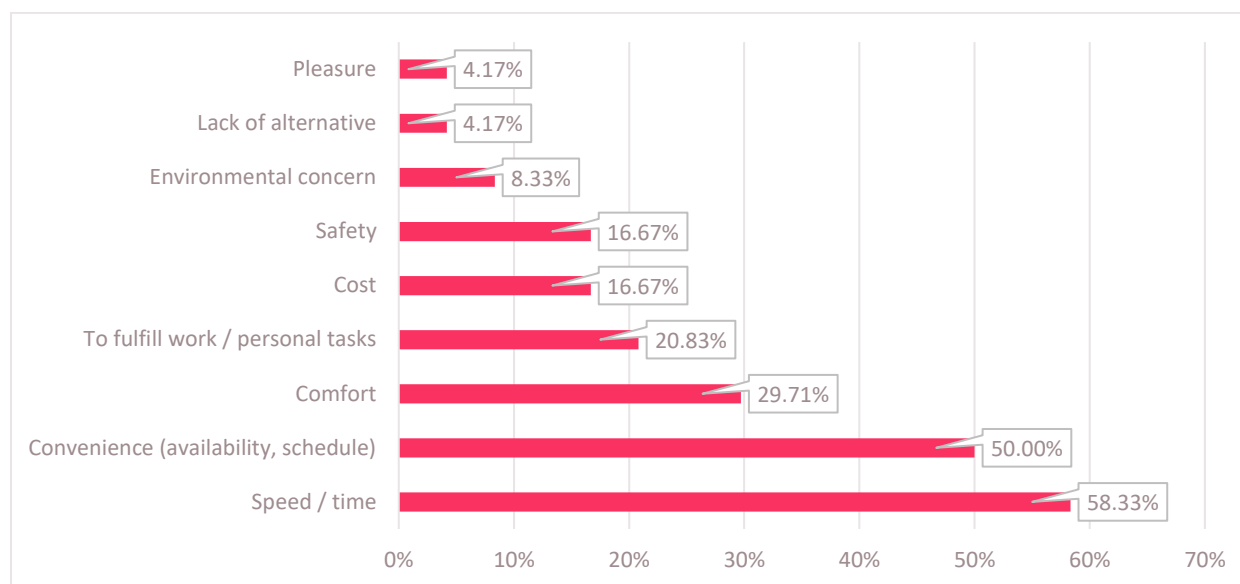


Figure 24 Main reasons for choosing daily mode of transport

The most frequently cited reason for primary mode choice was speed or time efficiency, accounting for 58.33% of all selections. This highlights travel time as the dominant determinant of modal choice, indicating that participants prioritize reaching their destination as quickly and predictably as possible. Efficiency therefore plays a central role in everyday mobility decisions.

Convenience, including factors such as availability and scheduling, was selected by 50% of respondents, followed by comfort at 29.71%. These findings suggest that, beyond speed, the overall ease and quality of the travel experience strongly influence transport preferences. Participants favor modes that are simple to access, well integrated into their daily routines, and physically comfortable, underscoring the importance of user experience and reliability.

A further 20.83% of participants indicated that their primary travel mode was chosen because it enables them to fulfill work or personal tasks. Cost and safety were each selected by 16.67% of respondents. The equal weighting of these two factors suggests that financial considerations and perceived safety are similarly influential, although both remain secondary to time efficiency and convenience in determining modal choice.

Only a small proportion of respondents cited environmental concern (8.33%), lack of alternatives (4.17%), or pleasure (4.17%) as motivating factors. This indicates that for a limited share of participants, travel decisions are shaped by environmental values, constrained choice, or enjoyment rather than purely functional considerations.

Overall, the findings demonstrate that transport mode choice is driven primarily by practical factors, particularly speed, convenience, and comfort. While cost and safety play a meaningful supporting role, environmental considerations and enjoyment influence only a small segment of users.

Respondents were also asked to identify factors that could encourage greater use of sustainable transport modes. One participant indicated that no factors would motivate such a shift in their case. The results indicate a strong preference for practical, system-level improvements, suggesting that infrastructure quality, network connectivity, and service performance are the most influential drivers of increased adoption.

	%
Better infrastructure (e.g. bike lanes, covered bus stops, sidewalks)	62.50%
Better connectivity / frequency	50%
Lower costs	37.50%
Time savings	20.83%
Improved safety	20.83%
Improved comfort	8.33%

Table 47 Factors encouraging greater sustainability

The most frequently cited factor is better infrastructure, selected by 62.50% of respondents. This highlights the importance of high-quality physical environments – such as dedicated bike lanes, continuous sidewalks, and well-designed transit stops – in encouraging the use of sustainable modes. The prominence of this factor suggests that many respondents perceive existing infrastructure as insufficient or unattractive, particularly when compared to the convenience and protection offered by private vehicles.

Closely following is better connectivity and higher service frequency (50%). This highlights the role of network coverage, coordination between modes, and reliable scheduling in shaping transport behavior. Limited service frequency, poor spatial coverage, or weak intermodal integration appear to constrain the usability of sustainable transport options, making them harder to integrate into everyday routines. Enhancing connectivity would therefore improve reliability and reduce the perceived effort associated with non-car travel.

Lower costs are selected by 37.50% of respondents, indicating that financial considerations play a meaningful, though not dominant, role in encouraging sustainable transport use. While affordability can act as an incentive, its position below infrastructure and connectivity suggests that cost reductions alone are unlikely to trigger widespread behavioral change unless accompanied by tangible improvements in service quality and accessibility.

Time savings and improved safety are each cited by 20.83% of respondents. The importance of time savings aligns with earlier findings showing that efficiency is a central concern in transport decision-

making. This suggests that sustainable modes are more likely to be adopted when they are perceived as competitive in terms of travel time. Safety concerns, meanwhile, indicate that perceived or actual risks remain a barrier for a notable minority of users. These concerns may be particularly relevant for walking and cycling, where exposure to traffic, insufficient separation from vehicles, or poor lighting can discourage use.

Improved comfort is mentioned by only 8.33% of respondents, making it the least frequently cited factor. While comfort-related improvements – such as shelter, seating, or smoother travel conditions – contribute positively to the travel experience, they appear to function more as supporting features rather than primary motivators for switching to sustainable modes.

Overall, the findings suggest that structural and functional improvements, particularly better infrastructure and stronger network connectivity, are the most effective levers for increasing sustainable transport use. Efficiency-related factors such as time savings further reinforce adoption, while cost, safety, and comfort play secondary but complementary roles.

Regarding participants' willingness to make compromises to increase environmental sustainability, the survey results indicate a generally high level of willingness among respondents to accept compromises in order to improve environmental sustainability in their city, such as reducing parking availability, increasing green spaces, or lowering speed limits (see Figure 25). Responses were measured on a scale from 0 (not willing at all) to 10 (very willing), and the distribution is clearly weighted toward the higher end of the scale.

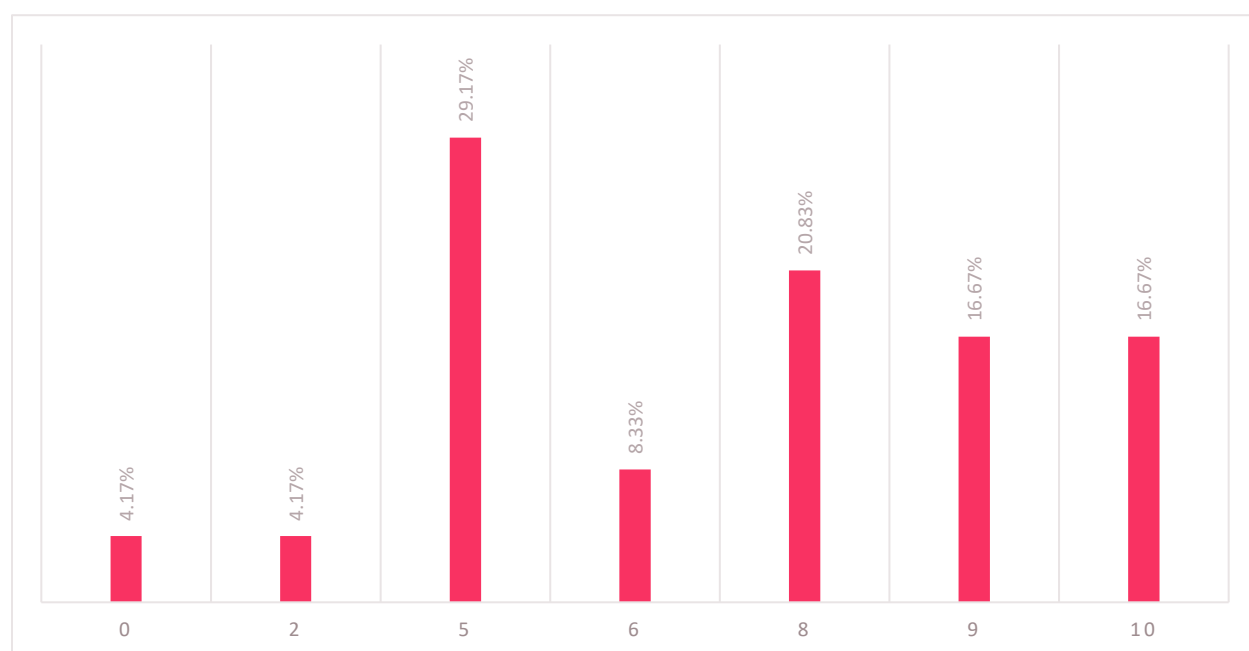


Figure 25 Willingness to accept trade-offs for environmental sustainability

Note: 0 = not at all, 10 = very willing

More than half of respondents (54.17%) reported high willingness to accept sustainability-related trade-offs, selecting values between 8 and 10 on the scale. The highest response values, 9 and 10, were each chosen by 16.67% of participants, while 20.83% selected 8. Together, these responses indicate that a substantial share of respondents express a strong readiness to support environmental measures, even when these involve personal or practical adjustments such as reduced parking availability, increased green space, or lower speed limits.

A further 37.50% of respondents indicated moderate willingness, selecting values between 5 and 6. Specifically, 29.17% selected 5 and 8.33% selected 6. This group appears generally supportive of sustainability initiatives but may hold reservations regarding their scope, implementation, or potential impacts on daily mobility. Their responses suggest conditional acceptance, which may depend on careful policy design, clear communication, and the mitigation of negative effects on accessibility and convenience.

A small minority of respondents (8.34%) expressed low willingness, selecting values below 5. This includes 4.17% who selected 0 and 4.17% who selected 2, indicating a limited degree of resistance to sustainability-related trade-offs within the sample. While not dominant, these responses suggest that a small segment perceives such measures as undesirable or incompatible with their needs.

Overall, the findings point to a broadly positive orientation toward urban sustainability measures. The majority of respondents demonstrate a willingness to accept personal or practical compromises in support of environmental goals, while a sizeable moderate group highlights the importance of thoughtful, inclusive implementation to maintain public support.

Lastly, participants were asked about a number of measures they believe can improve mobility and quality of life in their city, both in everyday situations and during crises (such as extreme weather or emergencies). Participants were free to select as many measures as wished.

	%
Improved walking/cycling infrastructures	70.83%
Reduced car use in busy or densely populated areas	66.67%
Better public transport	66.67%
More greenery	50.00%
Car-free areas	45.83%
Digital traffic management	41.67%
Increased shared mobility options	33.33%
30 km/h speed limit zones in residential or city areas	29.17%
Shared streets	25.00%
Fewer parking spaces	20.83%

Congestion charges	8.33%
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Table 48 Measures perceived as necessary to be antifragile

The results indicate a clear preference for measures that enhance active and sustainable mobility. Improved walking and cycling infrastructure is the top priority (70.83%), followed closely by reducing car use in busy or densely populated areas and better public transport (both 66.67%). These findings suggest that respondents value interventions that make sustainable modes safer, faster, and more convenient, directly supporting daily mobility needs.

Environmental and urban quality measures, such as more greenery (50%) and car-free areas (45.83%), also receive strong support, reflecting the desire for healthier, more livable urban spaces. Technology-driven solutions like digital traffic management (41.67%) and flexible mobility options such as shared vehicles (33.33%) have moderate appeal, indicating openness to innovative approaches.

Measures perceived as restrictive or limiting personal convenience – 30 km/h speed limits (29.17%), shared streets (25.00%), fewer parking spaces (20.83%), and congestion charges (8.33%) – are less popular. This suggests that acceptance of regulatory or cost-based interventions is lower unless paired with clear, tangible benefits.

Overall, the findings highlight that investments in infrastructure, accessibility, and high-quality active and public transport options are likely to receive broad support, while measures primarily focused on restrictions or financial penalties may require careful framing and complementary incentives to gain public acceptance.

6.2.3 Post-survey insights

This section synthesizes post-forum survey findings and qualitative feedback to evaluate the representativity, process design, expert contributions, deliberative interactions, participant takeaways, and attitudes toward mobility-related digital tools. A total of 11 valid responses were collected (n = 11). Where this number is different, this is explicitly mentioned.

6.2.3.1 Evaluation of the process

Perceptions of representativity are mixed among respondents (n = 9). According to the survey data, 33.33% indicated that certain groups were missing, 22.22% did not perceive any groups as missing, and 44.45% selected “don’t know.” Despite the relatively balanced distribution, open-ended responses suggest that perceived gaps primarily concerned specific mobility needs and user perspectives. Frequently mentioned were people with disabilities and reduced mobility, described for example as “disabled people, elderly,” “the elderly.” Participants also referred to the absence of younger age groups, such as “children,” and “young people.” One participant further commented that

“political leadership was absent”, as well as “residents of degraded areas,” noting that “people from 25-60, i.e., of productive age, participated, the rest were absent.”

	%
Yes	33.33%
No	22.22%
Don’t know	44.45%

Table 49 Perceived representativity

Participants also evaluated the structure, pacing, and duration of the meetings. A majority (54.55%) considered the duration appropriate. However, a considerable proportion (36.36%) felt the process was too long, while 9.09% regarded it as too short. This distribution suggests that while the overall design was broadly acceptable, there is room for fine-tuning the time allocation in future iterations to better balance depth of discussion with participant fatigue.

	%
Yes	54.55%
No, too long	36.36%
No, too short	9.09%

Table 50 Perceptions duration of process

This ambivalence is also reflected in participants’ willingness to remain engaged. When asked whether they would participate in a longer process, 63.64% responded positively, whereas 36.36% indicated they would not be willing to extend their engagement. The findings point to a generally high level of commitment, though not without limits in terms of time investment.

	%
Yes	63.64%
No	36.36%

Table 51 Willingness to participate in a longer process

Expert contributions were evaluated very positively (see Figure 26). A large majority agreed that experts provided diverse viewpoints (54.55% strongly agree; 36.36% rather agree), reinforcing perceptions of balance and credibility. All participants agreed that the information was clear and easy to understand (54.55% strongly agree; 45.45% rather agree) and that it was useful (63.64% strongly agree; 36.36% rather agree).

Regarding potential shortcomings, opinions were more differentiated. For example, views were evenly distributed concerning whether the information was too limited (27.27% strongly agree; 27.27% rather agree; 27.27% rather disagree; 18.18% strongly disagree). Similarly, perceptions of the information being too specific or too technical were mixed, though the largest shares tended toward disagreement. Importantly, no respondent reported feeling overwhelmed by the amount of information, indicating that the pacing and structure of expert input were appropriate. One participant noted that *“the issues were simple and understandable, visual presentations would help more.”*

	Strongly agree	Rather agree	Rather disagree	Strongly disagree
Experts provided diverse viewpoints	54.55%	36.36%	9.09%	0%
Information was clear and easy to understand	54.55%	45.45%	0%	0%
Information provided was useful	63.64%	36.36%	0%	0%
Information was too limited	27.27%	27.27%	27.27%	18.18%
Information was too specific	36.36%	27.27%	36.36%	0%
Information was too technical	27.27%	36.36%	36.36%	0%

Table 52 Perceptions on experts and information

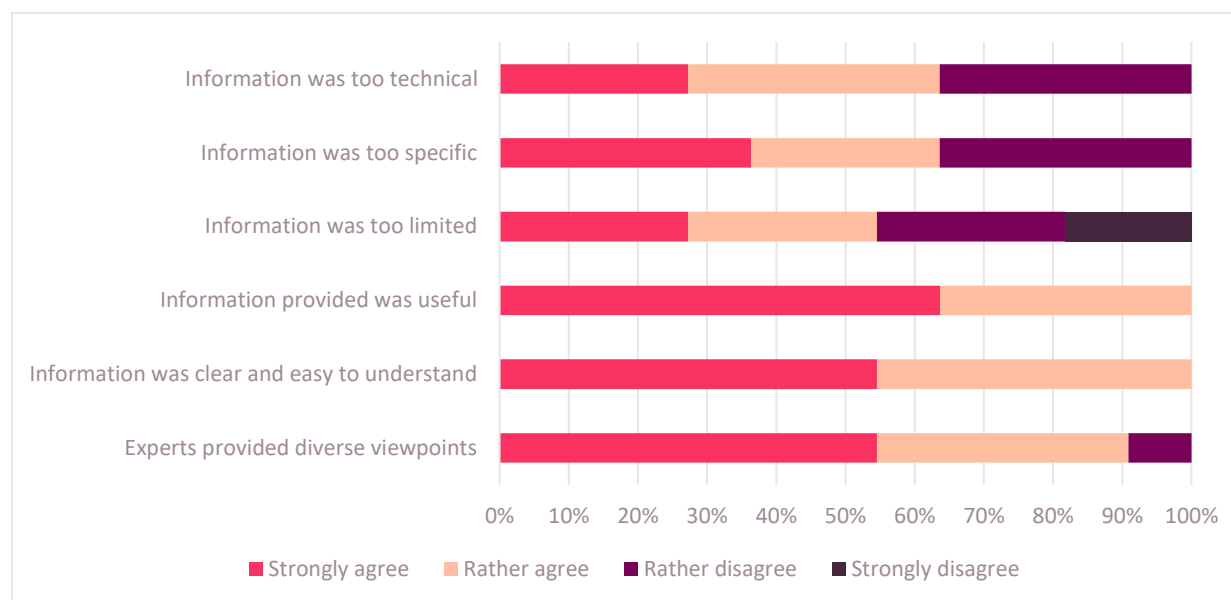


Figure 26 Perceptions of expert contributions

6.2.3.2 Evaluation of deliberation

The quality of deliberative interactions was assessed with a focus on diversity of perspectives and inclusivity. All participants reported encountering differing views within the group: 36.36% stated that most participants held different views, 36.36% indicated that several did, 18.18% said about half did, and 9.09% selected “don’t know.” These results demonstrate that participants were exposed to a range of differing opinions, suggesting that the forum succeeded in fostering pluralism within discussions.

	%
Most of them	36.36%
About half of them	18.18%
Several of them	36.36%
Don’t know	9.09%

Table 53 Perceptions on diversity

Participants then evaluated core dimensions of deliberative quality, including freedom of expression, equality of speaking opportunities, willingness to listen, respectfulness, and perceived dominance (see Figure 27). The findings point to a largely positive assessment. A substantial majority (81.82%)

strongly agreed that they were able to express their opinions freely, while the remaining 18.18% somewhat agreed. Equal opportunities to speak were rated even more positively (90.10% strongly agree; 9.09% somewhat agree).

Respectful interaction was also widely reported: 63.64% strongly agreed and 36.36% somewhat agreed that discussions were respectful. Similarly, 63.64% strongly agreed and 27.27% somewhat agreed that others were willing to listen, though 9.09% somewhat disagreed, indicating isolated experiences of insufficient responsiveness.

Perceptions of dominance were more ambivalent. While 27.27% strongly agreed and 45.45% somewhat agreed that some participants dominated the discussions, 27.27% somewhat disagreed. These findings suggest that although dominant voices were noticed, they coexisted with an overall perception of fairness and inclusivity. The presence of stronger personalities did not appear to undermine participants' broader evaluation of deliberative quality.

	Strongly agree	Somewhat agree	Somewhat disagree	Strongly disagree
Expressed opinions freely	81.82%	18.18%	0%	0%
Equal opportunities to speak	90.10%	9.09%	0%	0%
Others were willing to listen	63.64%	27.27%	9.09%	0%
Discussions were respectful	63.64%	36.36%	0%	0%
Some participants dominated	27.27%	45.45%	27.27%	0%

Table 54 Perceptions on deliberations

Taken together, these results indicate that deliberations were largely inclusive, respectful, and characterized by exposure to diverse viewpoints. Participants generally felt heard and able to articulate their perspectives, even if certain individuals were perceived as more vocal than others.

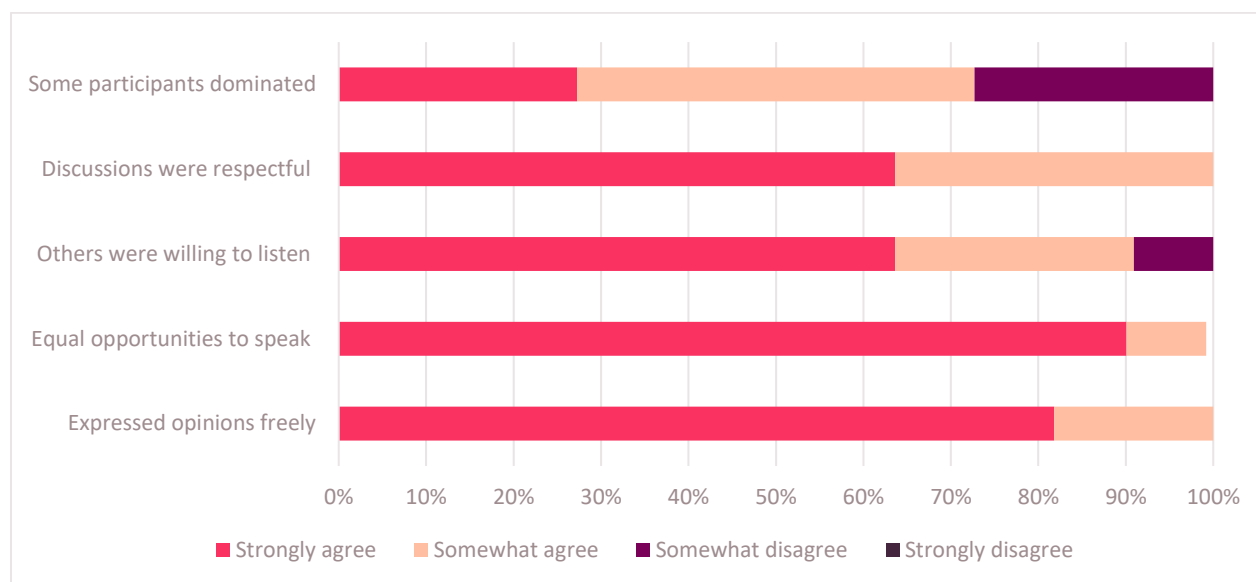


Figure 27 Perceived deliberative quality

6.2.3.3 Impact on participants

Beyond evaluating procedural aspects, the post-survey examined the perceived impact of participation (see Figure 28). Participants assessed how useful they found the forum in three areas: reflecting on their own mobility habits, learning new perspectives, and feeling empowered to contribute. 63.64% rated the forum as very useful for reflecting on personal mobility habits and 36.36% as somewhat useful. Identical shares were reported for learning new perspectives. Regarding empowerment, 54.55% found participation very useful, 36.36% somewhat useful, and 9.09% not useful. Overall, these results indicate that the forum was perceived as beneficial across multiple personal dimensions.

	Very useful	Somewhat useful	Not useful
Reflecting on personal mobility habits	63.64%	36.36%	0%
Learning new perspectives	63.64%	36.36%	0%
Feeling empowered to contribute	54.55%	36.36%	9.09%

Table 55 Perceived usefulness of the process

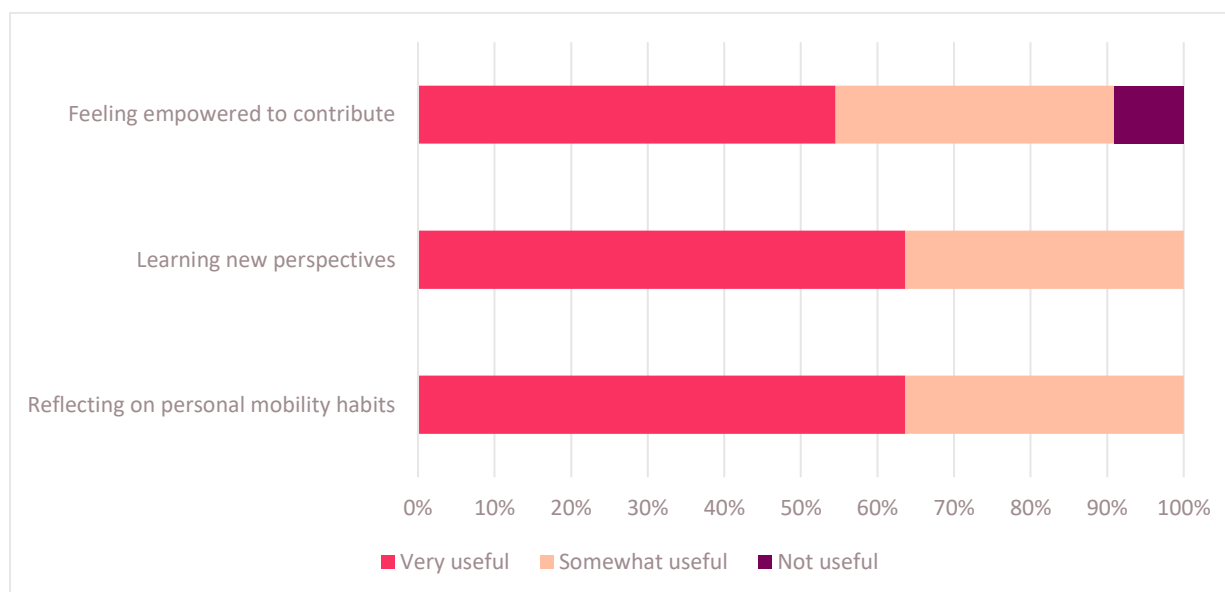


Figure 28 Impact on participants

Learning-related statements further underline this impact. Regarding the statement “I gained more arguments and perspectives to support my own opinion about the issue,” 36.36% strongly agreed, 36.36% agreed, and 27.27% rather disagreed. For “my understanding of others’ opinions of the issue became clearer through this process,” 45.45% strongly agreed, 36.36% agreed, and 18.18% rather disagreed. These findings suggest substantial cognitive gains for most participants, though not universally.

	Strongly agree	Agree	Rather disagree
I gained new arguments and perspectives	36.36%	36.36%	27.27%
I developed a better understanding of others’ opinions	45.45%	36.36%	18.18%

Table 56 Impact on participants

Participants also reported increased understanding across thematic dimensions of urban mobility (see Figure 29). For urban mobility overall, 72.73% strongly agreed and 27.27% agreed that their understanding improved. Identical overall agreement (100%) was observed for accessibility and inclusivity (63.64% strongly agree; 36.36% agree).

Learning gains related to crisis preparedness were somewhat lower but still substantial (45.45% strongly agree; 36.36% agree; 18.18% disagree). Similarly, increased understanding of diversity and redundancy in transport systems was reported by 90.91% (63.64% strongly agree; 27.27% agree), while 9.09% disagreed. For sustainable and low-carbon transport, 63.64% strongly agreed and 18.18% agreed, whereas 18.18% disagreed.

	Strongly agree	Agree	Disagree	Strongly disagree
Urban mobility	72.73%	27.27%	0%	0%
Accessibility & inclusivity	63.64%	36.36%	0%	0%
Crisis preparedness	45.45%	36.36%	18.18%	0%
Diversity & redundancy	63.64%	27.27%	9.09%	0%
Sustainable & low-carbon transport	63.64%	18.18%	18.18%	0%

Table 57 Perceived increase in knowledge

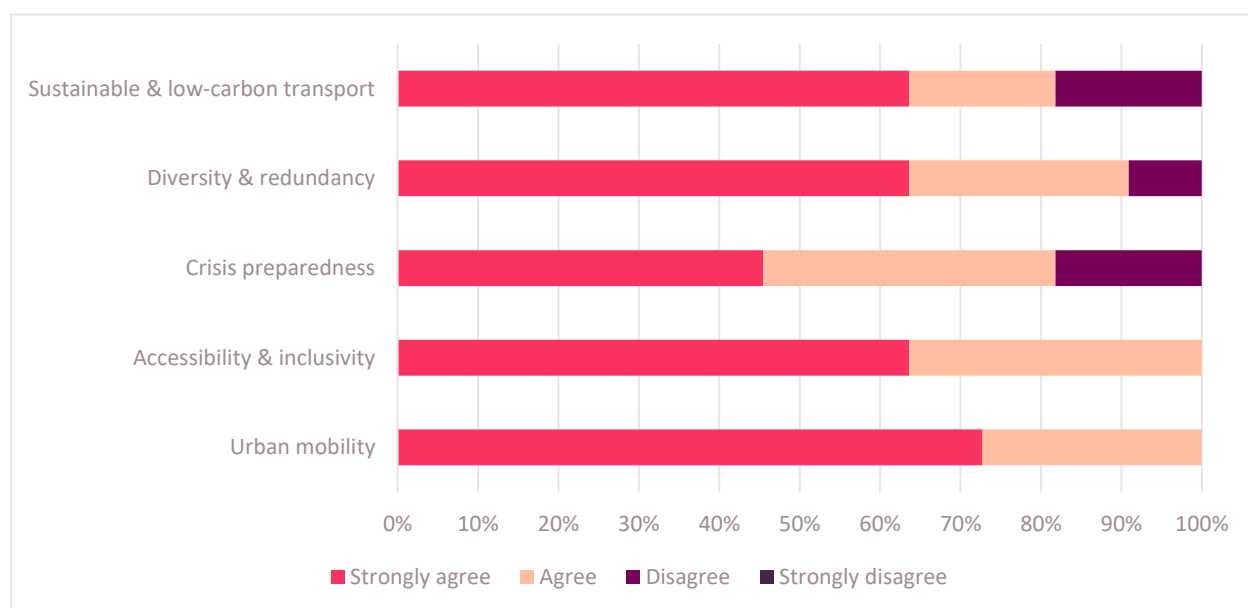


Figure 29 Participants' learning gains

Finally, participants were asked about anticipated behavioral changes. The most frequently selected response was “I will use more active forms of transport (walking/cycling)” (63.64%). Smaller shares indicated intentions to try shared mobility services (9.09%), advocate for improved transport

(9.09%), use public transport more (9.09%), or expected no change (9.09%). These results suggest that the forum primarily encouraged reflection and intentions toward active mobility, while more transformative behavioral commitments remained limited.

	%
I will use more active forms of transport (walking/cycling)	63.64%
I will try shared mobility services	9.09%
I will advocate for improved transport	9.09%
I will use public transport more	9.09%
No change expected	9.09%

Table 58 Perceived behavioral changes after participation

6.2.3.4 Open-ended feedback

Participants were invited to share their main takeaway and propose one next step for the city of Larissa. Only one participant responded.

Main insight	Next steps for the city of Larissa
The planning is at a high level, the intentions get stuck and stick to the political will that is absent	Limitation of parking, accessibility through durability and vice versa

Table 59 Open-ended feedback

6.2.3.5 Mobile application

In line with the broader objectives of the AntifragiCity project, the post-survey explored perceptions of a potential mobility application. 54.55% strongly agreed and 27.27% agreed that a mobility app could help the city respond better during crises or emergencies, while 18.18% disagreed. Support was even stronger for participatory functions: 63.64% strongly agreed and 27.27% agreed that residents should be able to share information through a mobility app. Furthermore, 54.55% strongly agreed and 36.36% agreed that non-digital channels should also be provided to ensure inclusivity, while 9.09% disagreed.

	Strongly agree	Agree	Disagree	Strongly disagree
A mobility app could help the city respond better during crises or	54.55%	27.27%	18.18%	0%

emergencies (such as heatwaves or flooding)				
Residents should be able to share information through a mobility app to help improve transport	63.64%	27.27%	9.09%	0%
The city should also provide non-digital ways to receive information and give feedback, so everyone can benefit	54.55%	36.36%	9.09%	0%

Table 60 Perceptions on a mobile application

Regarding specific functionalities, real-time public transport and traffic information was the most attractive feature (63.64% very likely; 27.27% likely; 9.09% unlikely). Route suggestions during disruptions were also well received (54.55% very likely; 27.27% likely; 18.18% unlikely). Travel suggestions supporting more sustainable choices were positively evaluated (63.64% very likely; 27.27% likely), as was accessibility information such as step-free routes, though with slightly more varied responses (63.64% very likely; 18.18% likely; 9.09% unlikely; 9.09% very unlikely). A feedback or reporting function was considered very likely to be used by 45.45% and likely by another 45.45%, with 9.09% unlikely.

	Very likely	Likely	Unlikely	Very unlikely
Real-time public transport and traffic information	63.64%	27.27%	9.09%	0%
Route suggesting during disruptions or emergencies	54.55%	27.27%	18.18%	0%
Travel suggestions that support more sustainable choices	63.64%	27.27%	9.09%	0%
Accessibility information (for example, step-free routes)	63.64%	18.18%	9.09%	9.09%
A way to give feedback or report problems	45.45%	45.45%	9.09%	0%

Table 61 Uses for mobile application

Qualitative feedback highlighted potential barriers to adoption, including *“if there is insufficient update and promotion of the application,”* and *“Its complex operation, the non-updating in a short time from the data that is in force, the non-commitment of the authorities to the adoption and promotion of an application. Coveting supplies, each municipal authority runs its own applications, as a result of which they only have traffic to the authorities and then are neglected.”*

Overall, participants expressed strong support for a mobility application, particularly for real-time information and participatory features, while simultaneously emphasizing the importance of usability, regular updates, institutional commitment, and complementary non-digital communication channels to ensure inclusive access.

6.2.4 Comparative insights

9 participants (n = 9) completed pre- and post-surveys assessing their attitudes toward urban mobility policies, civic participation, and their perceived role in transport decision-making. Both surveys included identical questions, allowing assessment of how participants’ views shifted after the process, with particular attention to:

- Attitudes toward participatory and inclusive decision-making
- Willingness to engage in future mobility initiatives
- Interest, knowledge, and confidence regarding mobility issues
- Support for specific mobility measures

6.2.4.1 Citizens’ participation in urban mobility

In both the pre- and post-surveys, participants were asked to indicate their level of agreement, ranging from strongly disagree to totally agree, with three statements: that public participation improves decision-making, that citizens’ everyday experiences are as important as expert knowledge, and that participation leads to fairer outcomes. Changes between the two survey points were categorized as more positive, less positive, or no change.

	More Positive	Less Positive	No Change
Public participation is important	55.56%	0%	44.44%
Citizens’ experiences equals expert knowledge	33.33%	33.33%	33.33%
Diversity leads to fairer outcomes	44.44%	0%	55.56%

Table 62 Changes in perceptions regarding citizens' participation

For the statement that public participation is important for good transport decisions, a majority of participants (55.56%) shifted in a more positive direction, while 44.44% reported no change. Notably, no participant became less positive on this item.

Perceptions regarding the value of citizens' everyday experiences relative to expert knowledge were more mixed. One-third of participants (33.33%) became more positive, one-third (33.33%) became less positive, and one-third (33.33%) remained stable. This balanced pattern suggests that the forum prompted reflection and reconsideration rather than uniform reinforcement.

For the statement that involving diverse residents leads to fairer mobility outcomes, over half of participants (55.56%) reported no change, while 44.44% shifted in a more positive direction. No participant became less positive on this item.

Taken together, these findings indicate that the forum primarily reinforced and, in some cases, strengthened already favorable attitudes toward participatory and inclusive governance. At the same time, the more differentiated shifts regarding experiential and expert knowledge suggest that participants engaged critically with these issues, leading some to reassess their positions in a more nuanced way.

Comparing participants' willingness to be involved in mobility initiatives before and after the forum reveals a shift overall toward less active engagement. Changes were classified as more active (movement toward more active involvement), less active (movement toward less active involvement), or no change.

	%
More active	22.22%
Less active	55.56%
No change	22.22%

Table 63 Direction of change

Just over half of participants (55.56%) shifted toward less active involvement. These shifts included movements from “yes, actively participate” to “maybe” or “occasionally,” as well as from “yes, occasionally” to “don't know.” In contrast, 22.22% moved toward more active engagement, for example from “no” or “maybe” to “actively” or “occasionally.” Another 22.22% reported no change in their stated willingness to engage.

Overall, this pattern suggests that while the forum may have strengthened attitudinal support for participation, it did not uniformly translate into a greater declared readiness for personal

involvement. Instead, for many participants, the process appears to have led to a more cautious or reconsidered assessment of their own level of engagement.

Comparing pre- and post-process responses, the overall pattern does not indicate a broad expansion of participation preferences. Instead, the results suggest a selective reconfiguration rather than a broad expansion of how participants imagine their role in urban mobility governance. While some participation modes gained prominence, others became less salient.

I. Selective shifts rather than overall expansion

The total number of selected participation options did not systematically increase across the sample. Instead, some participants reduced the number of modes they selected, while others maintained or slightly diversified their choices. This suggests a process of prioritization and clarification rather than simple expansion.

II. Declining interest in deliberative forums

Interest in participation through public consultations or citizens' forums decreased. This indicates that deliberative forums became less central for some participants. Rather than rejecting participation altogether, however, several appear to have shifted their interest toward other forms of engagement.

III. Stability in app-based engagement

Testing mobility-related applications remained stable. This consistency suggests that hands-on, practice-oriented engagement retains steady appeal. However, the absence of growth indicates that the process did not significantly expand interest in this mode.

IV. Slight decline in community-based participation

Engagement in community or neighborhood mobility initiatives decreased modestly. This suggests that while local involvement remains relevant for some, it did not gain broader traction through the process.

V. Increased preference for surveys as a feedback channel

Surveys appear to have gained attractiveness as a participation mode, possibly reflecting a preference for structured, lower-commitment forms of engagement. In several cases, surveys were selected alongside other options, indicating their role as a complementary channel.

Taken together, the findings do not point to a generalized broadening of participation preferences. Instead, they suggest a recalibration of engagement modes. Deliberative and community-based

formats became less prominent for some participants, while surveys gained relative importance and app-based testing remained stable. Rather than expanding participation across all fronts, the process appears to have prompted participants to differentiate more clearly between modes of involvement and to gravitate toward those that align with their capacities or preferences.

6.2.4.2 Participants and urban mobility

In line with the questions on citizen participation and the topic of urban mobility, the surveys also measured three aspects of respondents' engagement with urban mobility: their general interest in urban mobility, how informed they feel about the city's mobility policies, and their confidence in contributing to urban mobility discussions.

A substantial share of participants reported feeling more informed after the process (see Figure 30), with 55.56% indicating an increase in perceived informedness. At the same time, one-third (33.33%) experienced a decrease, while 11.11% reported no change.

	%
Increased	55.56%
Decreased	33.33%
No change	11.11%

Table 64 Direction of change



Figure 30 Comparison of participants' perceptions of feeling informed

The relatively high proportion of decreases suggests that, for some participants, the process may have heightened awareness of the complexity of urban mobility issues or revealed gaps in their prior understanding. Rather than indicating a simple loss of knowledge, this pattern points to more reflective and critical self-assessments. Thus, the process appears to have generated both informational gains and, for a notable minority, a more nuanced recognition of the limits of one's own expertise.

A similar but more clearly positive pattern emerges with regard to participants' confidence (see Figure 31). Two-thirds (66.67%) reported increased confidence following the process, while 22.22% indicated a decrease and 11.11% experienced no change.

	%
Increased	66.67%
Decreased	22.22%
No change	11.11%

Table 65 Direction of change

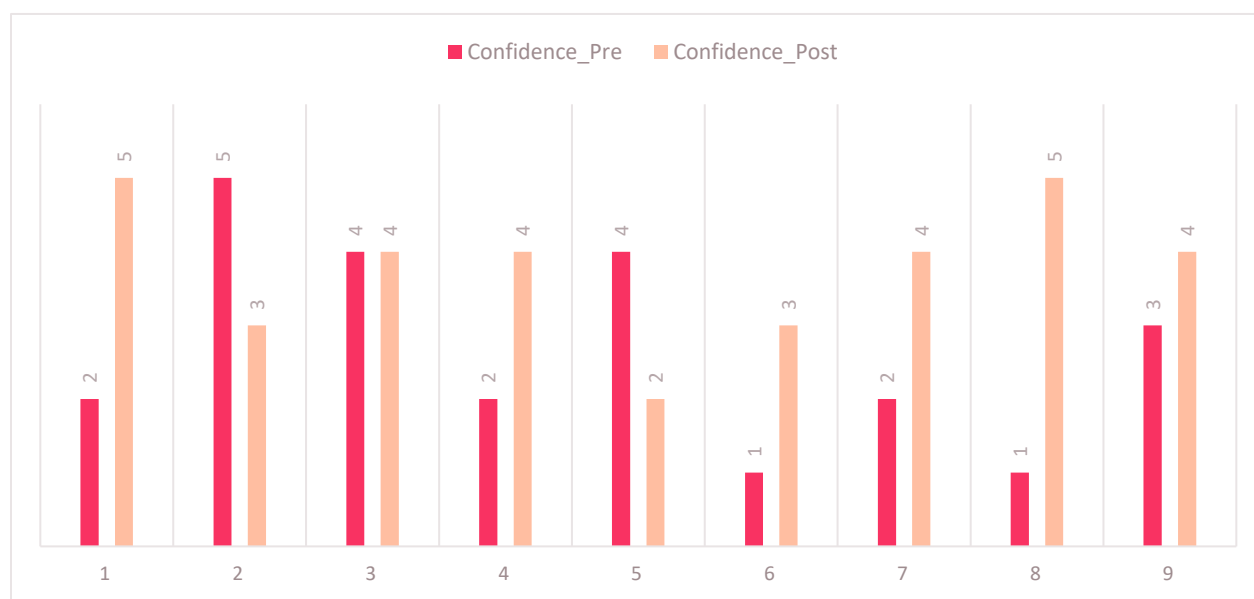


Figure 31 Comparison of participants' willingness to contribute to urban mobility policies

Overall, the dominant trend is one of strengthened self-assurance. Although a small group reassessed their confidence downward, potentially reflecting more critical self-evaluation, the prevailing effect of the process was to reinforce participants' sense of confidence in engaging with the topic.

6.2.4.3 Mobility habits and opinions

In both surveys, participants were also asked to identify factors that would encourage greater use of sustainable transport. The comparison of responses shows a marked reordering of priorities rather than a simple expansion of preferred measures.

	Pre (%)	Post (%)	Δ (Post-Pre)
Improved walking/cycling infrastructures	70.83%	55.56%	-15.27%
Reduced car use in busy or densely populated areas	66.67%	100%	+33.33%
Better public transport	66.67%	44.44%	-22.23%
More greenery	50.00%	33.33%	-16.67%
Car-free areas	45.83%	55.56%	+9.73%
Digital traffic management	41.67%	44.44%	+2.77%
Increased shared mobility options	33.33%	11.11%	-22.22%
30 km/h speed limit zones in residential or city areas	29.17%	22.22%	-6.95%
Shared streets	25.00%	11.11%	-13.89%
Fewer parking spaces	20.83%	11.11%	-9.72%
Congestion charges	8.33%	22.22%	+13.89%

Table 66 Measures perceived as necessary to be antifragile

Most strikingly, support for reducing car use in busy or densely populated areas increased from 66.67% to 100% (+33.33 percentage points), making it the most widely endorsed measure in the post-survey. This suggests a strengthened recognition of the need to actively manage car traffic in order to promote sustainable mobility.

By contrast, several infrastructure-focused measures declined in support. Endorsement of better public transport fell from 66.67% to 44.44% (-22.23), and improved walking and cycling infrastructures decreased from 70.83% to 55.56% (-15.27). Similarly, more greenery (-16.67), shared streets (-13.89), fewer parking spaces (-9.72), and 30 km/h speed limit zones (-6.95) all saw reduced support. These shifts may indicate that, after deliberation, participants differentiated more clearly between desirable long-term urban improvements and measures they view as most critical for immediate change.

Some measures gained moderate support. Car-free areas increased from 45.83% to 55.56% (+9.73), while congestion charges rose from 8.33% to 22.22% (+13.89), suggesting a greater openness to regulatory or restrictive instruments than before the process. Digital traffic management remained relatively stable (+2.77), indicating consistent but moderate endorsement. In contrast, support for increased shared mobility options declined sharply (–22.22), pointing to reduced enthusiasm for this strategy relative to others.

Overall, the post-survey results reflect a consolidation of priorities around structural measures that directly limit car dominance, particularly in dense urban areas, while enthusiasm for certain complementary or softer measures declined. Rather than broadening their selection across the board, participants appear to have sharpened their focus, converging more strongly on targeted interventions aimed at reducing car use as a central lever for advancing sustainable urban mobility.

6.2.4.4 Synthesis

The pre–post comparison suggests that participants’ responses indicate a sharpening and, in some areas, a recalibration of their views on urban mobility challenges, policy priorities, and their own role in governance.

With regard to policy measures, preferences did not broadly expand but became more focused. Most notably, support for reducing car use in busy or densely populated areas increased to unanimous endorsement in the post-survey. At the same time, support declined for several infrastructure- and amenity-oriented measures, such as better public transport, walking and cycling infrastructure, greenery, and shared mobility options. This pattern suggests a shift toward prioritizing structural interventions that directly address car dominance, rather than a general increase in support across all sustainability measures. Participants appear to have differentiated more clearly between what they consider desirable improvements and what they see as the most decisive levers for change.

Civic self-perception evolved in a more complex way. A majority reported increased confidence in contributing to urban mobility discussions, indicating that the process strengthened participants’ sense of self-assurance. However, willingness to engage personally in future mobility initiatives declined for more than half of the sample. This divergence suggests that while participants may feel more capable of engaging in discussions, they also reassessed, perhaps more realistically, their readiness to commit time and effort to concrete initiatives. Similarly, although more than half felt better informed, a substantial minority reported decreased perceived informedness, pointing to heightened awareness of the complexity of mobility governance rather than simple knowledge loss.

Attitudes toward participatory governance remained consistently positive. Support for public participation in decision-making strengthened further, and no participant became less positive on this dimension. Views on the role of diversity in achieving fairer outcomes also shifted in a positive direction for many, while perspectives on the equivalence of experiential and expert knowledge became more differentiated, reflecting critical engagement rather than uniform endorsement.

Preferences for modes of participation did not expand across the board but were selectively reconfigured. Interest in deliberative forums and community-based initiatives declined somewhat, survey-based feedback gained importance, and app-based testing remained stable. Rather than converging toward highly active, multi-modal engagement profiles, participants appear to have clarified and prioritized forms of participation that align with their capacities and preferences.

Overall, the findings point to the development of more reflective and system-aware perspectives on urban mobility. Participants emerged with clearer priorities, greater confidence in discussion, and a more differentiated understanding of both policy trade-offs and their own potential role within governance processes – even if this was accompanied, for some, by a more cautious stance toward personal engagement.

6.2.5 Recommendations

The citizens' forum in Larissa produced a coherent and prioritized set of recommendations that can be clearly organized around the four overarching thematic areas: liveable cities, crisis-responsive mobility, resilient and diverse mobility, and sustainable mobility transition. Across all themes, participants consistently emphasized predictability, safety, coordination, and trust as the core enablers of an antifragile urban mobility system.

1. Liveable cities

Citizens articulated a shared vision of a city with safe, shaded, and accessible streets, where walking and cycling are comfortable and intuitive choices. They emphasized the importance of continuous, connected walking and cycling corridors linking neighborhoods with schools, everyday services, and the city centre, allowing people to move safely without reliance on cars, even during periods of stress or disruption.

Key recommendations included:

- Pocket parks and small-scale green interventions to provide shade, cooling, and rest points along walking routes
- Smart parking and logistics management to reduce congestion and reclaim street space for people

These conclusions reflect a pragmatic citizen perspective: improving everyday liveability does not require radical restructuring, but targeted interventions that directly address heat, space competition, and comfort in daily movement.

2. Crisis-responsive mobility

Citizens described an ideal situation in which municipal authorities, transport operators, and civil protection services coordinate quickly and effectively. Central to this vision is the availability of reliable, real-time public information, as well as clear and tailored provisions for vulnerable users, ensuring that no group is left without support during emergencies.

Recommended measures included:

- Real-time disruption management systems, including dashboards and live status overviews
- Incident reporting mechanisms accessible to users
- Dynamic messaging to communicate changes, risks, and instructions clearly and promptly

These solutions reflect citizens' expectation that crises should be managed through clarity, speed, and coordination rather than improvised or ad hoc responses.

3. Resilient and diverse mobility

Citizens envisioned a balanced “mobility mix”, in which people can reliably choose between walking, cycling, public transport, and micro-mobility depending on their needs and circumstances. Diversity of options, rather than reliance on a single mode, was identified as the basis of resilience.

Participants converged on one core direction:

- Multimodal hubs with seamless transfers, particularly park-and-ride facilities, enabling easy switching between modes

This reflects a practical understanding that resilience depends not only on infrastructure quality, but on how well different modes are connected.

4. Sustainable mobility transition

The preferred future is a low-emission city in which most short journeys are made on foot, by bicycle, or using public transport, supported by infrastructure that makes these choices normal, convenient, and reliable.

Recommended features included:

- Reallocation of street space to support bus priority, protected cycling infrastructure, and pedestrian zones
- Parking reform and demand management, including pricing, controls, and the strategic use of park-and-ride systems

These conclusions indicate that citizens view the sustainable mobility transition as a matter of structural rebalancing of space and incentives, rather than individual behavior change alone.

6.2.5.1 Prioritization / triage of citizens' recommendations

No formal prioritization or triage of recommendations was conducted during the citizens' forum in Larissa. This was a deliberate decision, taken in response to the composition of participants on the second day of the forum. A significant lower number of participants were present during the second day. Accordingly, proceeding with voting, scoring, or ranking under these conditions would not fairly represent the full group's views, recognizing that prioritization implies collective judgment and carries normative weight.

Instead, the group agreed that the available time would be better used to clarify, refine, and consolidate the recommendations that had emerged from earlier discussions. The focus shifted toward improving the internal coherence of proposals, ensuring shared understanding of identified gaps and desired futures, and avoiding premature hierarchies that could obscure unresolved nuances or trade-offs.

As a result, the recommendations presented here should be understood as non-ranked, collaboratively developed conclusions, reflecting areas of convergence rather than ordered preferences. The absence of triage does not indicate indecision or lack of commitment; rather, it reflects a procedurally cautious and inclusive stance.

6.2.5.2 Synthesis for antifragile mobility planning

The citizens' forum in Larissa provides a clear picture of how residents conceptualize desirable urban mobility interventions, even in the absence of formal prioritization or voting. While no explicit hierarchy of measures was established, the structure and content of the recommendations reveal shared judgments about legitimacy, necessity, and system relevance. Public acceptance, as expressed through deliberation, appears to be shaped less by isolated measures and more by whether interventions contribute to a coherent, understandable, and people-centered mobility system.

Across all four thematic areas, participants consistently gravitated toward structural and enabling interventions rather than short-term or ad hoc fixes. Proposals related to street liveability, multimodal connectivity, crisis coordination, and spatial rebalancing were framed as foundational conditions for a functioning system. This suggests that citizens perceive long-term improvements to public space, information systems, and network integration as legitimate and necessary, even when they imply changes to established travel patterns or the reallocation of street space.

Safety and basic usability emerged as implicit threshold conditions for acceptance. Although not formally ranked, the emphasis on shaded routes, accessible streets, reliable information during crises, and seamless transfers indicates that interventions are only seen as meaningful if they reduce physical strain, uncertainty, and vulnerability. In this sense, safety, comfort, and accessibility function as prerequisites: without them, broader goals such as sustainability or modal shift lose credibility in the eyes of participants.

The forum discussions also highlight the importance of coordination and clarity as determinants of acceptability. In the crisis-responsive mobility theme, the focus on real-time information, dashboards, and dynamic messaging reflects an expectation that authorities should act in a visible, predictable, and organized manner during disruptions. Similarly, the emphasis on multimodal hubs and park-and-ride systems points to a citizen logic in which resilience depends on how well different parts of the system are connected, rather than on the expansion of any single mode.

Governance and implementation were addressed indirectly through the way solutions were framed. Participants did not call for isolated behavioral change, but for structural adjustments to space, information, and system design, such as parking management, street space reallocation, and integrated hubs. This framing suggests that citizens view mobility outcomes as the product of system-level choices, and that social acceptance increases when responsibility is placed on institutions rather than individuals.

Finally, the absence of formal triage itself provides insight into citizens' understanding of legitimacy. Participants consciously avoided ranking measures when representation was incomplete, prioritizing fairness and collective authorship over procedural completeness. This indicates that, for citizens, how decisions are made is as important as what decisions are made. Taken together, the Larissa forum suggests that socially acceptable mobility interventions are those that visibly improve everyday conditions, enhance safety and reliability, are coordinated across actors, and are developed through processes perceived as inclusive and procedurally just.

6.2.6 Social acceptability of mobility interventions: Insights from Larissa

The findings from the Larissa AntifragiCity Citizens' Forum – drawing on pre- and post-surveys responses, and the recommendations developed by the participants – provide insights into how residents perceive and evaluate mobility interventions aimed at improving both everyday urban

mobility and crisis resilience. Through the forum process, participants considered a broad range of potential measures, including infrastructure improvements, traffic management policies, governance arrangements, and digital tools. The combination of survey results and deliberative discussions therefore offers a nuanced view of citizens' priorities, expectations, and evolving perspectives on mobility governance in Larissa.

The pre-participation survey indicates that participants entered the forum with a strong general interest in urban mobility but comparatively lower levels of knowledge about local mobility policies and moderate confidence in their ability to contribute to policy discussions. While most respondents expressed openness to public involvement in mobility decision-making, their preferred forms of engagement tended to be pragmatic and low-threshold, such as testing mobility applications or providing feedback through surveys, rather than sustained participation in deliberative forums. This pattern suggests a participation culture that values practical contribution and flexible engagement formats over continuous civic involvement.

Participants' mobility habits and preferences reflect the existing structure of mobility in Larissa. Private car use remains the dominant daily transport mode among respondents, although walking also plays an important role. Decisions about travel modes are primarily driven by functional considerations – particularly speed, convenience, and comfort—rather than environmental motivations. Correspondingly, respondents identify structural improvements such as better infrastructure and stronger connectivity as the most important factors that could encourage greater use of sustainable transport modes. These findings indicate that everyday mobility choices are shaped primarily by the practical performance of the transport system rather than by normative sustainability considerations.

At the same time, participants express a relatively high willingness to accept environmental trade-offs in principle. More than half of respondents report strong readiness to accept measures such as reduced parking availability, increased green space, or lower speed limits in order to improve environmental sustainability. However, when specific policy measures are evaluated, support varies considerably depending on perceived practicality and impact. In the pre-survey, measures such as improved walking and cycling infrastructure, reduced car use in busy or densely populated areas, and better public transport receive the highest levels of support, while more restrictive or cost-based measures, such as congestion charges, are less widely endorsed. This pattern suggests that support for sustainability policies is strongest when interventions visibly improve daily mobility conditions rather than primarily imposing restrictions.

The comparison between pre- and post-survey responses suggests that the deliberative process prompted a re-evaluation of priorities rather than a simple expansion of support for all measures. Most notably, support for reducing car use in busy or densely populated areas increased from two-thirds of respondents to unanimous endorsement in the post-survey, indicating a strengthened recognition of the need to manage car traffic as a central lever for improving urban mobility. At the

same time, support declined for several infrastructure-oriented or complementary measures, such as improved public transport or walking and cycling infrastructure. This pattern suggests that participants sharpened their focus on structural interventions addressing car dominance, rather than broadly endorsing a wide range of policy options.

Changes are also visible in attitudes toward participation. Although support for participatory decision-making remained consistently high, declared willingness to personally engage in future mobility initiatives decreased for more than half of the respondents. This shift suggests that the forum may have led some participants to reassess more realistically the time and effort required for sustained civic engagement. Similarly, while a majority reported feeling more confident discussing mobility issues after the process, a notable minority indicated lower perceived informedness, likely reflecting a heightened awareness of the complexity of urban mobility governance. In this sense, the deliberative process appears to have generated both learning effects and more critical self-assessments among participants.

Process-related dynamics further shaped the outcomes of the Larissa forum. Attendance declined significantly on the second day of the event, which influenced procedural choices during the deliberation phase. In particular, the group decided not to conduct a formal prioritization or ranking of recommendations. Participants considered that voting or scoring measures with a reduced group could distort the representation of collective preferences. Instead, the available time was used to refine and consolidate the recommendations that had emerged from earlier discussions. The resulting outputs therefore represent areas of convergence rather than a formally ranked set of priorities.

Qualitative feedback also points to broader governance challenges perceived by participants. The open-ended response highlights a perceived gap between planning ambitions and political implementation capacity in the city, noting that planning is at a high level but that intentions often depend on political will that is perceived as lacking. Although based on limited qualitative input, this observation reflects a concern that the success of mobility reforms ultimately depends on institutional commitment and sustained policy follow-through.

Taken together, the Larissa forum findings illustrate how citizen perspectives on urban mobility evolve through structured deliberation. Participants demonstrate strong interest in improving urban mobility and increasing sustainability, but their preferences are shaped by practical considerations related to daily mobility patterns, perceived feasibility of interventions, and confidence in institutional implementation. The process itself contributed to greater reflection, clearer prioritization of policy directions – particularly regarding the management of car use – and a more nuanced understanding of both policy trade-offs and the realities of civic participation in mobility governance. In this respect, the forum highlights how participatory processes can support more informed and system-aware discussions about mobility transitions, even within relatively small citizen groups.

7 Thessaloniki

This section of the report presents the findings from the Thessaloniki AntifragiCity Forum. The objectives of this evaluation encompasses two main aspects:

- Learning about the Thessaloniki Forum proceedings: The evaluation assesses its success as a deliberative process, gaining valuable insights into its strengths and areas for improvement
- Understanding the Thessaloniki Forums' impact: The evaluation seeks to delve into its impact on its members, as well as delve into the recommendations

7.1 Introduction

The Thessaloniki AntifragiCity Forum was held on Saturday 14 February 2026, from 10:00 – 16:00.

The recruitment strategy was led by LISER in close collaboration with AUTH, leveraging their institutional networks and established best practices in citizen engagement.

A public call for participation was issued in December 2025 and disseminated through a coordinated, multi-channel outreach strategy. This included announcements via AUTH and the AntifragiCity project social media platforms, a press release published in a local newspaper, promotional posters displayed across the AUTH campus, and targeted announcements delivered to student audiences. This strategy led to 28 individuals registering to participate in the Forum. Given this limited number of pre-registrations, a strategic decision was made to extend invitations to all registered individuals and additionally allow participation from individuals who had not pre-registered. This adaptive approach aimed to maximize turnout while maintaining an inclusive and open deliberative environment.

The first day of the Forum welcomed 11 participants.² Given this number of participants, a strategic decision (also based on the lessons learned from the Larissa forum the week before) was taken to conduct the process over one day instead of two days. The participants were also asked for confirmation and many were satisfied by this decision and agreed to then stay as long as needed on Saturday the 14th of February. Despite this change, the Thessaloniki AntifragiCity Forum effectively met its objectives: allowing meaningful participant-expert interaction, hearing from citizens regarding urban mobility, and the production of high-quality recommendations.

² Due to a last-minute decision by the University of Thessaloniki, no events were allowed to occur over the weekend of 14-15 February 2026 at the University. This decision was communicated to the AntifragiCity's AUTH team on Friday the 13th. The team immediately found a different venue and personally contacted all registered individuals. Nevertheless, this very last minute change might have impacted the number of participants.

7.2 Evaluation

7.2.1 Representativity

Given that no sortition occurred for the Thessaloniki AntifragiCity Forum, this section first briefly summarizes the diversity and representativity of the respondents to the pre-survey. Notably, all participants in the TAF filled in the pre- and post-survey.

From the 11 respondents, 6 were female and 5 were male, ensuring a gender balanced process.

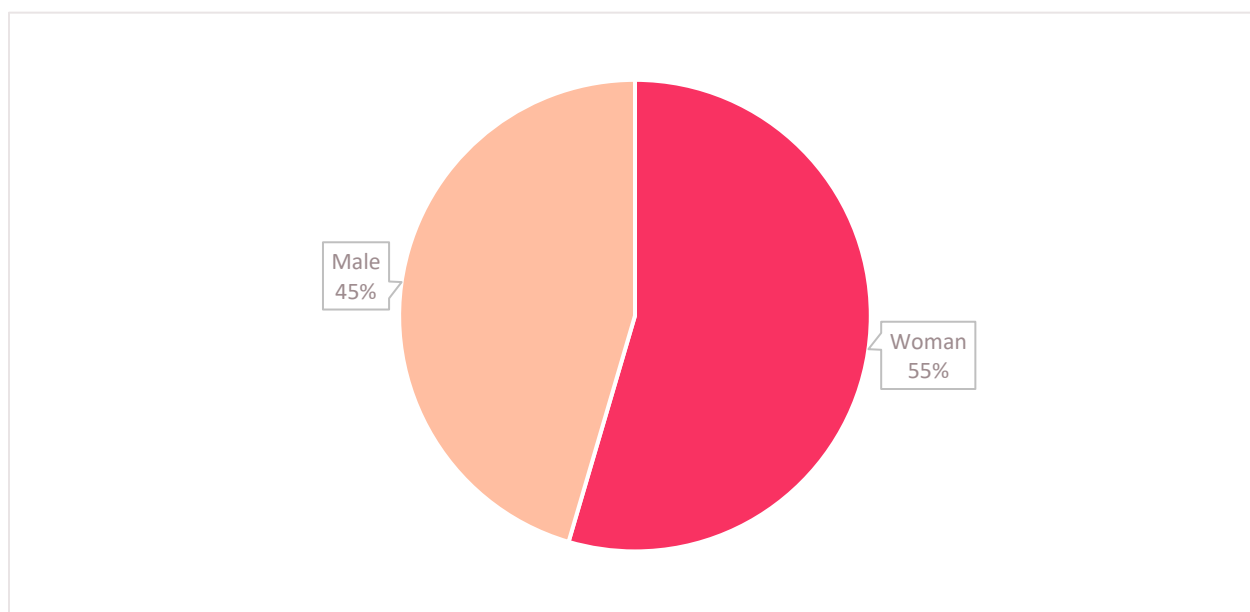


Figure 32 Participants by gender

When it comes to age, participants were between 18 and 54 years old.

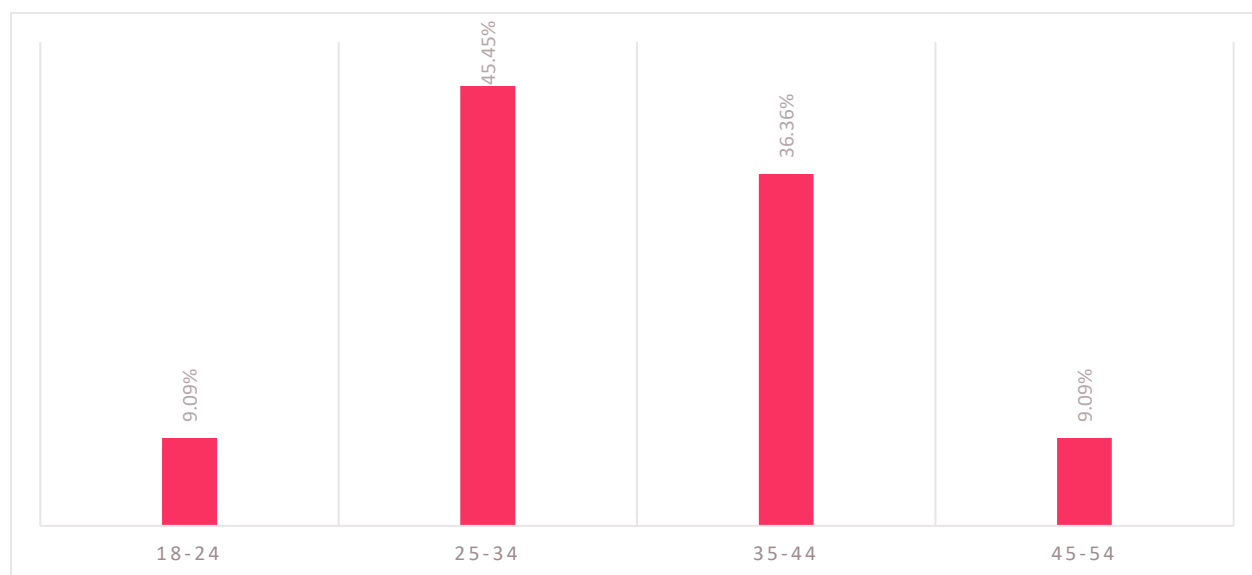


Figure 33 Participants by age groups

As typical in deliberative processes, the participants are highly educated. The vast majority of participants has at least a bachelor's degree, with all participants have a higher degree than secondary education.

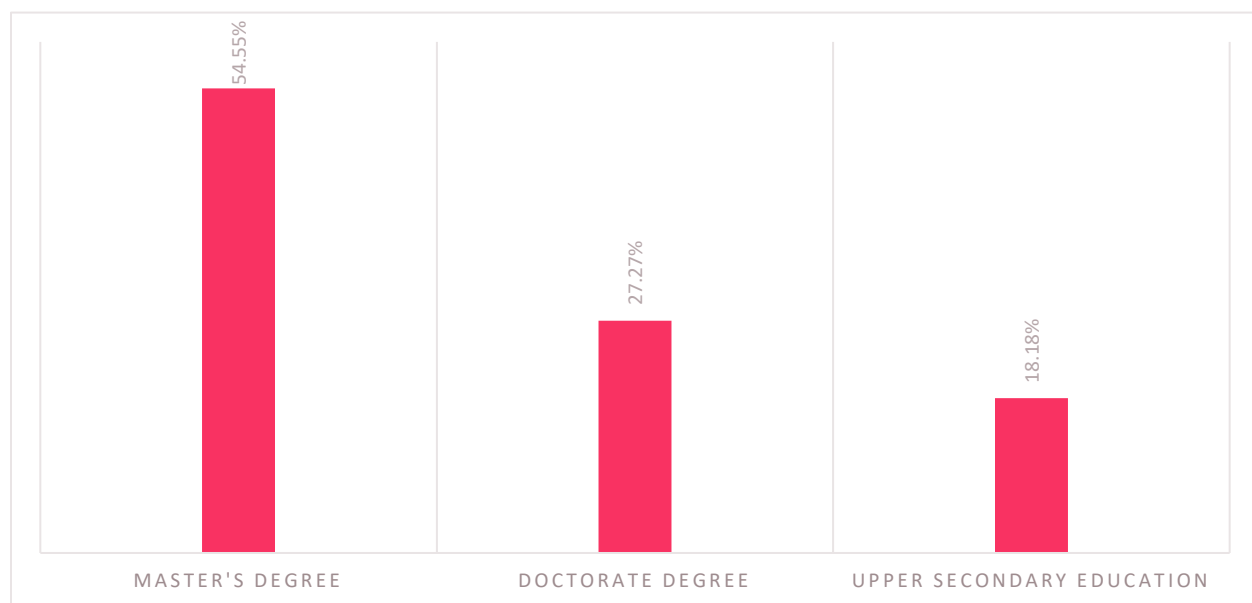


Figure 34 Participants by educational level

Considering participants' employment status, the majority are in paid employment ($n = 8$), while three participants are still studying.

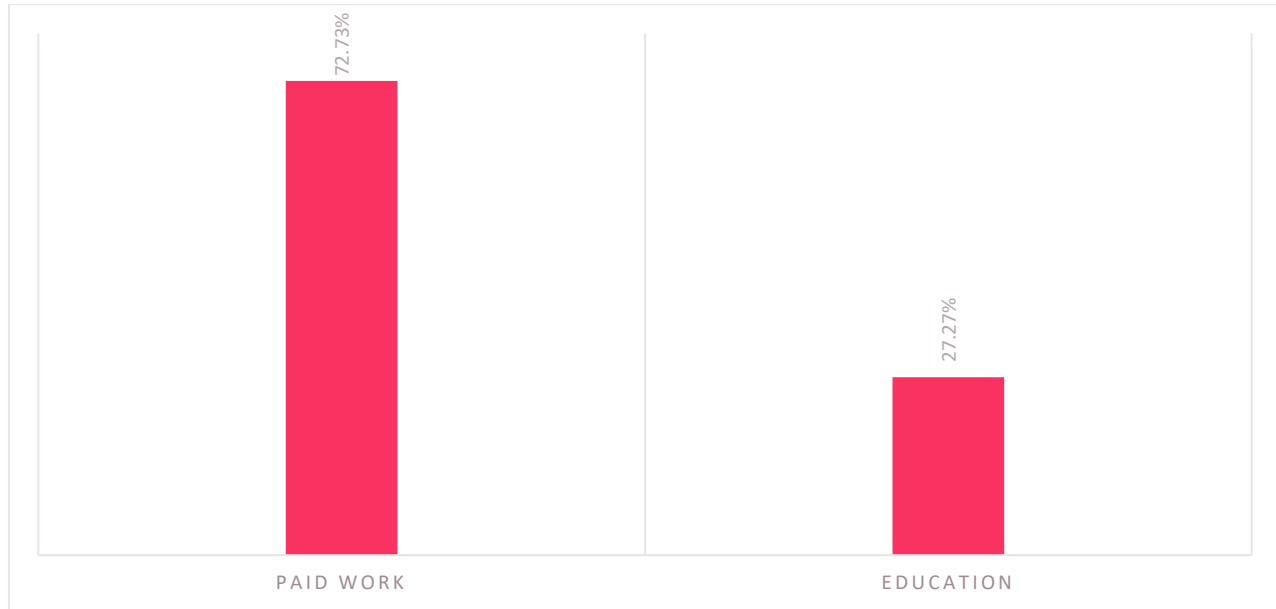


Figure 35 Participants by employment status

7.2.2 Pre-survey insights

This section summarizes the findings of the pre-participation survey conducted among participants of the Thessaloniki AntifragiCity Forum. The survey aimed to understand participants' current travel behaviors, motivations, attitudes toward sustainable urban transport, and their interest in contributing to urban mobility initiatives. A total of 11 valid responses were collected ($n = 11$). Were this number of responses is different, this is acknowledged.

7.2.2.1 Citizens' participation in urban mobility

Respondents were asked whether they would like to be involved in shaping mobility initiatives in their city (see Figure 36).

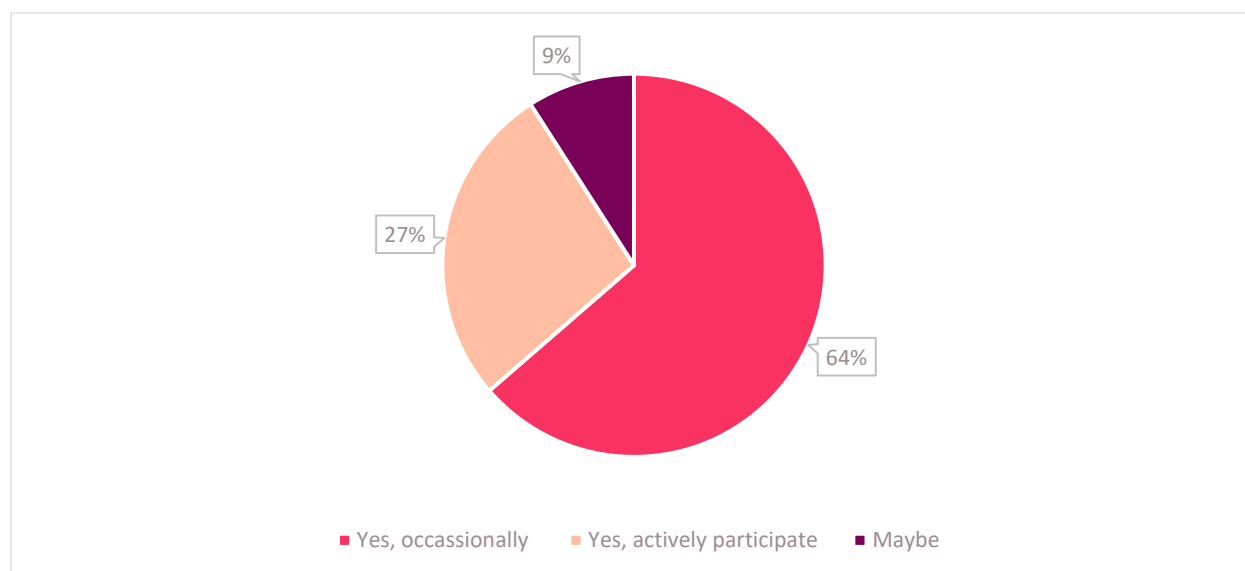


Figure 36 Willingness to shape mobility initiatives in city

A majority of respondents (63.64%) indicated that they would like to participate occasionally, suggesting a willingness to contribute on a flexible or periodic basis. A smaller but still notable share (27.27%) expressed interest in actively participating, reflecting a highly motivated group ready to engage regularly or take a hands-on role in mobility initiatives. 9.09% of respondents answered “Maybe”, indicating some conditional openness to involvement, while no respondents explicitly declined participation. Overall, the data indicate that a strong majority of participants are open to engagement, whether on a casual or active basis.

To better understand the forms of involvement preferred by participants, respondents were next asked how they would like to be engaged in shaping mobility initiatives in their city. Four main options were offered: participation in public consultations or citizens’ forums, testing mobility-related applications, engaging with community or neighborhood mobility projects, and providing feedback via surveys.

	%
Engaging with community or neighborhood mobility projects	63.64%
Providing feedback via surveys	63.64%

Testing mobility-related applications	54.55%
Participation in public consultations/citizens' forums	45.45%

Table 67 Participation preferences

The results indicate diverse but clearly structured participation preferences. Engaging with community or neighborhood mobility projects and providing feedback via surveys are the most popular options, each selected by 63.64% of respondents. This suggests that participants are motivated both by hands-on, localized initiatives that allow them to contribute to tangible improvements in their immediate environment, and by flexible, low-effort channels that permit input without significant time or logistical commitment. Testing mobility-related applications is selected by 54.55% of respondents, reflecting a strong interest in innovative, practical, and digitally oriented modes of participation. Participants appear willing to engage with experimental or trial-based initiatives, particularly those that allow direct interaction with new tools or technologies related to mobility. Participation in public consultations or citizens' forums is slightly less preferred, cited by 45.45% of respondents, highlighting the continued importance of structured, formal channels for civic engagement. These forums offer visibility, opportunities for dialogue, and a sense of direct involvement in decision-making processes, which remain appealing to a substantial portion of participants.

A notable observation is the variation in the number of participation modes selected per respondent. The distribution is as follows:

	%
1 Mode	9.09%
2 Modes	63.64%
3 Modes	18.18%
4 Modes	9.09%

Table 68 Number of participation modes selected (*per participant*)

In total, 90.91% of respondents expressed interest in at least one form of participation, with the majority (63.64%) willing to engage through two modes. This highlights the potential for multi-channel engagement, allowing participants to combine different forms of involvement depending on their availability, interest, and preferred level of commitment.

Overall, the findings demonstrate a strong willingness among respondents to engage with urban mobility initiatives, with preferences distributed across a variety of participation formats. There is a slight leaning toward interactive and innovative approaches, such as testing mobility-related applications and participating in structured consultations, which are perceived as engaging, tangible, and digitally accessible. These modes are generally more appealing than time-intensive, hands-on, community-based activities, which, while valued, require greater personal commitment.

Respondents were also asked to indicate their agreement with three statements regarding public involvement in urban mobility planning: that public participation is essential for good transport decisions, that citizens' everyday experiences are as important as expert knowledge, and that involving diverse residents leads to fairer mobility outcomes. The distribution of responses is summarized below:

	Strongly Agree	Somewhat Agree	Somewhat Disagree	Strongly Disagree
Public participation is essential	36.36%	63.64	0%	0%
Citizen experience as valuable as expert opinion	45.45%	54.55%	0%	0%
Engagement of diverse residents leads to fairer outcomes	45.45%	54.55%	0%	0%

Table 69 Opinions on citizens' participation

The majority of respondents strongly agree that public participation is essential for effective transport policy (36.36%), while a substantial portion (63.64%) somewhat agree. No respondents expressed disagreement, reflecting broad recognition of the value of citizen input in shaping urban mobility decisions.

A similarly high level of support is observed for the statement that everyday citizen experiences are as important as expert opinions, with 45.45% strongly agreeing and 54.55% somewhat agreeing. This indicates that respondents overwhelmingly value practical, on-the-ground knowledge as a critical complement to technical expertise in designing transport policies.

Strong agreement is also evident for the statement that engaging a diverse range of residents leads to fairer mobility outcomes, with 45.45% strongly agreeing and 54.55% somewhat agreeing. No respondents disagreed, demonstrating strong consensus on the importance of inclusivity and representation to ensure equity in transport planning. Overall, the findings indicate that respondents are supportive of participatory mechanisms in urban mobility planning.

7.2.2.2 Participants and urban mobility

Alongside questions on participation, the survey assessed three key aspects of respondents' engagement with urban mobility: their general interest in the topic, how informed they feel about the city's mobility policies, and their confidence in contributing to discussions about urban mobility (see Figure 37).



Figure 37 Knowledge urban mobility

Note: 1 = low interest, knowledge, or confidence; 5 = high interest, knowledge, or confidence

The data show that respondents have a strong general interest in urban mobility. The largest proportion selected the highest rating of 5 (45.45%), and a further 27.27% chose 4. Moderate interest (3) was reported by 18.18%, with 9.09% selecting 2, and no participants chose the lowest level of 1. Overall, this indicates that the majority of participants consider urban mobility an engaging and relevant topic, even though a small minority show limited interest.

By contrast, respondents appear less confident in their knowledge of city mobility policies. Self-assessed familiarity is distributed across the scale, with the largest share at a moderate level of 3 (36.36%), followed by 18.18% at 4, and another 18.18% at 2. Only 9.09% rated themselves as highly informed (5), and 18.18% reported very low awareness (1). This suggests a noticeable gap between general interest and specific understanding of the policies and frameworks governing urban mobility in the city.

Respondents' confidence in participating in discussions about urban mobility is similarly moderate. Nearly half of participants (45.45%) rated themselves at level 3, with 18.18% at 4 and 9.09% at 2.

High confidence (5) was reported by 9.09%, and low confidence (1) by 18.18%. These results indicate that while some respondents feel capable of contributing meaningfully, a significant portion remains hesitant or uncertain about their ability to engage in public discussions effectively.

Taken together, these findings highlight an imbalance between interest, knowledge, and confidence. While there is a clear enthusiasm for urban mobility, this is not fully matched by understanding of policy details or confidence to participate actively in discussions.

7.2.2.2.1 Mobility habits and opinions

Participants were asked to identify their primary mode of daily travel, selecting only one option (see Figure 38). Among the respondents, 46% report using a private car as their main means of transportation, highlighting the dominance of private vehicles in their everyday mobility. A substantial proportion (36%) indicated that they rely primarily on public transport, reflecting its role as a key alternative for commuting and other daily trips. Walking and motorcycle use were each selected by 9% of participants, representing smaller shares of the sample.

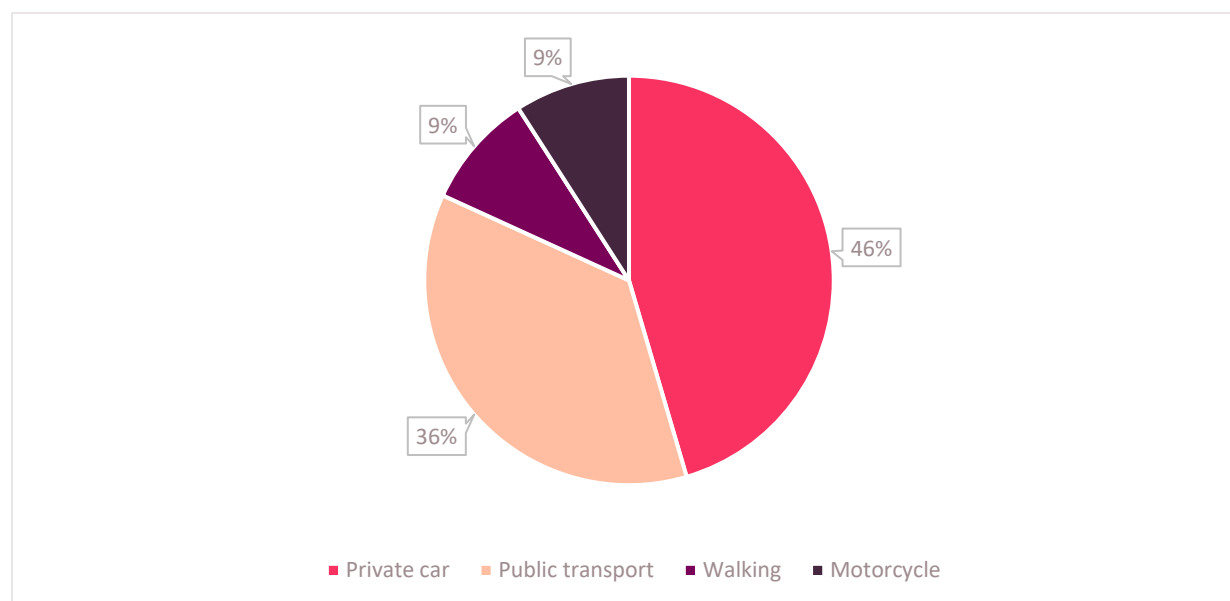


Figure 38 Most used daily transport mode

Respondents were asked to indicate the primary reasons for selecting their main mode of transport, with the option to choose multiple factors (see Figure 39). The results presented here reflect the aggregated frequency of selections across all participants, meaning the percentages show how often each factor was chosen overall. The findings reveal a strong emphasis on practical and functional considerations, such as speed and cost, which clearly outweigh environmental or purely experiential motivations. This underscores that everyday usability and efficiency are the dominant drivers of

travel behavior, with convenience, personal obligations, and safety playing a more limited role, and environmental concerns currently having minimal influence on mode choice

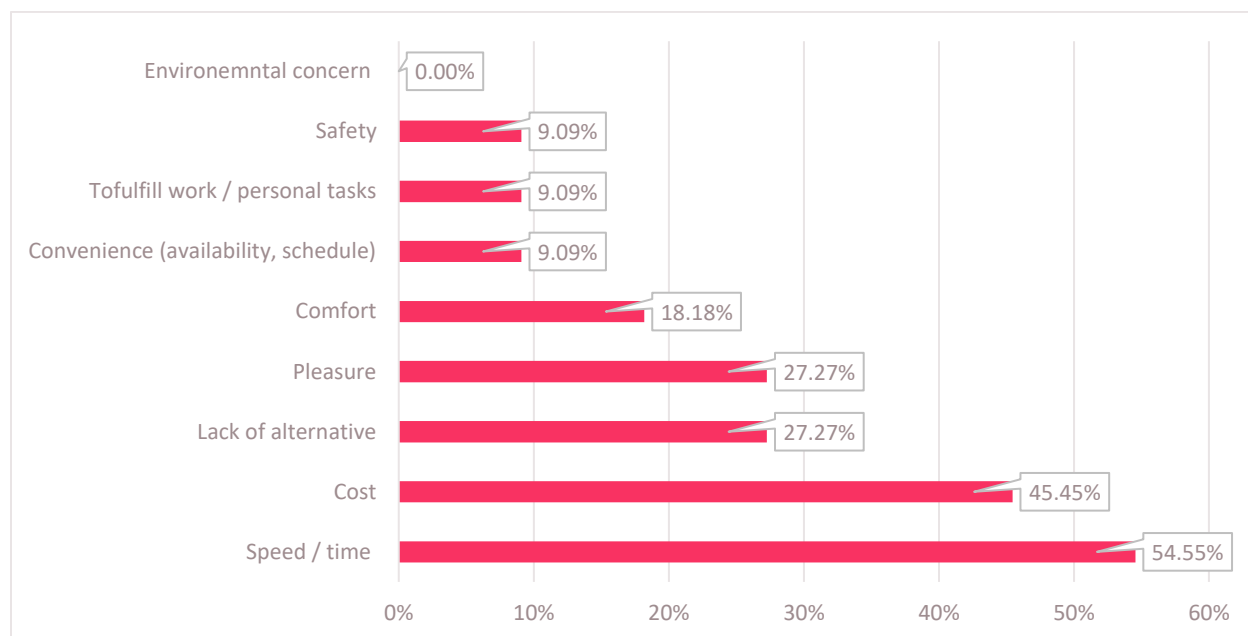


Figure 39 Main reasons for choosing daily mode of transport

Speed and travel time emerge as the most influential factor, cited by 54.55% of respondents, indicating that many participants prioritize efficiency and the ability to reach destinations quickly. Cost is the second most frequently mentioned factor (45.45%), reflecting the significance of financial considerations in shaping transport decisions.

A smaller but notable share of respondents point to the lack of alternatives and pleasure (each 27.27%) as reasons for their modal choice. The former highlights that some modes may be chosen by necessity rather than preference, while the latter indicates that enjoyment or personal satisfaction plays a modest role for certain users. Comfort is cited by 18.18% of participants, suggesting that the physical experience of travel, such as seating, space, or smoothness, can influence mode choice but is secondary to efficiency and cost.

Convenience, the ability to fulfill work or personal obligations, and safety are each selected by 9.09% of respondents, indicating that these considerations are relevant for a minority of users but do not dominate decision-making. Notably, environmental concerns were not cited by any participants, suggesting that sustainability is currently a negligible factor in influencing the primary mode of daily travel within this sample.

Overall, these findings demonstrate that time efficiency and affordability are the main drivers of travel behavior, while other factors such as comfort, convenience, safety, or environmental values play a more limited role.

Respondents were also asked to identify factors that could encourage greater use of sustainable transport modes. The results reveal a clear preference for practical improvements, suggesting that infrastructure quality, network connectivity, and service performance are the most influential drivers of increased adoption.

	%
Better infrastructure (e.g. bike lanes, covered bus stops, sidewalks)	81.82%
Better connectivity / frequency	81.82%
Time savings	45.45%
Improved comfort	18.18%
Improved safety	9.09%
Lower costs	0%

Table 70 Factors encouraging greater sustainability

The most frequently cited factors are better infrastructure and improved connectivity, each selected by 81.82% of respondents. This underscores the critical role of high-quality physical environments – such as continuous sidewalks, dedicated bike lanes, and well-designed transit stops – in encouraging sustainable travel. The prominence of these factors suggests that many respondents perceive current infrastructure and service networks as inadequate, particularly compared with the convenience, flexibility, and comfort offered by private vehicles.

Time savings, selected by 45.45% of participants, represents the next most important driver. This aligns with previous findings showing that efficiency is a central concern in travel behavior.

Sustainable modes are more likely to be adopted when they can compete with private cars in terms of travel time and reliability.

Improved comfort and safety play more secondary roles, cited by 18.18% and 9.09% of respondents, respectively. Comfort-related improvements – such as better shelter, seating, or smoother travel surfaces – enhance the travel experience but are not primary motivators. Safety considerations, while relevant for a smaller share of users, remain a concern, particularly for active modes like walking and cycling, where exposure to traffic or poorly lit routes can discourage use.

Interestingly, lower costs were not mentioned by any respondents, indicating that affordability is currently not a key factor driving the adoption of sustainable modes within this sample. This highlights that financial incentives alone may be insufficient to shift behavior unless paired with tangible improvements to infrastructure and service quality.

Overall, the findings indicate that structural and operational improvements – particularly enhancements to infrastructure and connectivity – are the most effective levers for promoting sustainable transport. Efficiency-related factors, such as time savings, reinforce adoption, while comfort and safety contribute as secondary enablers.

Regarding participants' willingness to make trade-offs for environmental sustainability, the survey results indicate a generally high level of readiness among respondents to accept personal or practical compromises in order to improve environmental conditions in the city, such as reducing parking, expanding green spaces, or lowering speed limits (see Figure 40). Responses were measured on a scale from 0 (not willing at all) to 10 (very willing), and the distribution is skewed toward the higher end of the scale.

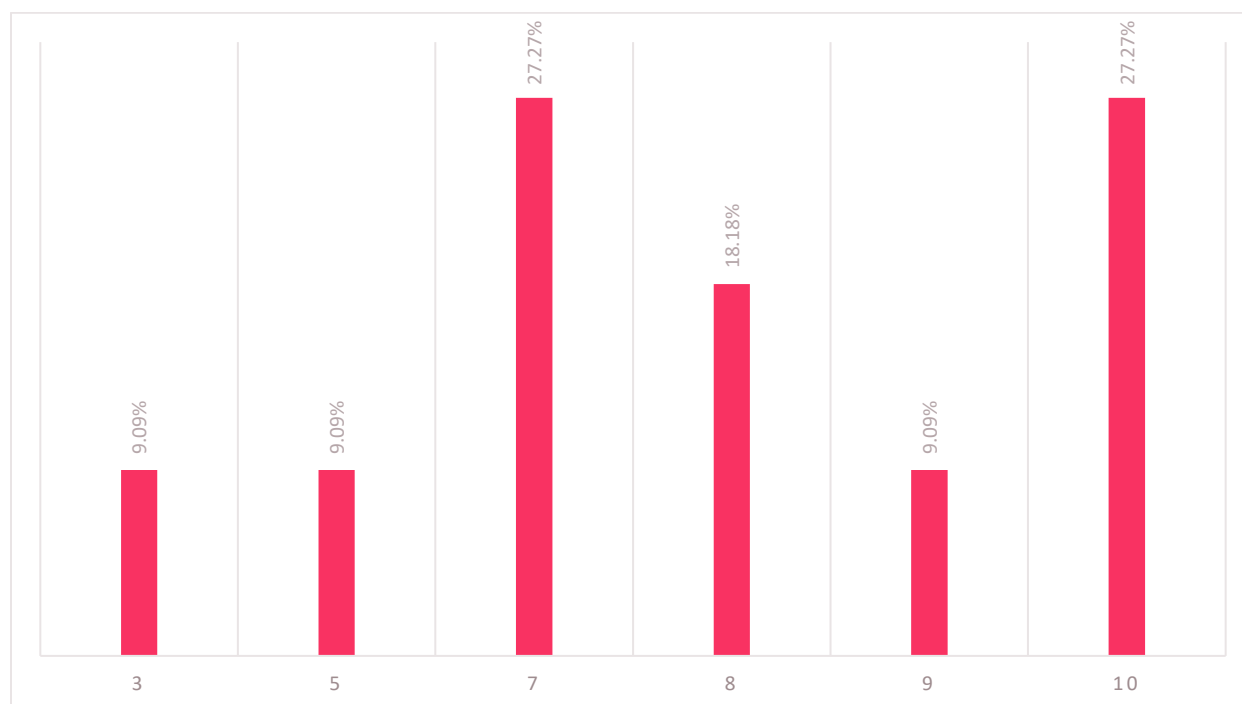


Figure 40 Willingness to accept trade-offs for environmental sustainability

Note: 0 = not at all, 10 = very willing

A substantial portion of respondents demonstrate strong willingness to accept sustainability-related trade-offs. Specifically, 27.27% selected the maximum value of 10, while 27.27% chose 7 and 18.18% selected 8. This indicates that nearly three-fifths of participants are highly receptive to initiatives that may require personal or practical adjustments, reflecting a pronounced commitment to environmental goals.

A smaller share of respondents report moderate willingness, with 9.09% selecting 5. This group appears generally supportive of sustainability measures but may have reservations regarding the implementation, scale, or possible impacts on daily life. Their responses suggest that conditional support can be maintained through careful policy design, clear communication, and attention to minimizing inconvenience.

A minority of participants (9.09%) expressed lower willingness, selecting 3, indicating a limited degree of reluctance toward making trade-offs. While not dominant, this group highlights that some residents may perceive certain measures as challenging or incompatible with their personal needs.

Overall, the findings point to a broadly positive attitude toward environmental sustainability, with the majority of respondents willing to make practical or personal compromises to support urban sustainability initiatives.

Lastly, participants were asked to indicate which measures they believe could improve mobility and quality of life in their city, both in everyday situations and during crises such as extreme weather or emergencies. Respondents were free to select as many initiatives as they considered relevant.

	%
Better public transport	90.91%
Reduced car use in busy or densely populated areas	63.64%
Digital traffic management	45.45%
Improved walking/cycling infrastructures	27.27%
More greenery	27.27%
Car-free areas	27.27%
30 km/h speed limit zones in residential or city areas	18.18%
Increased shared mobility options	9.09%
Shared streets	9.09%
Congestion charges	9.09%
Fewer parking spaces	0%

Table 71 Measures perceived as necessary to be antifragile

The responses reveal a strong preference for measures that enhance sustainable and public transport options. Better public transport is overwhelmingly supported (90.91%), highlighting the importance respondents place on reliable, frequent, and accessible transit systems for improving urban mobility. Reducing car use in busy or densely populated areas also ranks highly (63.64%), reflecting recognition that lowering traffic density can contribute to a more efficient and livable city. Measures related to digital traffic management (45.45%) also receive notable support, indicating openness toward technological solutions aimed at optimizing traffic flows and improving accessibility.

Measures supporting active mobility and environmental quality, such as improved walking and cycling infrastructure, increased greenery, and the creation of car-free areas (all 27.27%), receive moderate support, suggesting interest in interventions that enhance safety, comfort, and overall urban livability. Measures perceived as more regulatory or restrictive, including the introduction of 30 km/h speed limit zones in residential or urban areas (18.18%), are less popular but still recognized by a portion of respondents.

Finally, only a small share of participants indicate support for increased shared mobility options, shared streets, and congestion charges (each 9.09%). While these measures are less frequently selected, they nonetheless point to niche preferences and some openness to alternative or more experimental approaches among a subset of respondents.

Finally, a small share of participants indicated interest in shared mobility options, shared streets, and congestion charges (each 9.09%). In contrast, reducing the number of parking spaces receives no support among respondents (0%), suggesting that measures perceived as directly limiting parking availability may face strong resistance. While the less frequently selected measures indicate niche preferences, they nonetheless point to some openness to alternative or more experimental approaches among a subset of respondents.

7.2.3 Post-survey insights

This section synthesizes post-forum survey findings and qualitative feedback to evaluate the representativity, process design, expert contributions, facilitation quality, deliberative interactions, recommendation outcomes, participant takeaways, and attitudes toward mobility-related digital tools. A total of 11 valid responses were collected ($n = 11$). Where this number is different, this is explicitly mentioned.

7.2.3.1 Evaluation of the process

Perceptions of representativity are mixed among respondents. A plurality of participants (44.45%) indicated that they felt certain groups were missing. The open-ended responses suggest that perceived gaps related primarily to specific mobility needs and user perspectives. Frequently mentioned were people with disabilities and reduced mobility, described for example as “disabled

and 3rd age,” and “disabled, elderly.” Participants also pointed to the absence of younger age groups, referring to “students.” By contrast, one-third of respondents (33.33%) reported that they did not perceive any groups as missing. Meanwhile, a substantial proportion (22.22%) selected “don’t know.”

	%
Yes	44.45%%
No	33.33%
Don’t know	22.22%

Table 72 Perceived representativity

Furthermore, participants assessed the structure, pacing, and duration of the meetings, with all respondents (100%) considering the duration and number of meetings to be appropriate. This unanimous assessment supports the decision to adjust the Thessaloniki Forum format from two days to a single day.

High satisfaction with the process design is further reflected in participants’ willingness to remain engaged: when asked about participating in a longer process, an overwhelming majority (90.91%) responded positively, while only 9.09% indicated they would not be willing to do so.

	%
Yes	90.91%
No	9.09%

Table 73 Willingness to participate in a longer process

Overall, these results point to more than a positive experience; they reflect a shared perception that participation was meaningful. Collectively, the findings suggest that the forum successfully fostered a participatory environment that participants valued and would be willing to re-enter, providing a strong foundation for sustained engagement in future activities.

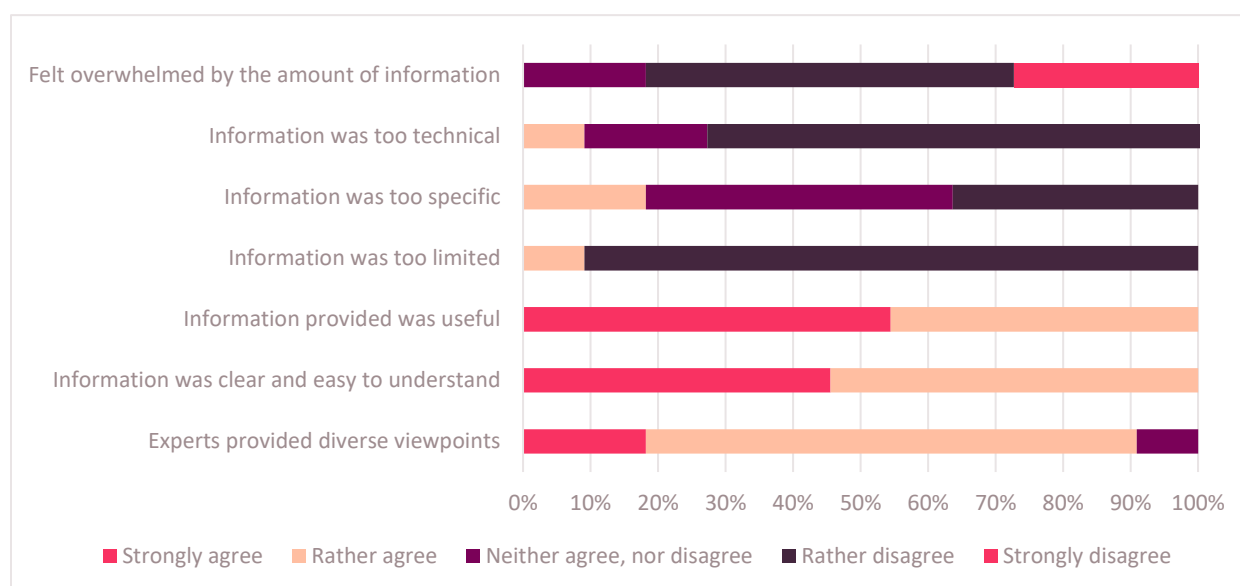
Another crucial part of any deliberative process is the contributions from experts (see Figure 41). Participants reported that expert input was generally clear, accessible, and useful. A large majority

agreed that experts provided diverse viewpoints (90.91% agreement), strengthening the perceived balance and credibility of the forum. Clarity of communication was rated particularly highly: 100% of participants agreed that the information provided was clear and easy to understand. Similarly, 100% considered the information useful, confirming that expert contributions were seen as relevant and valuable to the discussions.

While some participants perceived the information as too specific (18.18%), the majority of participants felt that information was not too limited nor too technical. Importantly, the volume of information was experienced as manageable: all respondents disagreed that they felt overwhelmed, indicating that the pacing and structure of expert sessions were appropriate for the attendees. One participant indeed claimed that *“our experts have provided excellent and informative information”*, with another participant, however, remarking that the information was *“useful though limited.”*

	Strongly agree	Rather agree	Neither agree, nor disagree	Rather disagree	Strongly disagree
Experts provided diverse viewpoints	18.18%	72.73%	9.09%	0%	0%
Information was clear and easy to understand	45.55%	54.45%	0%	0%	0%
Information provided was useful	54.45%	45.55%	0%	0%	0%
Information was too limited	0%	9.09%	0%	90.91%	0%
Information was too specific	0%	18.19%	45.45%	36.36%	0%

Information was too technical	0%	9.09%	18.19%	90.91%	0%
Felt overwhelmed by the amount of information	0%	0%	18.18%	54.55%	27.27%

Table 74 Perceptions on experts and information**Figure 41** Perceptions of expert contributions

7.2.3.2 Evaluation of deliberation

The evaluation examined the quality of deliberative interactions, focusing on both the diversity of perspectives and the inclusivity of discussions. Participants (n = 10) were first asked how many people in their group held views different from their own. All participants reported encountering differing perspectives: 80% indicated that several participants held different views, while 20% noted that about half of the group did. This suggests that the forum successfully fostered a range of viewpoints, creating space for meaningful debate while still allowing participants to identify common ground.

	%
About half of them	20%
Several of them	80%

Table 75 Perceptions on diversity

Participants were then asked to evaluate key elements of deliberative quality, including the ability to express opinions freely, equality of opportunity to speak, willingness to listen, respectfulness, and potential dominance in discussions (see Figure 42). Responses indicate strong support for inclusive and open dialogue. For instance, 90% of participants strongly agreed that they were able to express their opinions freely, and 77.78% strongly agreed that all participants had equal opportunities to speak. Respectful interaction was widely reported, with 90% strongly agreeing and 10% somewhat agreeing that discussions occurred respectfully. All participants agreed that others were willing to listen to their contributions.

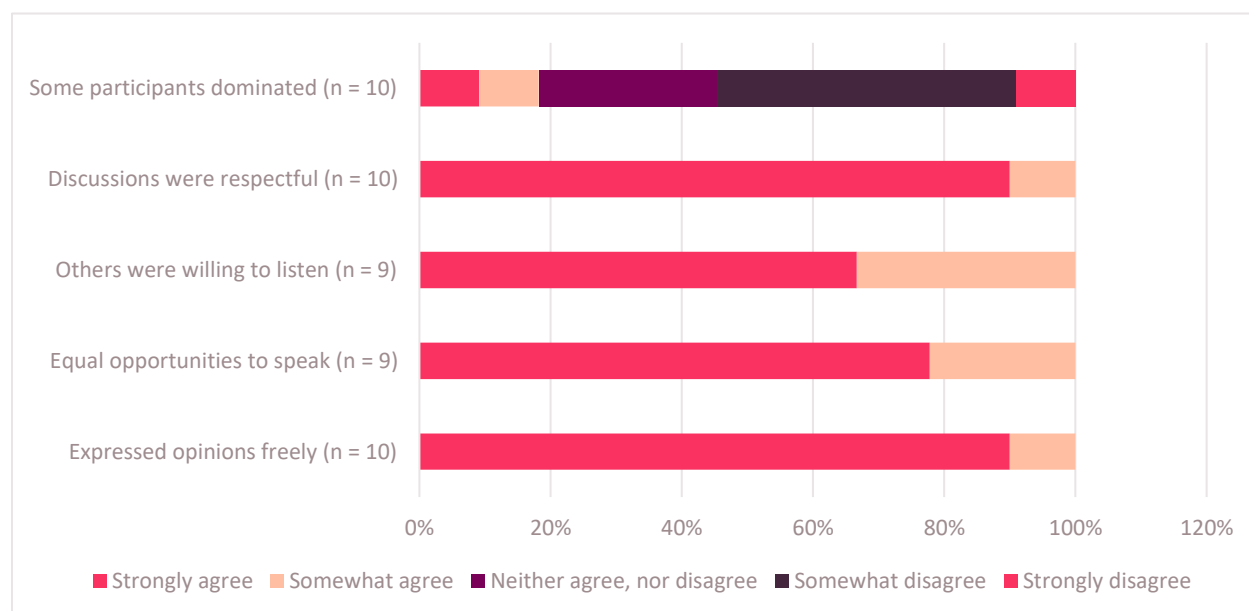
	Strongly agree	Somewhat agree	Neither agree, nor disagree	Somewhat disagree	Strongly disagree
Expressed opinions freely (n = 10)	90%	10%	0%	0%	0%
Equal opportunities to speak (n = 9)	77.78%	22.22%	0%	0%	0%
Others were willing to listen (n = 9)	66.67%	33.33%	0%	0%	0%
Discussions were	90%	10%	0%	0%	0%

respectful (n = 10)					
Some participants dominated (n = 11)	9.09%	9.09%	27.28%	45.45%	9.09%

Table 76 Perceptions on deliberations

As is common in deliberative processes, some concerns about dominance were noted. While 9.09% strongly agreed and a further 9.09% somewhat agreed that certain participants dominated discussions, the majority (54.54%) disagreed, and 27.28% reported no opinion. Importantly, this did not appear to compromise the overall quality of dialogue.

Taken together, these results indicate that deliberations were largely inclusive, respectful, and characterized by a broad range of viewpoints. Participants generally felt confident in expressing their perspectives and engaging in discussion, while the presence of dominant voices was acknowledged but did not undermine the deliberative process. Overall, the forum effectively balanced open expression with structured interaction, enabling meaningful dialogue among participants with diverse backgrounds and mobility experiences.

**Figure 42** Perceived deliberative quality

7.2.3.3 Evaluation of recommendations

The survey also explored participants' perceptions of the final recommendations, particularly whether their contributions were reflected in the outcomes. Most respondents felt that their input had been meaningfully incorporated, with 36.36% strongly agreeing and 63.64% agreeing that their contributions were considered. Notably, no participants disagreed. These findings indicate that participants perceived the process as responsive and felt that their perspectives were taken seriously, reinforcing both the sense of ownership and the legitimacy of the recommendations.

	%
Strongly agree	36.36%
Agree	63.64%
Disagree	0%
Strongly disagree	0%

Table 77 Agreement with overall recommendations

Participants were further asked to consider the recommendations from a decision-maker perspective, assessing how many they would adopt. About one-quarter (27.27%) reported they would adopt all recommendations, while 72.73% would adopt most of them. This suggests that the recommendations were seen as largely relevant and actionable.

	%
All	27.27%
Most of them	72.73%

Table 78 Perceptions on decision-makers' uptake of recommendations

Taken together, these findings highlight two key points. First, participants felt that the process was participatory and that their contributions influenced the outcomes, supporting the credibility of the recommendations. Second, while there was strong overall agreement with the final proposals, the

distribution of responses reflects the forum’s ability to balance diverse perspectives, producing recommendations that are broadly acceptable without requiring unanimous endorsement. This pattern aligns with the dynamics of deliberative processes involving heterogeneous stakeholders, demonstrating both inclusivity and the practical realities of negotiating shared outcomes.

7.2.3.4 Impact on participants

Beyond evaluating the process, the quality of deliberations, and the recommendations, the post-survey also examined the perceived impact of participation on individuals (see Figure 43). Participants assessed how useful they found the forum in three key areas: reflecting on their own mobility habits, learning new perspectives, and feeling empowered to contribute. Across all three dimensions, responses were uniformly positive. Nearly three-quarters of participants (72.73%) rated the forum as very useful in each area, while the remaining respondents found it somewhat useful. No participants indicated that the forum was not useful in any of these respects.

	Very useful	Somewhat useful	Not useful
Reflecting on personal mobility habits	72.73%	27.27%	0%
Learning new perspectives	72.73%	27.27%	0%
Feeling empowered to contribute	72.73%	27.27%	0%

Table 79 Perceived usefulness of the process

These results demonstrate that the forum effectively encouraged personal reflection, supported learning, and strengthened participants’ confidence in contributing to discussions. The consistency across all three dimensions suggests that the process not only conveyed information but also reinforced participants’ sense of agency in relation to urban mobility issues.

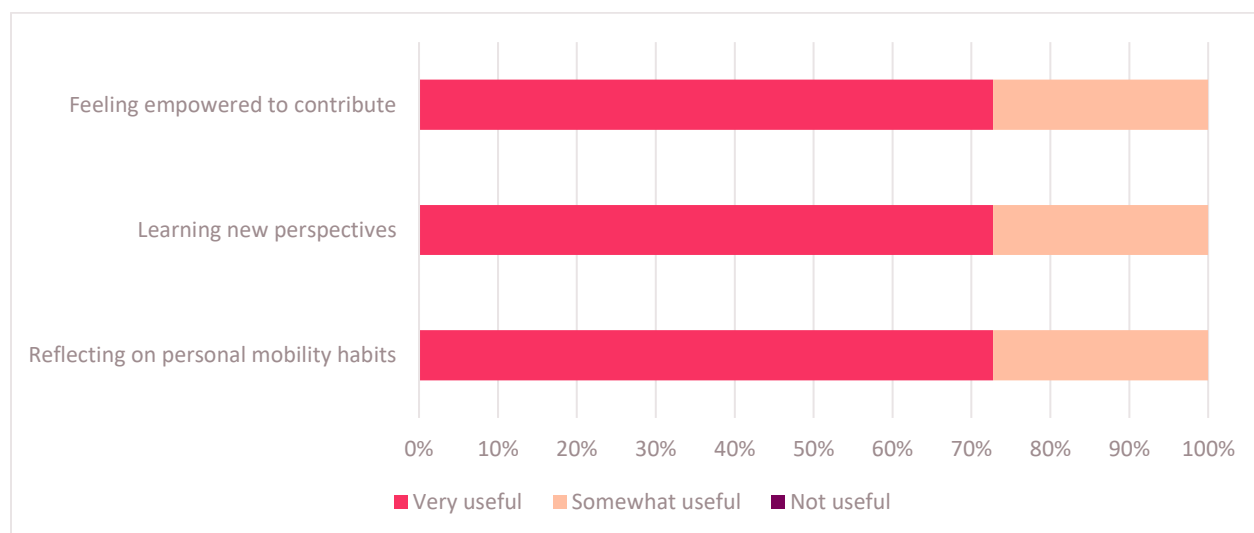


Figure 43 Impact on participants

This impact is further reflected in participants' assessments of learning and openness to new perspectives. All participants (100%) indicated that they gained new arguments and perspectives through participation, with 54.55% *strongly agreeing* and 45.45% *agreeing*. Similarly, 100% of respondents reported that the process helped them consider alternative viewpoints, with 36.36% *strongly agreeing* and 63.64% *agreeing*. In addition, nearly all participants (96.88%) stated that they developed a better understanding of others' opinions, again split between strong agreement (36.36%) and agreement (63.64%).

	Strongly agree	Agree
I gained new arguments and perspectives	54.55%	45.45%
The process helped me consider alternative viewpoints	36.36%	63.64%
I developed a better understanding of others' opinions	36.36%	63.64%

Table 80 Impact on participants

Together, these findings point to a strong contribution to participants' deliberative capacity. The forum supported not only the acquisition of new information, but also constructive engagement with diverse perspectives, enabling participants to reflect on, reconsider, and better understand differing viewpoints.

In addition to these general effects, participants reported increased understanding across several thematic dimensions of urban mobility (see Figure 44).

	Strongly agree	Agree	Neither agree, nor disagree	Disagree	Strongly disagree
Urban mobility	27.27%	72.73%	0%	0%	0%
Accessibility & inclusivity	27.27%	72.73%	0%	0%	0%
Crisis preparedness	45.45%	45.45%	9.09%	0%	0%
Diversity & redundancy	18.18%	54.55%	27.27%	0%	0%
Sustainable & low-carbon transport	36.36%	45.45%	18.18%	0%	0%

Table 81 Perceived increase in knowledge

Improved understanding of overall urban mobility was reported by 100% of respondents, with 27.27% *strongly agreeing* and 72.73% *agreeing*. Similar results were observed for accessibility and inclusivity, where again 100% reported improved understanding (27.27% strongly agree; 72.73% agree). Learning gains related to crisis preparedness were reported by 90.9% of participants (45.45% strongly agree; 45.45% agree), while 9.09% expressed a neutral position. Perceived increases in understanding of system diversity and redundancy were reported by 72.73% of respondents (18.18% strongly agree; 54.55% agree), with 27.27% remaining neutral. Understanding of sustainable and low-carbon transport also improved for 81.81% of participants (36.36% strongly agree; 45.45% agree), while 18.18% neither agreed nor disagreed.

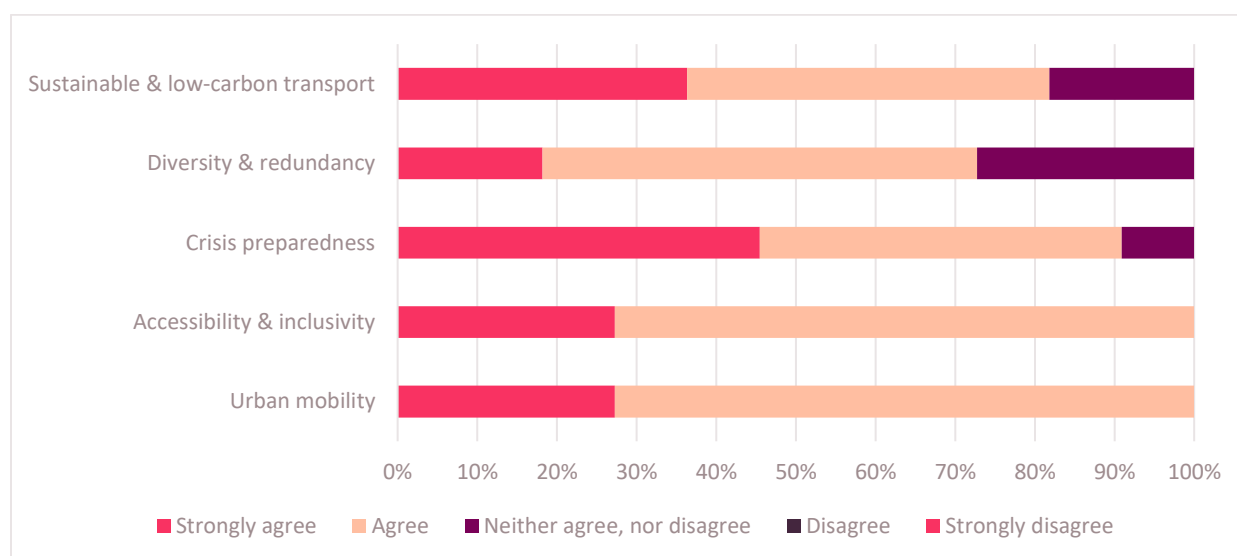


Figure 44 Participants' learning gains

To place these findings, participants were asked which elements of the process had the greatest influence on them. Small working group discussions were identified as influential by all respondents (100%), while expert presentations were highlighted by 45.45% and plenary sessions by 27.27%. This distribution underscores the importance of interactive and dialogical formats, which together played a central role in shaping participants' learning experiences.

	%
Small working group sessions	100%
Expert presentations	45.45%
Plenary sessions	27.27%

Table 82 Perceptions on various elements of process

Beyond learning outcomes, participants were also asked about anticipated changes in their own mobility-related behavior following the forum. The most frequently cited expected change was advocating for improved transport, reported by 45.45% of participants. This was followed by intentions to use more active forms of transport, such as walking or cycling (27.27%). Smaller shares

of respondents indicated plans to try shared mobility services (9.09%), use public transport more (9.09%), or reduce car use (9.09%).

	%
I will advocate for improved transport	45.45%
I will use more active forms of transport (walking/cycling)	27.27%
I will try shared mobility services	9.09%
I will use public transport more	9.09%
I will reduce car use	9.09%

Table 83 Perceived behavioral changes after participation

Taken together, these findings reveal a dual orientation among participants: a strong emphasis on civic engagement and advocacy, combined with more moderate intentions for direct behavioral change. The prominence of advocacy (45.45%) compared to lower rates of intentions to reduce car use (9.09%) or adopt shared mobility (9.09%) suggests that while the forum effectively stimulated awareness and reflection, additional supportive measures – such as infrastructure improvements, incentives, or follow-up initiatives – may be needed to translate deliberative learning into broader and more sustained changes in mobility behavior.

7.2.3.5 Open-ended feedback

Participants were invited to share their main takeaways from the forum and suggest one action they would like the city of Thessaloniki to undertake next. Analysis of these responses reveals a mix of positive experiences and actionable priorities.

Several participants described the forum as an *“interesting experience,” “great experience, I hope it is repeated,”* or *“excellent,”* reflecting high overall satisfaction with the process. Others noted specific gaps, such as the *“inadequacy of meeting citizens’ travel needs,”* signaling areas for improvement.

In terms of next steps, participants proposed a range of actionable measures for the city. Suggestions included putting discussions into practice, creating forums for citizen participation and co-decision, pedestrianization and tree planting initiatives, better informing citizens, and piloting the measures discussed during the forum. These recommendations indicate a strong desire for the city to translate dialogue into concrete actions while maintaining ongoing engagement with residents.

Main insight	Next steps for the city of Thessaloniki
Interesting experience	To put into practice all that was discussed
Inadequacy of meeting citizens' travel needs	Actions to involve citizens in mobility decision-making
Great experience, I hope it is repeated!	Pedestrianization and tree planting
Excellent	Citizens' forums to discuss and co-decide on improving mobility in the city of Thessaloniki
Very good and constructive	Informing citizens
	Pilot implementations of the measures discussed today

Table 84 Open-ended feedback

7.2.3.6 Mobile application

In line with the broader objectives of the AntifragiCity project, the post-survey explored participants' perceptions of the potential role of mobile applications in urban mobility, particularly for crisis response, everyday transport management, and inclusive participation. Overall, responses indicate a strong recognition of the value of digital tools, alongside a clear awareness of accessibility considerations.

When asked whether a mobility app could help the city respond more effectively during crises or emergencies, such as heatwaves or flooding, all participants either strongly agreed (45.45%) or agreed (54.55%). Similarly, an equal majority (45.45% strongly agree, 45.45% agree) endorsed the idea that residents should be able to share information through a mobility app to improve transport. Participants also highlighted the importance of inclusive access: 54.55% strongly agreed and 45.45% agreed that the city should provide non-digital channels to receive information and give feedback, ensuring that all residents can participate. These findings suggest that participants perceive mobile applications as potentially valuable for enhancing responsiveness, coordination, and citizen engagement, while emphasizing the need for complementary non-digital communication channels to maintain inclusivity.

	Strongly agree	Agree	Neither agree, nor disagree	Disagree	Strongly disagree
A mobility app could help the city respond better during crises or emergencies (such as heatwaves or flooding)	45.45%	54.55%	0%	0%	0%
Residents should be able to share information through a mobility app to help improve transport	45.45%	45.45%	9.09%	0%	0%
The city should also provide non-digital ways to receive information and give feedback, so everyone can benefit	54.55%	45.45%	0%	0%	0%

Table 85 Perceptions on a mobile application

Participants were then asked about their personal likelihood of using specific app functionalities, with (very) likely 81.82% to use it. Real-time public transport and traffic information emerged as the most highly valued feature, with all participants (100%) indicating they were very likely to use it. A feedback and reporting function was also highly attractive, with 71.88% very likely and 18.75% likely to use it. Route suggestions during disruptions or emergencies were considered relevant (54.55% very likely, 27.27% likely). Features supporting sustainable travel choices (72.73% very likely) and accessibility information, such as step-free routes (45.45% very likely), were viewed as useful but showed somewhat lower adoption likelihood, suggesting that participants prioritize immediate

operational information and participatory functionalities over longer-term sustainability or accessibility features.

	Very likely	Likely	Unlikely	Very unlikely
Real-time public transport and traffic information	100%	0%	0%	0%
Route suggesting during disruptions or emergencies	54.55%	27.27%	18.18%	0%
Travel suggestions that support more sustainable choices	72.73%	27.27%	0%	0%
Accessibility information (for example, step-free routes)	45.45%	27.27%	18.18%	9.09%
A way to give feedback or report problems	71.88%	18.75%	0%	9.38%

Table 86 Uses for mobile application

Qualitative feedback identified potential barriers to app adoption, including “*incomplete information and not seeing tangible results from its use*”, not being used to working with “*technology*”, and the “*the lack of will to innovative changes*.”

Taken together, these findings indicate a strong perceived value of a city mobility app for both crisis management and citizen engagement, particularly for real-time information and feedback. At the same time, participants emphasized the need for inclusive and integrated design, accounting for digital accessibility and complementary non-digital communication channels.

7.2.4 Comparative insights

All participants (n = 11) completed pre- and post-surveys assessing their attitudes toward urban mobility policies, civic participation, and their perceived role in transport decision-making. Both surveys included identical questions, allowing assessment of how participants' views shifted after the process, with particular attention to:

- Attitudes toward participatory and inclusive decision-making
- Willingness to engage in future mobility initiatives
- Willingness to accept sustainability-related trade-offs
- Interest, knowledge, and confidence regarding mobility issues
- Support for specific mobility measures

7.2.4.1 Citizens' participation in urban mobility

In both the pre- and post-surveys, participants rated their level of agreement (from strongly disagree to totally agree) with three statements: that public participation improves decision-making, that citizens' everyday experiences are as important as expert knowledge, and that participation leads to fairer outcomes. Changes between the two survey points were categorized as more positive, less positive, or no change.

	More Positive	Less Positive	No Change
Public participation is important	54.55%	0%	45.45%
Citizens' experiences equals expert knowledge	36.36%	9.09%	54.55%
Diversity leads to fairer outcomes	27.27%	27.27%	45.45%

Table 87 Changes in perceptions regarding citizens' participation

Support for participatory and inclusive governance was already high prior to the forum and largely remained stable. Nevertheless, a substantial share of participants strengthened their agreement across all three statements, most notably regarding the value of public participation in improving decision-making. Shifts in a less positive direction were limited and, where present, tended to move toward more neutral positions rather than outright disagreement. Taken together, these patterns suggest that the process primarily reinforced existing positive attitudes toward citizen participation, while also encouraging some participants to adopt a more reflective and nuanced view, particularly with regard to fairness outcomes.

Comparing participants' willingness to be involved in mobility initiatives before and after the forum shows a clear shift toward more active engagement. Changes were classified as more positive (movement toward more active involvement), less positive (movement toward less active involvement), or no change.

	%
More active	63.64%
Less active	9.09%
No change	27.27%

Table 88 Direction of change

Nearly two-thirds of participants (63.64%) moved toward more active involvement, either shifting from “maybe” to “participate occasionally” or from “participate occasionally” to “participate actively.” Just under one-third (27.27%) reported no change, maintaining their initial level of willingness, while only a small minority (9.09%) shifted toward less active participation. Overall, this pattern indicates that the process was effective in translating attitudes into a stronger declared willingness to engage in real-world mobility initiatives.

Comparing pre- and post-process responses reveals notable shifts in how participants imagine their role in urban mobility governance. Rather than a simple substitution of participation modes, the data point to a broadening and reconfiguration of engagement preferences. Several key trends can be identified.

I. Overall expansion of participation preferences

Across the sample, participants tended to select more participation options after the process than before. While pre-survey responses often reflected one or two preferred modes of engagement, post-

survey responses more frequently included three or even all four options (forums, apps, community initiatives, and surveys). This suggests that the process encouraged participants to envision a more active and multifaceted role in mobility governance.

II. Strengthened interest in deliberative forums

Interest in participation through public consultations or citizens' forums increased notably. While fewer than half of participants selected forums before the process, nearly three-quarters did so afterward. Importantly, forums were rarely the sole option selected post-process. Instead, they were increasingly combined with other forms of engagement, indicating that participants continue to value deliberative spaces but now see them as part of a broader participation ecosystem rather than as a standalone mechanism.

III. Strong growth in hands-on and practice-oriented engagement

Two participation modes showed particularly strong post-process growth: testing mobility-related applications and involvement in community or neighborhood mobility initiatives. Both options were selected by a large majority of participants after the process, including several who had not chosen them beforehand. This pattern points to a shift toward more practical, experiential forms of participation, where citizens contribute not only through opinions and feedback but also through testing, co-creation, and local action.

IV. Repositioning of surveys as a complementary tool

Providing feedback through surveys remained a common and stable participation option across both time points. However, its role appears to have evolved. Post-survey, surveys were more often selected alongside other modes of engagement rather than on their own. This suggests that participants increasingly view surveys as a useful but supplementary form of participation, complementing more interactive or hands-on involvement.

V. Increased convergence toward multi-modal participation

Taken together, the post-process responses indicate a convergence toward multi-modal participation profiles. Fewer participants relied on a single channel, and more combined deliberative, digital, community-based, and feedback-oriented roles. This reflects a more differentiated understanding of participation depth and a readiness to engage at multiple levels, depending on context and opportunity.

Overall, these trends suggest that the process broadened participants' perceptions of what participation in urban mobility governance can entail. Rather than replacing existing preferences, it layered additional forms of engagement onto them, fostering a more comprehensive and active conception of citizen involvement.

7.2.4.2 Participants and urban mobility

In line with the questions on citizen participation and the topic of urban mobility, the surveys also measured three aspects of respondents' engagement with urban mobility: their general interest in urban mobility, how informed they feel about the city's mobility policies, and their confidence in contributing to urban mobility discussions.

Interest levels were already high at baseline, creating a ceiling effect. nearly half of the participants showed an increase in interest following the process, while an equally large share reported stable interest levels, often remaining at already high values (see Figure 45). Only one participant reported a decrease in interest, indicating that the process generally maintained or strengthened participants' engagement rather than diminishing it.

	%
Increased	45.45%
Decreased	9.09%
No change	45.45%

Table 89 Direction of change



Figure 45 Comparison of participants' interest in urban mobility

Notably, a majority of participants reported feeling more informed after the process, indicating a positive learning effect (see Figure 46). However, a notable minority experienced a decrease in perceived informedness, suggesting that for some participants the process may have introduced greater awareness of complexity or knowledge gaps. This pattern points not only to information gains but also to more reflective and nuanced self-assessments of knowledge.

	%
Increased	54.55%
Decreased	27.27%
No change	18.18%

Table 90 Direction of change

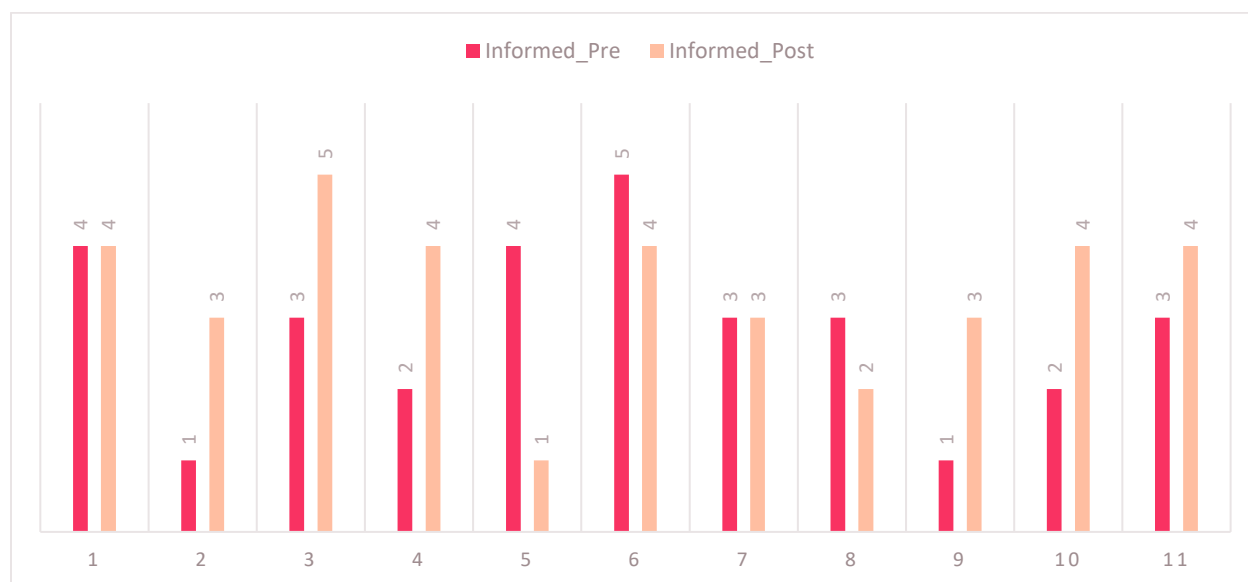


Figure 46 Comparison of participants' perceptions of feeling informed

The large majority of participants reported an increase in confidence following the process, indicating that participation substantially strengthened their sense of confidence (see Figure 47). The small share of decreases suggests that while the process may have led some participants to reassess their confidence more critically, the dominant effect was clearly positive, reinforcing participants' self-assurance in relation to the topic addressed.

	%
Increased	72.73%
Decreased	18.18%
No change	9.09%

Table 91 Direction of change

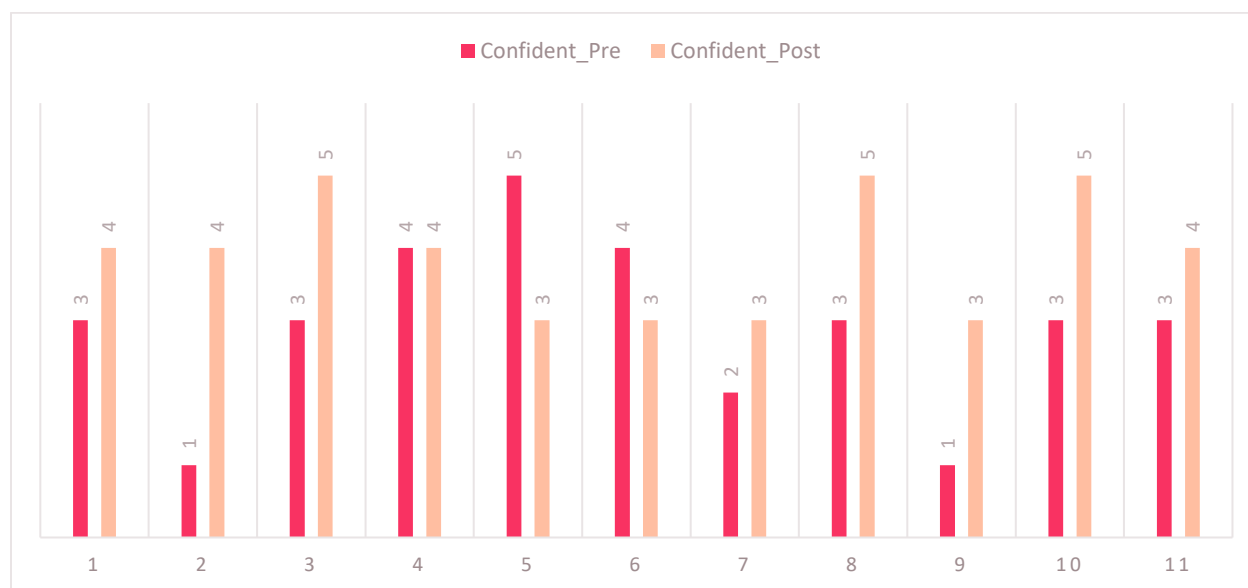


Figure 47 Comparison of participants' willingness to contribute to urban mobility policies

7.2.4.3 Mobility habits and opinions

In both surveys, participants rated their willingness to accept trade-offs related to sustainable mobility measures (scale 0–10) (see Figure 48).

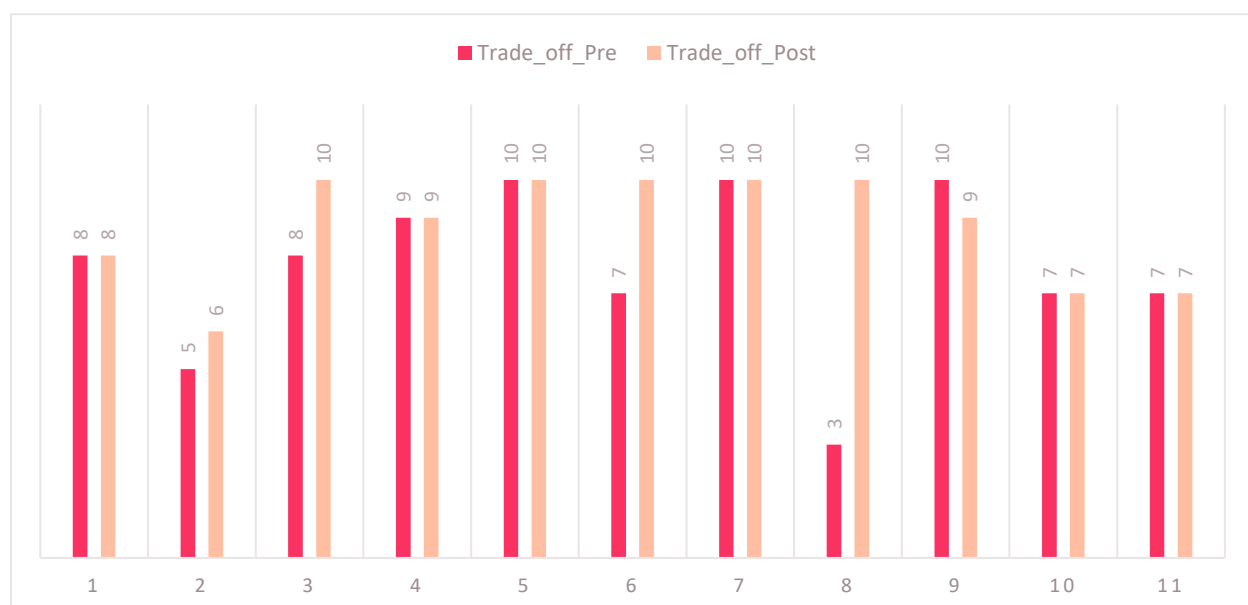


Figure 48 Comparison of participants' willingness to accept trade-offs

	%
Increased	36.36%
Decreased	9.09%
No change	54.55%

Table 92 Direction of change

A substantial share of participants (36.36%) became more willing to accept trade-offs or compromise on mobility measures, reflecting increased openness to systemic or regulatory interventions such as car restrictions or reallocation of public space. A small minority (9.09%) reported a decrease in willingness, while the majority (54.55%) maintained their original stance. Overall, the pattern indicates that the process tended to reinforce or enhance participants' readiness to consider trade-offs, supporting a more flexible and pragmatic approach to urban mobility planning.

In a related question, participants were also asked to identify factors that would encourage greater use of sustainable transport.

	Pre (%)	Post (%)	Δ (Post-Pre)
Better public transport	90.91%	63.64%	-27.27%
Reduced car use in busy or densely populated areas	63.64%	72.73%	+9.09%
Digital traffic management	45.45%	27.27%	-18.18%
Improved walking/cycling infrastructures	27.27%	54.55%	+27.28%
More greenery	27.27%	45.45%	+18.18%

Car-free areas	27.27%	45.45%	+18.18%
30 km/h speed limit zones in residential or city areas	18.18%	18.18%	0%
Increased shared mobility options	9.09%	45.45%	+36.36%
Shared streets	9.09%	36.36%	+27.27%
Congestion charges	9.09%	0%	-9.09%
Fewer parking spaces	0%	9.09%	+9.09%

Table 93 Measures perceived as necessary to be antifragile

The comparison of pre- and post-survey responses reveals a clear reconfiguration of priorities rather than a wholesale replacement of preferences. Overall, participants tended to select a broader set of measures in the post-survey, suggesting a shift away from single-solution thinking toward combinations of interventions addressing infrastructure, regulation, technology, and urban space simultaneously.

Better public transport remained the most highly valued option, though support declined from 90.91% pre-survey to 63.64% post-survey (-27.27%). This decline may indicate that while participants still view public transport as central, they also began to consider complementary measures to improve overall mobility. Support for improved walking and cycling infrastructures increased substantially from 27.27% to 54.55% (+27.28%), while support for more greenery rose from 27.27% to 45.45% (+18.18%). Similarly, car-free areas gained support from 27.27% to 45.45% (+18.18%). These changes highlight a stronger post-survey emphasis on sustainable, people-centered, and environmentally oriented urban interventions.

The largest relative gains occurred in measures promoting flexible and shared mobility. Support for shared streets increased from 9.09% to 36.36% (+27.27%), and increased shared mobility options rose from 9.09% to 45.45% (+36.36%). This indicates growing recognition of participation in and adaptation to innovative mobility solutions beyond traditional infrastructure improvements.

Reduced car use in busy or densely populated areas also saw a modest increase from 63.64% to 72.73% (+9.09%).

By contrast, support for digital traffic management declined from 45.45% to 27.27% (-18.18%). Congestion charges also decreased from 9.09% in the pre-survey to 0% in the post-survey (-9.09%), suggesting limited acceptance of this measure among respondents. In contrast, reducing the number of parking spaces increased slightly from 0% to 9.09% (+9.09%), indicating that although this measure was initially not selected by any participants, a small number considered it relevant in the post-survey. Support for 30 km/h speed limit zones remained unchanged at 18.18%, indicating stable but limited endorsement.

Overall, post-survey responses show participants adopting a more integrative approach, combining core public transport priorities with interventions that promote sustainable mobility, active travel, greenery, and shared or community-oriented solutions. While some regulatory or system-management measures saw declines, the pattern suggests a shift toward practical, tangible, and people-centered measures, reflecting a nuanced understanding of how multiple strategies can work together to improve urban mobility.

7.2.4.4 Synthesis

The pre-post comparison indicates that the process did more than reinforce general support for mobility or sustainability policies. Participants showed a deeper understanding of urban mobility challenges and the complexities of policy implementation, along with a clearer sense of their own role in governance.

Many participants became more willing to accept sustainability trade-offs, reflecting an increased openness to measures with tangible implications for daily travel or urban space. At the same time, policy preferences expanded and became more integrated, with growing support for combinations of interventions that include regulation, infrastructure, technology, and environmental improvements.

Civic self-perception also shifted positively. Confidence in contributing to mobility discussions increased for a majority, while willingness to engage in real-world initiatives grew. Notably, some participants reported lower perceived knowledge, suggesting that greater awareness of complexity did not reduce their sense of legitimacy as contributors, but rather accompanied more reflective engagement.

Support for participatory governance remained high. Participants continued to value public involvement, the importance of lived experience, and diversity in decision-making, reinforcing the legitimacy of inclusive approaches to mobility planning.

Preferences for participation became more differentiated and multi-modal. Post-process, participants increasingly saw themselves not only as respondents to surveys or forums, but also as testers of mobility applications, collaborators on local projects, and actors contributing at multiple levels.

Overall, the findings suggest that participants developed more reflective, engaged, and system-aware perspectives on urban mobility. They emerged better equipped to understand policy trade-offs, consider integrated solutions, and situate themselves as active contributors within broader governance processes.

7.2.5 Recommendations

The citizens' forum in Thessaloniki produced a structured set of recommendations per thematic area.

7.2.5.1 Liveable cities

1. Education, awareness, and cultivation of mobility culture

Participants emphasized that building a culture around shared and active mobility is crucial to improve the use of public transport, cycling, walking, and other sustainable modes.

Key recommendations include:

- Launch education campaigns on the benefits of active and shared mobility for health, environment, and community life
- Develop school and community programs that teach safe, inclusive, and respectful mobility practices
- Promote best practices from other cities and local pilot projects to inspire adoption of sustainable mobility behaviors

These measures aim to create lasting cultural change, making sustainable and shared mobility a default choice for all citizens.

2. Activation of citizens and local initiatives

Participants noted that empowering citizens to co-create solutions strengthens neighborhoods, improves quality of life, and ensures that local needs are addressed.

Recommended measures included:

- Facilitate neighborhood assemblies or citizen councils to discuss local mobility and urban space issues

- Support local-scale initiatives, such as community-maintained pocket parks, shared bikes, or street events, that respond to specific neighborhood needs
- Encourage active participation in urban planning to foster trust, co-responsibility, and a sense of belonging

This approach leverages collective action to complement formal planning, ensuring solutions are practical, inclusive, and community-driven.

3. Traffic-free zones in the historical center

Reducing or restricting motorized traffic in the city center emerged as a key experimental solution to enhance safety, walkability, and social interaction.

Key recommendations included:

- Establish car-free streets or limited-access zones in Thessaloniki's historical center
- Implement temporary pilot initiatives (weekends, festival, or seasonal experiments) to test and refine traffic restrictions
- Complement restrictions with enhanced pedestrian and cycling infrastructure to maintain accessibility and mobility options

This measure prioritizes people over cars, fostering safer streets, vibrant public spaces, and healthier urban life.

7.2.5.2 Crisis-responsive mobility

4. Integrated crisis mobility governance

Participants highlighted that fragmented coordination and lack of integrated governance reduce the effectiveness of emergency response. Effective crisis management requires collaboration across authorities, transport operators, and emergency services.

Recommended features included:

- Establish integrated crisis mobility governance framework connecting city authorities, mobility operators, and emergency services in real time
- Develop standardized protocols and coordination mechanisms for rapid decision-making during disruptions
- Conduct regular simulations and joint exercises to test crisis plans and refine inter-agency collaboration
- Encourage co-development with communities and stakeholders, ensuring local needs are considered in emergency planning

This approach ensures mobility systems are responsive, coordinated, and capable of handling cascading disruptions during crisis.

5. Inclusive emergency mobility planning

Participants stressed that vulnerable populations – elderly, disabled, digitally excluded – are often overlooked in crisis planning. Emergency mobility must be accessible, equitable, and adaptable to all citizens' needs.

Citizens recommended:

- Integrate evacuation protocols and accessibility-by-design measures into mobility planning
- Provide targeted support for vulnerable groups, including priority transport access and personalized guidance during emergencies
- Invest in soft mobility infrastructure that supports walking, cycling, and low-emission transport modes during disruptions
- Implement education and awareness programs to prepare citizens for safe and efficient emergency mobility

These measures aim to ensure that no group is left behind during disruptions, enhancing overall urban resilience.

6. Real-time data sharing and intelligent mobility systems

Participants recognized that limited information flows exacerbate crises, causing delays, confusion, and cascading failures. Smart, data-driven systems can significantly improve situational awareness and operational flexibility.

Recommendations included:

- Deploy real-time data-sharing platforms linking transport operators, authorities, and emergency services
- Use AI and predictive tools for early warning, crisis detection, and adaptive routing
- Develop “one-stop-shop” mobility applications providing personalized information, optimal routes, and predictive alerts
- Implement adaptive traffic control and mobility management systems to respond dynamically to emerging disruptions.

These systems enable anticipatory, flexible, and efficient responses, minimizing the impact of crises on urban mobility.

7.2.5.3 Resilient and diverse mobility

7. Development of multimodal mobility hubs and integrated MaaS platforms

Participants emphasized that weak interconnections between transport modes and fragmented mobility systems reduce efficiency, convenience, and attractiveness of public and shared mobility.

Key recommendations included:

- Create multimodal mobility hubs connecting buses, metro, trams, cycling, e-scooters, and shared mobility services in accessible and user-friendly locations
- Implement Mobility-as-a-Service (MaaS) platforms offering unified journey planning, booking, payment, real-time updates, and predictive routing through a single interface
- Promote seamless integration of transport modes, ensuring minimal waiting times and convenient transfers between services
- Encourage interoperability between operators and digital systems, including standardized ticketing and pricing policies across mobility modes

This approach fosters a fully connected, multimodal ecosystem, enabling intuitive, efficient, and sustainable travel for all users.

8. Inclusive infrastructure and mobility services

Participants highlighted persistent accessibility challenges, particularly for people with disabilities, elderly citizens, parents with strollers, and other vulnerable groups. Ensuring equity in mobility is essential for social inclusion and urban resilience.

Recommended measures included:

- Upgrade and expand infrastructure with accessibility-by-design standards, including ramps, tactile guidance, elevators, and inclusive signage
- Improve walking and cycling networks to make soft mobility safe, continuous, and convenient
- Expand mobility services in underserved neighborhoods to reduce car-dependency and spatial inequalities
- Introduce personalized support measures, such as mobility assistance, priority access during peak times or disruptions, and inclusive digital guidance

These measures ensure that all citizens can access and benefit from multimodal mobility systems, reducing exclusion and promoting equity.

9. Coordinated governance and data-driven integration

Participants noted that fragmented governance and lack of shared data prevent integrated management of mobility systems and reduce resilience to disruptions. Unified coordination and communication are critical for smooth operations.

Key recommendations included:

- Establish coordinated governance frameworks connecting authorities, transport operators, and service providers
- Deploy shared data platforms enabling real-time monitoring, predictive analytics, and adaptive operational responses
- Standardize communication channels to provide unified, accurate, and timely information to travelers
- Promote co-design and collaboration with citizens, vulnerable user groups, and community actors to ensure mobility solutions are practical, inclusive, and widely accepted

This approach strengthens system-wide integration, operational flexibility, and real-time responsiveness, making urban mobility more reliable and resilient.

7.2.5.4 Sustainable mobility transition

10. Extension and densification of pedestrian, cycling, and public transport networks

Participants emphasized that the current dominance of private cars and limited mobility alternatives reduces sustainability, accessibility, and urban livability.

Recommended functions included:

- Expand and densify pedestrian walks and cycling networks, ensuring continuous and safe routes along major city arteries
- Extend public transport networks (metro, bus, tram, and maritime transport) to provide diverse and reliable alternatives to private cars
- Improve interconnectivity between modes, facilitating seamless transfers and multimodal journeys
- Prioritize active and sustainable mobility corridors in urban planning, ensuring accessibility and comfort for all users

This approach encourages modal shift toward low-carbon travel, reducing congestion, emissions, and dependence on private vehicles.

11. Familiarization with technology and commuting applications

Participants highlighted that technology can facilitate sustainable mobility choices but requires citizen engagement and cultural adoption.

Citizens highlighted:

- Introduce digital applications and platforms for journey planning, real-time updates, booking, and payment across multiple modes
- Promote car-sharing, e-scooter sharing, and other shared mobility services, integrating them into digital platforms for seamless use
- Conduct public awareness and training campaigns to familiarize citizens with sustainable mobility technologies and benefits
- Implement incentive systems that reward the use of public transport, active mobility, and shared mobility solutions

These measures aim to change travel behavior, making sustainable mobility convenient, attractive, and culturally accepted.

12. Introduction of additional public transport modes

Participants noted that reliance on a limited set of transport modes restricts flexibility and coverage, contributing to private car preference.

Key recommendations included:

- Introduce new transport modes, such as tram lines and maritime urban transport (boats/ferries), to expand travel options
- Integrate new modes with existing infrastructure, ensuring multimodal connectivity
- Focus on convenience, reliability, and safety to increase public trust and adoption of alternative transport
- Support local neighborhood centers with improved transport access, reducing the need for car trips to distant destinations

This ensures greater modal diversity, improves accessibility, and support a transition to low-carbon urban mobility.

7.2.5.5 Prioritization / triage of citizens' recommendations

To support the translation of citizens' inputs into actionable guidance, the recommendations generated during the Thessaloniki citizens' forum were prioritized based on the voting scores assigned by participants. Each proposal received positive ("plus") and, in some cases, negative ("minus") points, reflecting collective judgments about importance, urgency, and perceived trade-

offs. Proposals without an explicit score were interpreted as neutral, indicating neither strong endorsement nor opposition.

The resulting prioritization does not represent a prescriptive implementation sequence, but rather a collective assessment of relative importance as expressed by participants. It highlights which interventions were perceived as foundational to the functioning of the urban mobility system, which were seen as important enabling measures, and which were considered situational or supportive. This triage provides insight into how citizens differentiate between structural investments, governance mechanisms, safety measures, and operational responses when considering urban mobility challenges and potential crisis conditions.

It is important to note that recommendations from the crisis-responsible mobility and resilient and diverse mobility thematic groups were not scored using pluses and minuses. Instead, participants reached consensus on their critical importance:

Crisis-responsible mobility

- Establishment of integrated crisis mobility governance frameworks
- Investment in inclusive emergency mobility planning
- Deployment of real-time data-sharing platforms

Resilient and diverse mobility

- Development of multimodal mobility hubs
- Development of new infrastructure and expansion of mobility networks
- Establishment of coordinated governance frameworks

For the remaining recommendations, the prioritization exercise resulted in the following triage:

Tier 1 - high priority: very strong collective support, perceived as foundational and urgent

- Education, awareness, and cultivation of culture that relies on mobility services (+6) → Seen as critical for shifting travel behavior and embedding a culture of sustainable, shared, and active mobility
- Extension and densification of pedestrian walks, cycling routes and public transport network (+4) → Recognized as structural interventions essential to enabling accessible, reliable, and multimodal urban mobility

Tier 2 – medium priority: strong support, considered significant conditions for a functioning system

- Familiarization with new technologies and applications for commuting (+3) → Supports adoption of integrated mobility tools and encourages behavioral change
- Activation of citizens, neighborhoods' assemblies and initiatives to dynamically provide local-scale solutions to quality-of-life challenges (+3) → Encourages participatory approaches and local problem-solving as complementary enablers
- Introduction of additional means of public transport (+3 / -1) → Enhances modal diversity, with minor concerns about feasibility or trade-offs

Tier 3 – low-neutral priority: recognized as important, but not strongly emphasized in voting

- Traffic-free zones (0) → Seen as a situational or experimental measure rather than a structural necessity

The prioritization reveals a clear preference among participants for foundational, system-defining interventions over incremental or situational measures. Recommendations related to infrastructure, public space, and cultural change received the strongest support, reflecting the belief that long-term investments in capacity, accessibility, and urban culture are more consequential than reactive or temporary adjustments.

Crisis-responsible mobility proposals were recognized as system-level priorities. These measures were unanimously seen as essential enablers of resilient and coordinated responses during disruptions or crises. Rather than focusing on incremental operational fixes, participants valued interventions that strengthen the city's capacity to manage emergencies proactively and inclusively, ensuring mobility continuity for all citizens, including vulnerable groups.

Similarly, in the resilient and diverse mobility domain, interventions such as the development of multimodal mobility hubs, expansion of infrastructure and networks, and coordinated governance frameworks were perceived as foundational investments. Participants highlighted that these interventions are critical for creating an integrated, reliable, and accessible mobility system that supports sustainable travel behavior and equitable access across neighborhoods.

The triage exercise demonstrates that participants differentiate clearly between foundational interventions, enabling measures, and situational responses. Structural interventions received the strongest support because they underpin system reliability, accessibility, and resilience. Enabling measures, including citizen engagement and technology adoption, were valued for enhancing usability and adoption, while situational interventions like traffic-free zones were seen as supplementary experiments rather than core necessities.

7.2.5.6 Synthesis for antifragile mobility planning

The citizens' forum in Thessaloniki provides a coherent picture of how different mobility interventions are perceived in terms of social acceptability, priority, and legitimacy. The voting scores and consensus-based triage indicate that public support is not driven by individual measures in isolation, but by how interventions fit into a consistent, integrated, and understandable mobility system.

Across the recommendations, participants showed strong acceptance of foundational and transformative interventions, particularly when these were framed as enhancing system capacity, resilience, accessibility, and reliability. Measures such as multimodal mobility hubs, expansion of pedestrian and cycling networks, public transport densification, and coordinated governance frameworks were perceived as essential structural investments rather than optional or experimental improvements. This suggests that citizens are willing to endorse long-term, visible, and broadly beneficial interventions, even when they require significant spatial, financial, or organizational commitments.

Safety, accessibility, and inclusivity emerged as central determinants of social acceptability. Participants consistently prioritized measures that visibly reduce vulnerability, enhance user confidence, and guarantee equitable access. This is evident in support for inclusive emergency mobility planning, accessibility-by-design infrastructure, and citizen-targeted support in crises. In this sense, safety and inclusivity function as threshold conditions: without them, other interventions – regardless of environmental, operational, or efficiency benefits – risk underuse or rejection.

Governance and coordination were closely linked to perceived legitimacy. Strong backing for integrated crisis mobility governance, coordinated operational frameworks, and shared data platforms indicates that participants value predictable, expert-led, and transparent decision-making structures. Interventions were more acceptable when seen as part of a long-term, system-level strategy, rather than reactive or politically driven actions. Citizens clearly recognize that fragmented or inconsistent governance undermines the performance, reliability, and resilience of urban mobility systems.

Information provision and communication were highlighted as essential enablers of acceptance, particularly during crises or disruptions. Participants expect timely, clear, and multi-channel communication that explains both the actions being taken and the reasoning behind them. The emphasis on one-stop-shop digital platforms, real-time alerts, and predictive routing tools reflects an understanding that uncertainty, confusion, and lack of guidance can erode trust and limit the effectiveness of otherwise well-designed interventions.

Taken together, the Thessaloniki citizens' forum suggests that socially acceptable mobility interventions share several key characteristics: they enhance safety and inclusivity, strengthen

structural and operational capacity, are embedded in coordinated governance frameworks, and are supported by transparent and timely communication. The findings indicate that citizens are prepared to support long-term prioritization, regulation, and transformative changes in the mobility system, provided these measures are visible, equitable, and clearly improve daily experiences for all urban residents.

7.2.6 Social acceptability of mobility interventions: Insights from Thessaloniki

The Thessaloniki AntifragiCity Citizens' Forum, combining pre- and post-surveys, deliberations, and participant-formulated recommendations, provides detailed insights into the social acceptability of mobility interventions, with a focus on sustainable, resilient, and crisis-ready urban transport. Participants evaluated a wide range of proposals – spanning infrastructure, governance, regulatory measures, digital tools, and operational adaptations – considering both everyday functionality and extreme-event scenarios.

Key determinants of social acceptability emerged consistently across the forum. Mobility measures gain support when they enhance reliability, predictability, safety, and inclusivity, and when they are embedded within coherent, transparent, and evidence-based governance frameworks. Legitimacy was reinforced when interventions were accompanied by clear planning, expert coordination, and structured participatory processes, particularly in crisis contexts where ad hoc or opaque measures could undermine trust.

Infrastructure and system robustness formed central criteria for acceptability. Participants emphasized high-quality public transport, continuous pedestrian networks, connected cycling routes, and well-integrated interchanges, perceiving these as foundational, widely beneficial, and resilient under stress. Infrastructure that delivered visible improvements in accessibility, continuity of service, and safety received the highest support. Measures supporting active mobility – including bike lanes, sidewalks, and pedestrianized areas – saw increased endorsement following the forum, reflecting a post-deliberation shift toward sustainable, people-centered interventions.

Safety, accessibility, and predictability dominated discussions. Both objective safety (e.g., well-lit and maintained public spaces, physically separated transport modes, signalized crossings) and perceived safety influenced participants' willingness to adopt interventions. Similarly, adherence to binding mobility plans and expert-led, apolitical coordination structures was regarded as essential to maintain trust, legitimacy, and compliance.

Communication and information provision emerged as essential enablers of acceptability. Participants highlighted the need for multi-channel, inclusive, and real-time information to guide behavior, explain the rationale behind interventions, and clarify alternative routes during disruptions. Digital tools, particularly mobile applications providing real-time public transport

information, feedback functions, and route guidance during crises, were strongly endorsed, though participants stressed the importance of complementary non-digital channels to ensure accessibility for all residents.

Participation and civic engagement strengthened acceptance and understanding of integrated interventions. Across the pre- and post-surveys, participants reported greater willingness to engage in multiple forms of involvement – including public forums, community projects, application testing, and surveys – indicating a broadening of perceived roles in governance. Civic engagement also translated into heightened openness to sustainability trade-offs, such as reducing car use, reallocating street space, and prioritizing active mobility. Participants increasingly endorsed combinations of infrastructure, regulatory, and environmental measures, reflecting a nuanced understanding that effective mobility interventions require integrated strategies.

The post-forum assessment confirmed that participatory processes improve reflection, knowledge, and confidence. Participants reported learning new perspectives, considering alternative viewpoints, and better understanding diverse opinions. They valued interactive small-group sessions and expert input, which together facilitated informed and inclusive deliberation. Perceived behavioral impacts included increased advocacy for improved transport, greater intentions to use active modes, and openness to shared mobility and reduced car use, suggesting a dual orientation toward both civic engagement and personal behavioral change.

Shifts in intervention preferences reveal important patterns: while better public transport remained central, support grew for walking and cycling infrastructure, car-free areas, shared streets, and flexible mobility options, reflecting a post-deliberation shift toward sustainable, community-oriented, and environmentally friendly solutions. Regulatory or abstract system-management measures, such as congestion charges and fewer parking spaces, remained less popular, suggesting participants favor visible, tangible, and experiential improvements.

In summary, socially acceptable mobility interventions in Thessaloniki share several characteristics:

- Robust infrastructure ensuring safe, reliable, and continuous mobility across modes, including active and shared transport
- Transparent and evidence-based governance, incorporating expert-led planning, binding mobility strategies, and participatory mechanisms
- Clear, multi-channel communication, providing timely guidance, rationale, and alternative options, including inclusive digital and non-digital channels
- Adaptive operational measures that maintain usability during disruptions and crises
- Equitable and visible trade-offs, where restrictions or prioritizations are justified, fairly applied, and understandable to citizens

These findings highlight that citizens' support for mobility interventions is enhanced through participatory processes, learning, and reflection. Participants emerged more engaged, system-aware, and willing to contribute, with greater recognition of trade-offs, integrated strategies, and innovative mobility solutions. The Thessaloniki forum demonstrates that social acceptability is dynamic, informed by deliberation, and strengthened when interventions are practical, transparent, and visibly improve daily urban mobility.

8 Overarching insights

This section synthesizes insights across the three forums, with a dual focus. First, it seeks to explain the substantial variation observed in the number of citizens registering interest to participate, as well as differences in actual forum engagement. This analysis draws on a combination of contextual and procedural factors, established findings from citizen participation and deliberative processes, and the pre-survey results collected among participants. Importantly, none of these elements in isolation can account for the observed variation; rather, they provide complementary and partial explanations that, when considered together, help illuminate the differing participation dynamics across the three cases.

Second, the section addresses two related but distinct areas. The first examines insights from both the pre- and post-surveys across the forums, highlighting similarities and differences in participants' perspectives and experiences. The second consolidates the recommendations emerging from the three forums, providing a comparative lens to identify broader patterns of social acceptability related to urban mobility and crisis-oriented solutions.

8.1 Explaining variations in forum engagement

The significant variation in forum engagement observed across Bratislava, Larissa, and Thessaloniki is likely the result of multiple, interacting factors rather than any single determinant.

The pre-survey findings, collected among participants across the three AntifragiCity Forums, cannot explain why individuals initially responded to outreach. However, they do help contextualize:

- How aligned the accepted participant group was with a deliberative forum format
- How stable participation proved once individuals had committed
- How participation may have plateaued despite open invitations

Essentially, the pre-survey results reveal distinct participation profiles across Bratislava, Larissa, and Thessaloniki.

These insights must be interpreted alongside structural conditions, cultural expectations of participation, communication strategies, and perceived legitimacy and impact of the process.

Importantly, from other deliberative processes, we know that response rates to deliberative invitations are structurally low, even under favorable conditions. When citizens are directly invited, for instance, through household invitations, response rates often range between 3% and 6%. Against this backdrop, the registration figures observed across all three cities are not anomalous in themselves. What requires explanation, rather, is the observable differences between Bratislava, Larissa, and Thessaloniki.

8.1.1 Bratislava

Bratislava represents a case in which several enabling conditions converged, producing a large pool of registrants and a highly stable final participant group.

First, the forum benefitted from exceptionally high visibility, notably through inclusion in the city newsletter delivered to all households. Such broad and official communication channels are known to increase perceived legitimacy, signal institutional seriousness, and reduce uncertainty about whether participation will have meaningful consequences.

Second, Bratislava deliberately minimized the use of the project label AntifragiCity, instead framing the event as the “Bratislava Forum.” From a participation perspective, this choice likely reduced cognitive and cultural barriers to entry. Abstract, technical, or unfamiliar project branding can suppress engagement, whereas locally grounded, intuitive, and place-based framing likely tends to increase uptake and perceived relevance.

Third, participation was financially reimbursed. Compensation directly addresses a well-documented structural barrier to participation: the unequal distribution of time, income, and caregiving responsibilities. Even modest reimbursement can function both as a practical enabler, offsetting participation costs, and as a symbolic signal that citizens’ time and experiential knowledge are valued.

From a broader contextual perspective, transportation is a highly salient and contested issue in Bratislava. Issue salience is a strong predictor of participation: citizens are more likely to engage when an issue affects everyday life, is publicly debated, and is perceived as unresolved. This salience was reinforced by the forum’s explicit linkage to the development of a new mobility plan, offering a credible promise of follow-up and accountability. Participation research consistently highlights

perceived impact, feedback loops, and institutional responsiveness as among the strongest motivators for engagement.

The pre-survey provides an additional, complementary explanation. Participants in Bratislava exhibited the strongest overall orientation toward formal participatory processes. A large majority expressed a willingness to participate either occasionally (50%) or actively (32.35%), indicating a high level of engagement capacity within the accepted group. This was coupled with a very strong preference for public consultations and citizens' forums, selected by 88.24% of respondents – by far the highest share across the three sites.

In addition, a substantial proportion of respondents (73.53%) strongly endorsed the view that public participation is essential. Taken together, these figures suggest a participant group that not only supports participation in principle, but also perceives deliberative forums as a legitimate and preferred mode of engagement. This alignment between participatory preferences and the forum format likely contributed to both the exceptionally high number of registrations and the stability of attendance once participants were selected.

8.1.2 Larissa

In Larissa, both registration and sustained participation levels were more modest than in Bratislava, reflecting structural and contextual differences rather than a lack of interest in urban mobility issues per se. Larissa lacked several amplification factors that proved influential elsewhere, most notably financial incentives and strong institutional signaling through explicit policy linkage. As a result, registrations likely captured most of the willing and reachable participants – often described in the deliberative democracy field as the “usual suspects” – but did not extend far beyond this group.

Pre-survey results further contextualize this pattern. Even among accepted participants, willingness to engage was more conditional and cautious. In Larissa, 29.17% of respondents expressed interest in occasional participation and 25% in active participation. Preferences for participation formats also differed: the most attractive option was testing mobility-related applications (58.33%), followed by public consultations and citizens' forums (45.83%). Consistent with this, only 58.33% of respondents strongly agreed that public participation is essential, indicating a more pragmatic or instrumental view of engagement.

These preferences help explain why participation declined sharply after the first day. The forum format – two-days and discussion-heavy – may not have aligned well with how many participants preferred to engage. Several participants indicated that engagement could have been strengthened through more practical and tangible elements, such as live demonstrations of tools or platforms supporting urban resilience. This aligns with the survey finding that hands-on, application-oriented participation was particularly attractive in Larissa and suggests that abstract deliberation alone may not be sufficient to sustain interest.

While initial curiosity and interest were sufficient to attract attendance on the first day, sustained engagement proved harder to maintain, particularly when the process required a high time commitment without compensation or immediate, visible impact. The forum was not clearly linked to an ongoing local consultation or decision-making process, such as civil protection planning or a sustainable urban mobility plan. Moreover, participants perceived that the outcomes were not binding for local authorities, which likely weakened incentives to invest time and effort – especially given that key local plans are already under implementation until 2027. The forum was also conceived as a stand-alone event, with no clear follow-up meetings or continuity mechanisms, reinforcing perceptions of limited impact.

Process design factors reinforced these dynamics. Although the overall number of registered participants and participants showing up on day 1 was satisfactory, actual attendance was limited in duration, with most participants attending only briefly on the first day and only five returning on the second. The total duration – approximately 10 hours across two days – was widely perceived as too demanding for volunteers. Participants explicitly suggested that a shorter format would be necessary to ensure stronger commitment and retention.

Scheduling also played a significant role. Holding the second session on a Saturday, a non-working day in Greece, proved unsuitable for much of the target audience and contributed significantly to the drop in attendance. This highlights the sensitivity of deliberative processes to local social rhythms and competing personal obligations.

At the same time, low attendance produced one unintended benefit. The small group size enabled deeper and more substantive discussions, allowing participants to contribute meaningfully across all thematic areas and fostering high-quality interaction. This suggests a trade-off between breadth and depth: while the process struggled to sustain broad participation, it succeeded in generating reflective, coherent inputs among those who remained engaged.

Overall, the Larissa case illustrates that participation levels are shaped not only by outreach and topic relevance, but by format design, time demands, institutional anchoring, and perceived impact.

8.1.3 Thessaloniki

In Thessaloniki, participation occurred under more fragile logistical conditions. The last-minute change of venue likely introduced uncertainty and friction, thereby affecting turnout.

The pre-survey results show unanimous support for participatory principles and strong normative endorsement of citizen involvement. A large majority of respondents expressed willingness to participate either occasionally (64%) or actively (27%), indicating a high degree of openness to engagement. However, preferences for participation modes reveal a weaker orientation toward

deliberative forums. Community or neighborhood projects were the most preferred option (63.64%), while public consultations and citizens' forums were the least preferred mode (45.45%).

Notably, only 36.36% of respondents strongly agreed that public participation is essential – the lowest share among the three sites. While this does not indicate opposition to participation, it may reflect a more instrumental or context-dependent understanding of citizen involvement, where participation is valued primarily when it is concrete, local, and action-oriented.

This combination – high stated willingness to engage but comparatively lower normative endorsement of participation and weaker preference for forums – may help explain why forum participation remained limited in scale, particularly when combined with logistical challenges

8.1.4 Synthesis

Overall, the variation in engagement across the three cities is best understood as the outcome of layered and mutually reinforcing influences, including:

- Structural and contextual factors (visibility, incentives, topic salience, accountability)
- Logistical conditions (such as venue stability)
- Institutional signaling and framing
- the engagement profiles of those who accepted participation, as revealed by the pre-surveys

Crucially, neither the survey insights nor the contextual and procedural factors discussed above provide certainty or a causal explanation. Rather, when considered jointly, they offer a plausible interpretive framework for understanding why registration numbers diverged so markedly and why participation scaled smoothly in some contexts while plateauing in others, despite the fact that a broadly similar outreach strategy and methodological approach were applied across all three AntifragiCity forums. In essence, while these patterns do not establish causality, they provide a plausible explanatory layer for why identical or similar participatory designs yielded markedly different engagement outcomes across the three contexts.

In sum, while site-specific conditions likely influenced recruitment strategies – this cannot be stated in absolute terms, given that individuals registering for the survey nor participants were explicitly asked how they found out about the forum or why they decided to sign up – the overarching strategy adopted in the AntifragiCity Forums successfully convened diverse groups with variation in socio-demographic profiles, mobility practices, and lived experiences in all three Forums, meeting the project's goal of generating meaningful quantitative and qualitative insights. Furthermore, a common methodological framework, facilitation guidelines, and forum structure ensured comparability across sites. In this view, local adaptations were treated as contextual enrichment rather than methodological deviation.

8.2 Lessons learned from the AntifragiCity Forums

Beyond explaining observed differences in registration and engagement, the three AntifragiCity Forums generate a set of broader lessons for the design and implementation of deliberative and participatory processes in EU research-oriented projects, particularly those addressing complex and potentially contentious policy domains such as urban mobility and crisis preparedness. These lessons should not be interpreted as prescriptive or universally applicable recommendations. Rather, they constitute context-sensitive insights derived from the implementation of the ACFs across three distinct local settings. Their value lies in illustrating how participatory designs perform under varying institutional, cultural, and operational conditions, thereby informing future applications in comparable research and policy environments.

In addition to these practical insights, the ACFs contribute to deliberative and urban mobility research by illustrating how CAs can be operationalized in multi-site, research-driven contexts while maintaining methodological robustness and adaptability. Whereas many citizens' assemblies are designed as large-scale processes with direct policy mandates, the AntifragiCity approach demonstrates that more flexible, resource-conscious, and context-sensitive formats can still produce meaningful deliberative outcomes and generate analytically valuable insights.

8.2.1 Contribution to Deliberative Urban Mobility Research

The ACFs make a threefold methodological contribution to deliberative urban mobility research by demonstrating how deliberative approaches can be integrated into applied, multi-site research settings, how social acceptability can be reconceptualized as a dynamic and deliberatively constructed process, and how deliberative engagement can be understood within a broader framework of adaptive and antifragile socio-technical systems.

First, the ACFs advance the integration of deliberative methods into the study of urban mobility as a socio-technical and experiential system. Rather than relying solely on stated preferences, survey responses, or behavioral indicators, the forums capture how participants interpret mobility challenges in relation to their everyday experiences, local contexts, and broader societal objectives. This approach enables the systematic inclusion of experiential knowledge, which is often underrepresented in conventional, technically oriented planning and modelling approaches. In doing so, the ACFs illustrate how deliberative settings can complement quantitative and model-based methods by providing insight into the social and contextual dimensions of mobility systems.

Second, the methodology contributes to a reconceptualization of social acceptability as a dynamic and deliberatively constructed process. Instead of treating acceptability as a fixed *ex ante* condition or a constraint to be measured prior to policy implementation, the forums demonstrate how citizens' evaluations evolve through exposure to information, interaction with others, and engagement with competing considerations. Acceptability in this sense emerges through processes of collective

reasoning, reflection, and negotiation, rather than as a simple aggregation of pre-existing attitudes. This perspective allows for a more nuanced understanding of how support, ambivalence, and resistance toward mobility interventions develop over time, and how these positions may shift within deliberative environments.

Third, the ACFs situate deliberative engagement within the broader conceptual framework of antifragility. By creating structured environments for dialogue, reflection, and knowledge exchange, the forums foster processes of social learning, perspective diversification, and adaptive reasoning. These processes are essential for navigating uncertainty in complex socio-technical systems, where robustness alone is insufficient and adaptive capacity becomes critical. From this perspective, participatory deliberation is not merely an auxiliary component of system design, but an integral mechanism through which systems can evolve, adapt, and improve in response to stressors, uncertainty, and changing conditions.

Beyond these three core contributions, the ACFs also demonstrate the methodological value of deliberative processes as both participatory and analytical instruments. On the one hand, they enable meaningful citizen engagement by providing structured opportunities for discussion, reflection, and collective judgment formation. On the other hand, they generate rich data on how participants reason about trade-offs, evaluate policy options, and construct arguments. This dual function positions deliberative formats as a valuable research approach for examining decision-making processes in complex policy domains such as urban mobility, where competing objectives – such as efficiency, sustainability, equity, and feasibility – must be balanced.

Finally, the AntifragiCity approach illustrates the feasibility and value of operationalizing deliberative methods in a multi-site, research-driven context. While many CAs are designed as large-scale processes with direct policy mandates, the ACFs demonstrate that more flexible, resource-conscious, and context-sensitive formats can still produce meaningful deliberative outcomes while generating analytically valuable insights. The multi-site implementation further highlights how a consistent methodological framework can be adapted to different local contexts, enabling both comparability and context-specific understanding.

Taken together, the methodological approach developed in AntifragiCity provides a replicable and adaptable framework for integrating deliberative engagement into applied research on urban systems. It demonstrates how deliberation can contribute not only to participatory inclusion but also to knowledge production, by capturing not only what citizens think, but how their perspectives are formed, articulated, and transformed through interaction.

The following section therefore complements this methodological perspective with empirically grounded lessons on the design, communication, and organization of deliberative processes, highlighting how these principles translate into practice under varying institutional and cultural conditions.

8.2.2 Lessons learned

Building on the methodological and conceptual contributions outlined in Section 8.3.1, the implementation of the ACFs across three distinct local contexts also generated a set of practical insights relevant to the design, communication, and organization of deliberative processes. Whereas the previous section focused on how the forums contribute to deliberative urban mobility research in analytical and conceptual terms, the following lessons are derived from their empirical application in real-world settings characterized by differing institutional, cultural, and operational conditions.

These lessons reflect observed patterns in participation, engagement, and process dynamics, illustrating how methodological principles translate into practice. In particular, they highlight the importance of aligning deliberative designs with local institutional frameworks, communication strategies, participant expectations, and resource constraints. Where participants clearly understood the scope of the forums and how their input would be used, engagement appeared more stable and meaningful. Conversely, misalignment between expectations and actual influence may affect trust, motivation, and perceived legitimacy.

At the same time, they underscore that participatory processes should not be viewed as uniform interventions, but as context-sensitive instruments whose effectiveness depends on how they are embedded and interpreted locally. While the core methodological framework remained consistent across locations, differences emerged in participation patterns, engagement dynamics, and preferences for participation formats. These variations reflect local institutional settings, cultural expectations, and participatory traditions. Rather than being viewed as inconsistencies, such differences provide valuable insight into how deliberative processes operate under different conditions, reinforcing the importance of interpreting results in a context-sensitive manner. Multi-site implementations thus offer not only comparative evidence but also a means of identifying which elements of participatory design are robust and which are context-dependent.

8.2.2.1 Visibility and institutional embedding

One clear lesson is that the institutional embedding of participatory processes and their public communication often matter as much as methodological rigor. In Bratislava, high visibility through official city channels and a clear connection to an ongoing policy process appeared to significantly support participation. Where institutional anchoring was weaker or less visible, participation tended to plateau earlier, even when the methodological design was comparable.

This suggests that participatory activities should, where possible:

- Be clearly connected to recognizable policy processes
- Be communicated through trusted, official channels
- Signal institutional ownership rather than appearing as standalone research exercises

8.2.2.2 Naming, framing, and symbolic accessibility

The AntifragiCity experience highlights that project branding can unintentionally act as a barrier to participation. The use of the locally grounded “Bratislava Forum” label, rather than the AntifragiCity project name, likely reduced cognitive distance and improved accessibility.

This reinforces the importance of framing participation in ways that:

- Are immediately understandable to citizens
- Use familiar and accessible language
- Position participation as a civic opportunity rather than a technical or research exercise

At the same time, this requires balancing project identity requirements with locally meaningful communication strategies.

8.2.2.3 Incentives and practical support

The provision of participant reimbursement in Bratislava illustrates that even modest incentives can help reduce structural barriers to participation. While incentives alone do not determine engagement, they can:

- Broaden the pool of participants
- Reduce socio-economic bias
- Signal respect for citizens’ time and contributions

In research-oriented participatory settings, such measures should be understood less as inducements and more as mechanisms that support inclusivity and equitable access. Importantly, their use also implies that deliberative processes require adequate financial and organizational resources, highlighting the need for realistic planning in line with participation objectives.

8.2.2.4 Format alignment with local participatory cultures

The pre-survey results across the three forums indicate that citizens relate differently to participatory formats. While deliberative forums aligned strongly with participant preferences in Bratislava, participants in Larissa and Thessaloniki showed relatively greater preference for more practical or low-effort formats, such as surveys or application-based engagement.

This suggests that deliberative processes are most effective when:

- They align with existing participatory practices and expectations
- The format of discussion and collective reasoning is culturally familiar
- They are combined with complementary, more action-oriented or flexible participation modes

These findings support the use of hybrid participatory designs rather than standardized, one-size-fits-all approaches.

8.2.2.5 Perceived impact and follow-up

Across all three cases, participants' expectations regarding tangible outcomes and follow-up emerged as a key factor influencing engagement. Where processes clearly communicated how input would be used and provided feedback after completion, participation appeared more sustained.

This reflects a broader lesson from participatory governance: perceived impact is closely linked to motivation. For research projects, this implies the need to:

- Clearly articulate the role of participants' input
- Manage expectations regarding influence and decision-making power
- Provide structured feedback and follow-up communication after the process

Such measures contribute to transparency, trust, and perceived legitimacy.

8.2.2.6 EU research projects

Finally, the ACFs illustrate a structural tension inherent in EU-funded participatory research. While such projects often promote experimentation, innovation, and cross-context learning, they may face challenges in:

- Establishing local legitimacy
- Maintaining relevance beyond the project timeframe
- Compete with participation fatigue among citizens

Addressing these challenges requires deliberate design choices that translate research objectives into locally meaningful civic experiences, rather than relying solely on methodological novelty to generate engagement.

8.2.3 Summary of lessons learned

In summary, the ACFs suggest that successful participatory processes – in EU research contexts – depend on:

- Strong institutional embedding and visibility
- Culturally and symbolically accessible framing
- Practical support and incentives
- Alignment between participation formats and local expectations
- Clear communication of purpose, expectations, and impact

- Careful adaptation to local institutional and cultural contexts
- Sensitivity to the limits and opportunities of EU-funded research settings

Taken together, these lessons reinforce the importance of integrating methodological design with contextual sensitivity, institutional alignment, and participant-centered communication in order to achieve meaningful, credible, and sustained engagement.

8.3 Socially acceptable mobility interventions

Across the three AntifragiCity Forums, participants engaged with a range of mobility interventions in relation to both everyday mobility and crisis-oriented scenarios. This aggregate perspective allows for the identification of shared patterns of acceptance, concern, and prioritization across the three forums, while reducing the influence of site-specific dynamics or isolated local factors. In doing so, it highlights common understandings of which types of interventions are perceived as legitimate, appropriate, and desirable in the context of urban mobility resilience – and antifragility.

Consequently, these aggregate findings should be interpreted as indicative of how citizens evaluate the social acceptability of alternative mobility measures. They illuminate which interventions are more readily associated with improved mobility and quality of life under conditions of stress and disruption, and which require additional justification or contextualization to be perceived as acceptable within crisis-oriented mobility planning.

8.3.1 Pre-surveys

The results presented in this section represent aggregate findings across all three AntifragiCity Citizens' Forums, comprising all participants from Bratislava, Larissa, and Thessaloniki who completed the pre-survey (n = 69).

The pre-survey results provide insight into how participants initially assessed different mobility interventions prior to participation in the citizen forums, that is, before expert input and facilitated discussion with other participants. As such, they capture participants' baseline perceptions and intuitive judgments regarding which types of interventions are considered legitimate, meaningful, and worth supporting when thinking about everyday mobility and crisis situations.

These results are particularly relevant for understanding the starting conditions. They reflect how citizens approach questions of sustainable and crisis-resilient mobility based on their existing experiences, assumptions, and values, rather than on arguments developed or refined through the deliberative process. In this sense, the pre-survey establishes a reference point against which subsequent discussions and shifts in perspective can be interpreted, and helps to identify which interventions already enjoy a degree of social acceptance.

8.3.1.1 Sustainability

Participants were asked which factors would increase their use of sustainable transport modes. The results provide direct insight into how citizens prioritize different intervention types when reflecting on their own mobility behavior (see Figure 49).

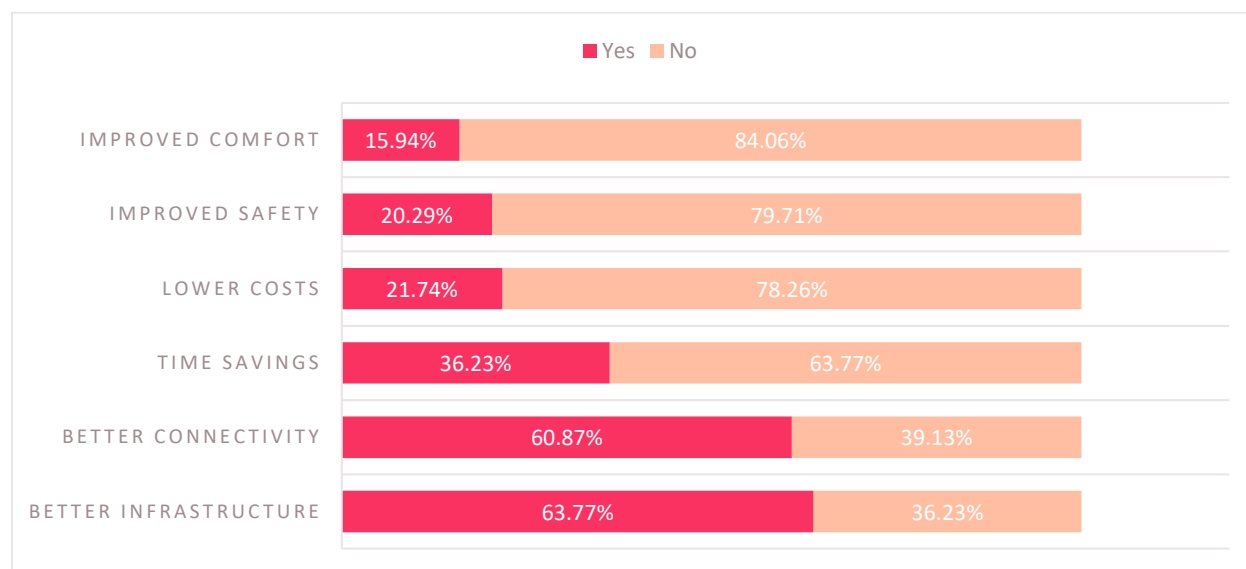


Figure 49 Sustainable measures

Across all three sites, participants most frequently identified better infrastructure (63.77%) and better connectivity or service frequency (60.87%) as the most important enablers of sustainable transport use. This indicates a strong and consistent preference for interventions that are visible, system-level, and directly experienced in everyday mobility. Improvements to physical infrastructure (such as cycling lanes, pedestrian networks, or public transport stops) and operational enhancements (such as more frequent or better-coordinated services) were clearly understood as measures that reduce practical barriers to sustainable mobility. These findings suggest that interventions delivering immediate and tangible improvements are most readily perceived as acceptable and worthwhile.

Other factors, such as time savings (36.23%), lower costs (21.74%), and improved safety (20.29%), were cited less frequently. While these elements remain important, particularly for specific population groups or travel contexts, they appear to function as secondary considerations within participants' overall assessments. This suggests that citizens may view individual efficiency or cost-related gains as contingent upon more fundamental improvements to the transport system as a whole. In this sense, infrastructure and service quality form the baseline upon which other benefits become meaningful.

Improved comfort (15.94%) was least frequently selected. This indicates that comfort-oriented measures, while desirable, are unlikely to be decisive drivers of sustainable mode adoption when considered in isolation. Comfort appears to be perceived as an added value rather than a core justification for intervention, particularly when weighed against more pressing functional concerns such as accessibility, reliability, or network quality.

Taken together, these results indicate that the social acceptability of sustainable mobility interventions is closely tied to their capacity to address practical, everyday mobility challenges. Interventions that operate primarily at a technical, incremental, or experiential level, without clearly improving how people move through the city, are less likely to be prioritized. From a social baseline perspective, citizens appear to evaluate sustainability measures primarily in terms of practical improvements that reduce barriers, increase accessibility, and make daily travel easier, rather than abstract environmental or behavioral arguments alone.

8.3.1.2 Antifragility

The pre-survey also asked participants to identify measures they believed would improve mobility and quality of life both in everyday situations and during crises, such as extreme weather events or emergencies (see Figure 50). This provides insight into how citizens perceive interventions that enhance the resilience and robustness of urban mobility systems.

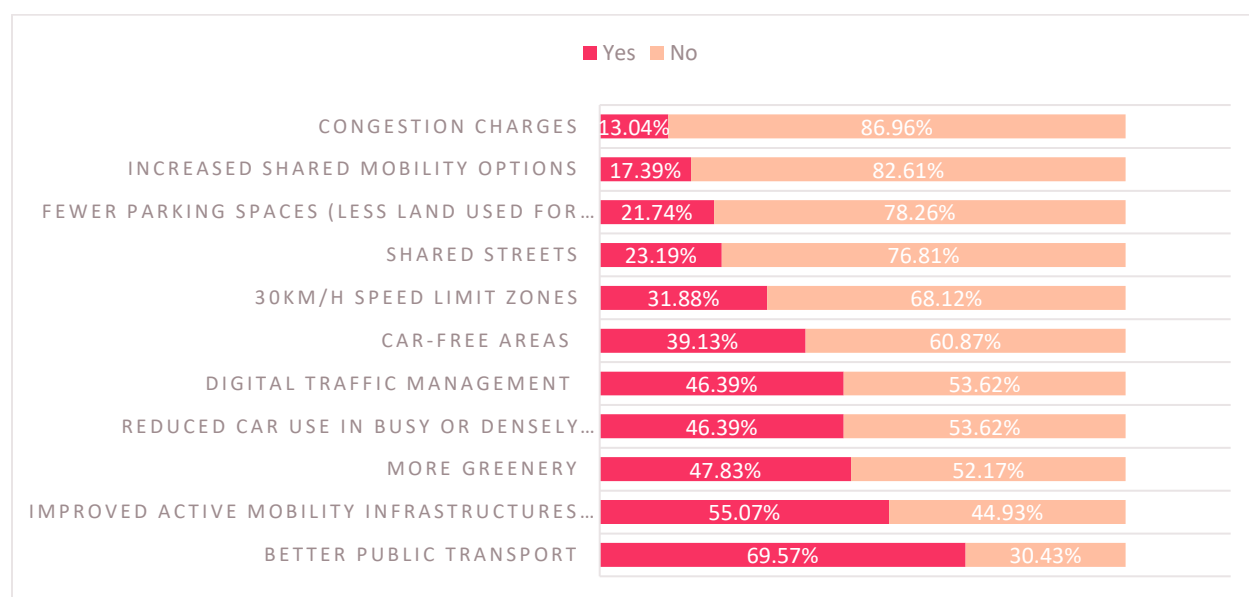


Figure 50 Antifragile measures

Participants expressed the strongest support for better public transport (69.57%), suggesting that it is widely understood as a foundational component of both everyday mobility and crisis response

capacity. Public transport appears to be perceived as a system that can provide collective reliability, scalability, and continuity under disrupted conditions, thereby supporting accessibility when individual modes or routes are constrained.

Similarly, improved active mobility infrastructure for walking and cycling (55.07%) received strong support, highlighting their role as resilient alternatives to motorized transport. These results indicate that participants recognize the importance of locally accessible, low-dependence modes in crisis situations. Walking and cycling infrastructure is implicitly understood as robust in the face of disruptions such as fuel shortages, congestion, or partial system failures, reinforcing its perceived contribution to mobility continuity during emergencies.

Measures with co-benefits beyond transport, including broader environmental and spatial benefits, such as more greenery (47.83%), were also positively evaluated. Participants appear to associate greenery not only with improved everyday quality of life, but also with enhanced resilience to extreme weather events, including heat stress or flooding. In this context, greenery is perceived as part of a supportive urban environment that indirectly strengthens mobility systems by improving comfort, safety, and environmental conditions during crises.

Interventions that restructure or regulate car use, including reduced car use in busy or densely populated areas (46.39%) and digital traffic management (46.39%), elicited more ambivalent responses. While a substantial share of participants recognized their potential benefits, these measures did not command majority support. This suggests that their contribution to crisis resilience is less immediately apparent to citizens and may be understood as context-dependent or conditional.

Support declined further for interventions involving strong spatial or behavioral constraints, such as car-free areas (39.13%), 30 km/h speed limit zones (31.88%), shared streets (23.19%), and fewer parking spaces (21.74%). Although such measures can enhance safety and adaptability, participants appeared less likely to associate them directly with improved mobility or quality of life during crisis situations. These interventions may be perceived primarily as everyday regulatory tools rather than as mechanisms for crisis response or system robustness.

Finally, demand-management or pricing measures, including increased shared mobility options (17.39%) and congestion charges (13.04%), received the lowest levels of support. This indicates that participants are least likely to perceive economic or access-based restrictions as contributing positively to mobility or quality of life under crisis conditions. Concerns related to fairness, accessibility, or exclusion during emergencies may underlie this limited acceptance.

Overall, the antifragility-related findings suggest that social acceptability is highest for interventions that are perceived to strengthen resilience through redundancy, flexibility, and universal accessibility, rather than through restrictions or financial disincentives. Interventions that are visible, inclusive, and crisis-robust are more likely to be endorsed, whereas those that rely on

behavioral enforcement or economic regulation require careful contextualization to be socially accepted.

8.3.2 Post-survey insights

The post-survey results (n = 50) provide insight into how participants assessed different mobility interventions after their participation in the ACFs, that is, after expert input and facilitated discussion with other participants.

8.3.2.1 Antifragility

Similar to the pre-survey, the post-survey also asked participants to identify measures they believed would improve mobility and quality of life both in everyday situations and during crises, such as extreme weather events or emergencies (see Figure 51). This provides insight into how citizens perceive interventions that enhance the resilience and robustness of urban mobility systems.

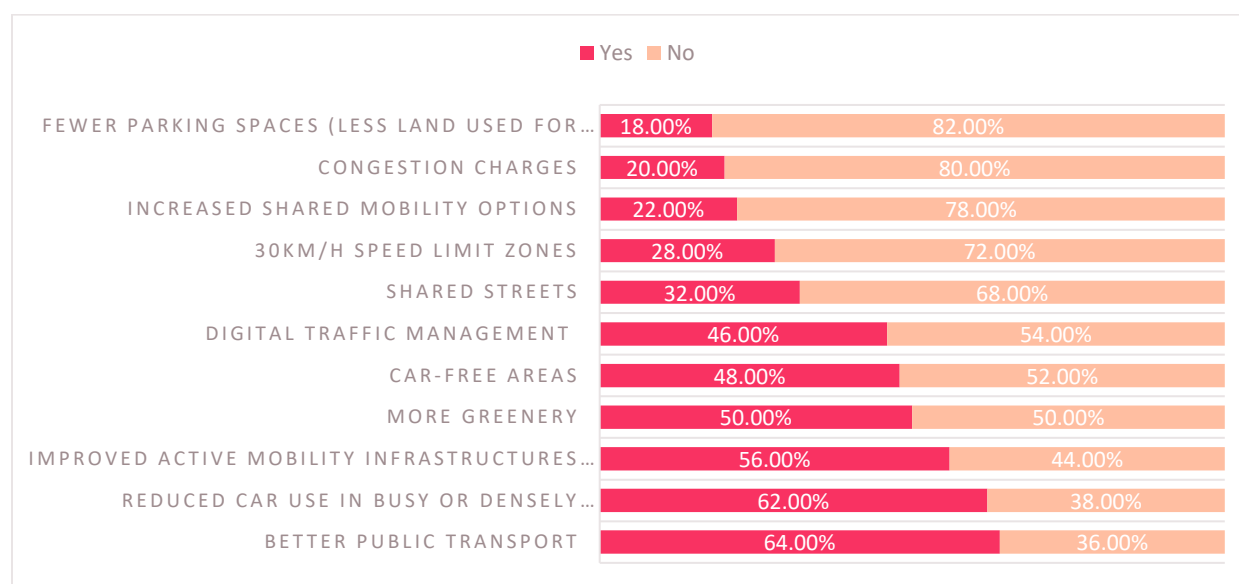


Figure 51 Antifragile measures

Participants expressed the strongest support for better public transport (64%), indicating that citizens widely perceive it as a foundational component of both everyday mobility and crisis response capacity. Public transport is understood as providing collective reliability, scalability, and continuity under disrupted conditions, thereby supporting accessibility when individual modes or routes are constrained.

Similarly, improved active mobility infrastructure for walking and cycling (56%) received strong backing, highlighting its role as a resilient alternative to motorized transport. These results suggest

that participants recognize the importance of locally accessible, low-dependence modes in crisis situations. Walking and cycling infrastructure is seen as robust in the face of disruptions such as fuel shortages, congestion, or partial system failures, reinforcing its contribution to mobility continuity during emergencies.

Measures with co-benefits beyond transport, including broader environmental and spatial benefits such as more greenery (50%), were also positively evaluated. Citizens appear to associate greenery not only with improved everyday quality of life but also with enhanced resilience to extreme weather events, such as heat stress or flooding. In this context, greenery is perceived as part of a supportive urban environment that indirectly strengthens mobility systems by improving comfort, safety, and environmental conditions during crises.

Interventions that restructure or regulate car use, including reduced car use in busy or densely populated areas (62%) and digital traffic management (46%), elicited more ambivalent responses. While a substantial share of participants recognized their potential benefits, these measures did not achieve overwhelming majority support. This suggests that their contribution to crisis resilience is less immediately apparent to citizens and may be understood as context-dependent.

Support declined further for interventions involving strong spatial or behavioral constraints, such as car-free areas (48%), 30 km/h speed limit zones (28%), shared streets (32%), and fewer parking spaces (18%). Although such measures can enhance safety and adaptability, participants appeared less likely to associate them directly with improved mobility or quality of life during crisis situations. These interventions may be perceived primarily as everyday regulatory tools rather than as mechanisms for crisis response or system robustness.

Finally, demand-management or pricing measures, including increased shared mobility options (22%) and congestion charges (20%), received the lowest levels of support. This indicates that participants are least likely to perceive economic or access-based restrictions as contributing positively to mobility or quality of life under crisis conditions. Concerns related to fairness, accessibility, or exclusion during emergencies may underlie this limited acceptance.

Overall, the antifragility-related findings suggest that social acceptability is highest for interventions perceived to strengthen resilience through redundancy, flexibility, and universal accessibility, rather than through restrictions or financial disincentives. Interventions that are visible, inclusive, and crisis-robust are more likely to be endorsed, whereas those relying on behavioral enforcement or economic regulation require careful contextualization to achieve public acceptance.

8.3.3 Cross-cutting interpretation

The comparison of pre- and post-participation responses provides a nuanced understanding of citizens' social acceptability of urban mobility interventions, particularly in the context of resilience and crisis preparedness.

Strong and Consistent Support for Core Mobility Measures

Across both pre- and post-participation stages, better public transport and active mobility infrastructure (walking and cycling) consistently received the highest levels of support. This suggests that citizens perceive these interventions as foundational for everyday mobility and for maintaining continuity under disrupted conditions. Public transport is valued for collective reliability, scalability, and system-wide continuity, while walking and cycling infrastructure is understood as robust, locally accessible alternatives that contribute to mobility resilience during crises.

- Pre-participation support: public transport 69.57%, active mobility 55.07%
- Post-participation support: public transport 64%, active mobility 56%

The small differences indicate that participation and deliberation slightly moderated expectations but did not fundamentally change citizens' priorities, reinforcing the central role of accessible, reliable, and flexible mobility systems.

Recognition of Environmental and Co-Benefit Measures

Measures with co-benefits beyond transport, such as more greenery, were positively evaluated at both stages (pre: 47.83%, post: 50%). Citizens appear to understand these interventions as enhancing both everyday quality of life and urban resilience, for instance by reducing heat stress, improving safety, or supporting emergency conditions. This demonstrates that social acceptability extends beyond purely mobility-focused outcomes to include integrated environmental and spatial benefits.

Mixed Responses to Car Use and Traffic Regulation Measures

Interventions that restructure or regulate car use, such as reduced car use in busy areas or digital traffic management, elicited ambivalent responses. While a notable portion of participants acknowledged their benefits, these measures generally did not achieve majority consensus, suggesting that their contribution to resilience is less intuitive or context-dependent. Interestingly, support for reduced car use increased post-participation (pre: 46.39%, post: 62%), implying that deliberation or exposure to system-wide perspectives may enhance acceptance when benefits are clarified.

Low Support for Strong Behavioral or Pricing Constraints

Interventions imposing strict spatial, behavioral, or economic constraints—including car-free areas, 30 km/h speed zones, shared streets, fewer parking spaces, congestion charges, or pricing mechanisms—received the lowest support at both stages. This reflects a perception that such measures are primarily regulatory or restrictive, rather than immediately contributing to crisis-resilient mobility. Concerns around fairness, accessibility, or exclusion likely underlie this limited acceptance. Minor fluctuations in percentages post-participation indicate that citizen deliberation does not dramatically shift perceptions of these restrictive measures.

8.3.3.1 Key Takeaways for Social Acceptability

The overall patterns reveal important insights:

1. **Visibility and inclusivity matter:** Interventions that are tangible, widely used, and support collective accessibility are more socially acceptable
2. **Redundancy and flexibility are valued:** Citizens prioritize measures that strengthen mobility continuity under disruption rather than those enforcing behavioral or financial compliance
3. **Deliberation slightly adjusts perception but reinforces core priorities:** The modest shifts in post-participation support (e.g., for reduced car use) suggest that informed discussion can enhance legitimacy for measures whose benefits are not immediately obvious
4. **Environmental and systemic co-benefits increase acceptability:** Measures like greenery are recognized for indirect contributions to resilience, indicating a holistic understanding of mobility within the urban system

In conclusion, social acceptability of urban mobility interventions is highest for measures that enhance resilience, accessibility, and system-wide functionality, while restrictive or pricing-based measures require careful framing and contextual justification to gain support. Deliberative participation can clarify benefits, slightly increase support for some measures, and reinforce the importance of visible, inclusive, and flexible interventions.

8.3.4 Recommendations

Across the AntifragiCity citizens' forums in Bratislava, Larissa, and Thessaloniki, a consistent set of insights emerged regarding how citizens evaluate, accept, and legitimize urban mobility interventions. Despite certain differences in scale, governance context, and deliberative formats, the forums collectively met their core aim: eliciting the social acceptability logic underpinning antifragile mobility planning.

Citizens did not focus on isolated measures; instead, they assessed interventions through a systemic lens. Social acceptability was shaped by whether measures contributed to a mobility system that is predictable, safe, coordinated, and trustworthy, especially under conditions of disruption or stress. Interventions were judged not only by their technical merits but by how they fit into a coherent whole and how fairly they were developed and communicated.

8.3.4.1 Core findings

1. Structural capacity and long-term investments are socially acceptable and often expected

Citizens support structural, transformative interventions framed as improving system-wide capacity, reliability, and fairness. High-capacity public transport, connected cycling and pedestrian networks, multimodal hubs, and integrated governance frameworks were prioritized. This challenges assumptions that the public prefers only incremental changes; visible, broadly shared, and resilience-linked investments are socially acceptable.

2. Safety, accessibility, and inclusivity function as threshold conditions

Across all three cities, safety – both objective and perceived – was a baseline condition for acceptance. Measures included lighting, crossings, separation of modes, barrier-free access, shaded walking routes, and safe interchanges. Vulnerable populations, particularly in crises, require inclusive planning. Without these, other goals such as sustainability or efficiency lose legitimacy.

3. Citizens prioritize system coherence over ad hoc or reactive measures

Citizens favored predictable, rule-based, coordinated systems over fragmented or improvised responses. Binding mobility plans, integrated crisis governance, coordinated traffic management, shared data platforms, and multimodal integration were highly valued. Operational crisis measures were seen as expected standards rather than discretionary improvements. Social acceptability depends on meeting baseline expectations and delivering meaningful enhancements.

4. Governance quality and decision-making processes shape legitimacy

Trust in institutions and transparent decision-making processes strongly influence acceptability. Apolitical expert platforms, binding mobility plans, and coordinated governance frameworks were prioritized. Citizens value procedural fairness and inclusive participation as much as policy outcomes.

5. Communication and information are critical enablers of acceptance, especially in crises

Clear, timely, multi-channel communication – including rationale, context, and real-time updates – strengthens compliance and trust, particularly during disruptions. Citizens expect authorities to explain intent and provide context for measures affecting daily routines.

6. Responsibility for change is placed on systems, not individuals

Citizens do not frame mobility transformation primarily as a matter of individual behavior change. Structural adjustments – street space reallocation, parking reform, integrated hubs, and accessible transport – are seen as legitimate levers for shaping behavior indirectly. Social acceptability rises when institutions visibly lead change, enabling citizens rather than pressuring them.

8.3.4.2 Overall insight

Taken together, the AntifragiCity forums demonstrate that socially acceptable mobility interventions share a common profile. They:

- Enhance safety, accessibility, and inclusivity
- Strengthen long-term system capacity and resilience
- Are embedded in coherent, coordinated governance frameworks
- Are accompanied by transparent, timely, and intelligible communication
- Are developed through processes perceived as fair, inclusive, and legitimate

Citizens are willing to support prioritization, regulation, spatial reallocation, and long-term transformation when measures improve daily life, protect vulnerable users, and build trust.

8.3.4.3 Actionable principles

Looking across the Bratislava, Larissa, and Thessaloniki forums, we can distill actionable principles that go beyond site-specific suggestions, which can directly inform a model for antifragile urban mobility. These emerge from repeated patterns in what citizens accepted, prioritized, or treated as essential.

Structural and System-Level Principles

1. Prioritize foundational infrastructure first (tram, metro, high-capacity buses, cycling/pedestrian networks; ensure multimodal connectivity and redundancy)
2. Enable multimodality and seamless transfers (hubs, MaaS platforms, ticketing systems)
3. Safety and accessibility as threshold conditions (barrier-free design, audits, inclusive standards)

Governance and Decision-Making Principles

4. Predictable, coordinated, expert-led governance (expert bodies, integrated crisis frameworks, standard protocols)
5. Procedural legitimacy and inclusivity (structured deliberation, iterative feedback, transparent documentation)

Information and Communication Principles

6. Transparent, timely, multi-channel communication (explanations, rationale, real-time updates)
7. Anticipatory and adaptive data use (dashboards, predictive tools, continuous flow monitoring)

Engagement and Cultural Principles

8. Engage citizens proactively and locally (participatory programs, community campaigns, pilot projects)
9. Responsibility is systemic, not individual (structural interventions with incentives for sustainable choices)

Crisis-Responsive Principles

10. Visible and coordinated resilience (real-time dashboards, incident reporting, emergency route planning)
11. Inclusive crisis planning (access for vulnerable populations, personalized support)

Integration and Holistic Principles

12. Link interventions to system logic (coordinated, mutually reinforcing infrastructure, governance, communication, and cultural initiatives)
13. Prioritize long-term capacity over short-term fixes (sequence interventions: infrastructure, governance, safety first; operational enhancements second)

9 Conclusions

This section of D2.2 presents the key conclusions in a twofold manner. First, it introduces the AntifragiCity Model Framework, derived from insights gathered through the ACFs. Second, it synthesizes the main empirical findings, highlighting the core patterns and lessons emerging from the analysis of participation, deliberation, and social acceptability across the three case studies.

9.1 Socially acceptable and antifragile urban mobility

The insights from the ACFs combined with the recommendations generated during the forums provide the empirical basis for the AntifragiCity Model Framework, which conceptualizes socially acceptable and antifragile urban mobility (see Figure 52). The design of this system is informed by two complementary sources: (i) pre- and post-participation responses, and (ii) deliberation-generated recommendations from the forums.

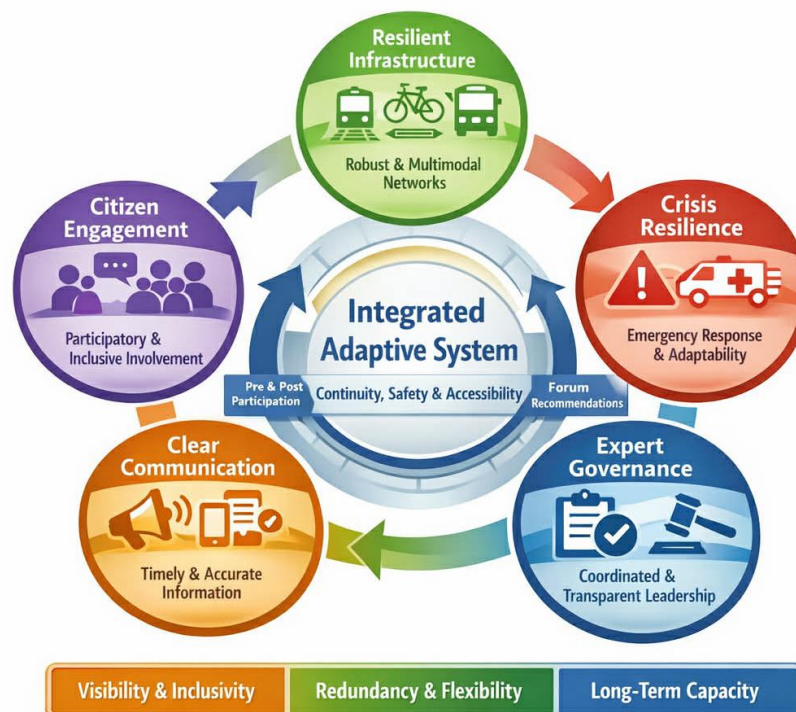


Figure 52 AntifragiCity Model Framework

The AntifragiCity Model Framework demonstrates that socially acceptable and antifragile urban mobility emerges from the integration of multiple interdependent domains into a coherent and adaptive system. Rather than operating in isolation, infrastructure, governance, crisis management, communication, and citizen engagement interact dynamically, forming feedback loops that support continuity, accessibility, and safety under both routine and disrupted conditions.

At the core of the framework lies an integrated adaptive system that continuously aligns operational performance with societal expectations. Deliberative engagement plays a central role in this process, as it reveals not only citizen preferences but also how these preferences evolve through interaction, reflection, and exposure to trade-offs. In this sense, the model reflects the broader insight that social

acceptability is not static but is shaped through deliberative processes, where legitimacy, understanding, and support are co-produced alongside policy ideas.

The framework is structured around five interdependent domains:

The first domain, **resilient infrastructure**, prioritizes robust and accessible mobility networks, including high-capacity public transport, active mobility infrastructure, and multimodal connectivity. These elements are designed with redundancy, safety, and inclusivity in mind, ensuring continuity of mobility services under both normal and disrupted conditions. Across all cases, such structural measures consistently received strong support, reflecting their perceived importance for everyday mobility as well as crisis preparedness.

The second domain, **crisis resilience**, focuses on the system's ability to respond flexibly to disruptions. This includes real-time adaptive management, incident response mechanisms, and inclusive emergency planning. Particular emphasis is placed on ensuring accessibility for vulnerable population groups. Citizens associate visible and well-coordinated resilience measures with trust and reliability, which in turn contributes to the social acceptability of interventions under uncertain or stressful conditions.

The third domain, **expert governance**, emphasizes transparent, coordinated, and predictable decision-making processes. Citizens value integrated mobility planning, clear institutional responsibilities, and coherent policy frameworks. Procedural legitimacy is reinforced through structured decision-making processes and participatory formats, including deliberative mechanisms that enable citizens to engage with information, question assumptions, and contribute to policy-relevant discussions. Governance is therefore not only a technical function but also a normative one, shaping trust and acceptance through fairness and clarity.

The fourth domain, **clear communication**, underlines the importance of accessible, timely, and multi-channel information. Citizens expect explanations of mobility measures, including their rationale, expected impacts, and real-time updates during disruptions. Effective communication supports informed engagement, enhances transparency, and strengthens public trust, particularly for measures whose benefits are not immediately visible or intuitive.

The fifth domain, **citizen engagement**, highlights the importance of proactive, inclusive, and locally grounded participation. Engagement formats such as community initiatives, pilot projects, and deliberative forums enable citizens to contribute experiential knowledge and contextual insights. Consistent with participant perspectives across the forums, mobility transformation is understood as a systemic responsibility shared between institutions and society, rather than as an individual behavioral burden. Structural interventions, such as reallocation of street space, development of multimodal hubs, and improvements in public transport accessibility, are perceived as legitimate enablers of sustainable mobility choices.

Across these domains, several cross-cutting principles emerge. First, social acceptability is highest for interventions that are visible, inclusive, and perceived as fair. Second, citizens consistently prioritize redundancy, flexibility, and reliability, particularly in the context of disruptions. Third, measures that generate co-benefits, such as improved environmental quality, public space, or urban livability, are more readily accepted, reflecting a holistic understanding of mobility within broader urban systems. Finally, deliberative engagement can contribute to shifts in perception by enabling participants to better understand trade-offs, contextual constraints, and the rationale behind policy measures.

Importantly, the AntifragiCity Model emphasizes long-term system capacity over short-term, isolated interventions. Effective transformation requires sequencing actions across domains, beginning with foundational infrastructure and governance structures, followed by operational enhancements and iterative refinements. The interaction between infrastructure, governance, communication, engagement, and crisis readiness creates continuous feedback loops, reinforcing both system performance and social legitimacy over time.

In summary, the AntifragiCity Model demonstrates that socially acceptable and antifragile urban mobility arises from systemic integration, participatory governance, and proactive adaptation. By aligning technical, institutional, and social dimensions within a coherent framework, cities can design mobility systems that are not only robust under disruption but also trusted, inclusive, and responsive to evolving societal needs.

9.2 To sum up

Through structured Social Acceptability Workshops implemented as citizens' forums in Bratislava, Larissa, and Thessaloniki, diverse residents were engaged in informed deliberation on mobility priorities, trade-offs, and adaptive responses. These forums generated both qualitative and quantitative insights, contributing empirically grounded knowledge on public perceptions, behavioral considerations, and normative expectations, providing a societal perspective essential for antifragile urban mobility planning. In line with the conceptual framing introduced earlier in this report, the deliberative format enabled not only the elicitation of opinions but also the observation of how these views are formed and refined through interaction, information exchange, and collective reasoning.

Importantly, the forums successfully convened diverse participant groups capable of high-quality deliberation, meeting the objective of generating meaningful insights despite differing local contexts and recruitment challenges. Variation in participation across the three cities reflects a combination of interacting factors, including institutional visibility, framing and communication strategies, incentive structures, local participatory cultures, and logistical conditions. These findings underscore that deliberative processes are highly context-sensitive and that their effectiveness depends not only

on methodological design but also on how they are embedded within local institutional and social environments. For EU research-oriented deliberative processes, key enabling conditions include strong institutional anchoring, accessible and locally meaningful framing, appropriate practical support for participants, alignment with existing participatory practices, and clear communication regarding follow-up and impact.

Across all three cases, participants consistently prioritized interventions that enhance accessibility, safety, inclusivity, and system reliability. Structural and system-level measures, such as high-capacity public transport, integrated cycling and pedestrian networks, multimodal hubs, and redundancy-oriented infrastructure, were generally associated with high levels of social acceptability. In contrast, measures that rely primarily on behavioral restrictions, spatial limitations, or financial disincentives tended to receive more cautious or mixed responses. This pattern highlights that citizens tend to evaluate mobility interventions holistically, emphasizing coherence, fairness, and institutional responsibility rather than focusing narrowly on individual behavior change.

At a broader level, the findings suggest that socially acceptable interventions share a common profile: they strengthen long-term system capacity, are embedded within coordinated governance frameworks, are supported by transparent and effective communication, and are perceived as inclusive and equitable. These characteristics align closely with the notion of antifragility, insofar as systems that are adaptable, participatory, and responsive to feedback are better able to evolve under uncertainty and stress.

From these insights, several actionable principles for antifragile urban mobility can be derived:

- Structural and system-level: prioritize robust, multimodal, safe, and accessible infrastructure
- Governance: ensure coordinated, transparent, and procedurally legitimate decision-making
- Information and communication: provide clear, timely, and multi-channel communication supported by reliable data
- Engagement and culture: foster inclusive participation and reinforce shared responsibility for mobility transitions
- Crisis responsiveness: implement visible, coordinated, and inclusive antifragile mechanisms
- Integration and long-term planning: align interventions across domains and prioritize long-term adaptive capacity over short-term fixes

Overall, the ACFs demonstrate that socially acceptable urban mobility interventions are those that are systemically integrated, context-sensitive, and grounded in both technical robustness and social legitimacy. By combining infrastructure development, governance structures, communication

strategies, and citizen engagement within a coherent adaptive framework, cities can better navigate the challenges of urban mobility transitions while maintaining public trust and support.

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